# T2K $6.914\times10^{20}$-POT Neutrino $4.011\times10^{20}$-POT Anti-Neutrino 3-Flavour Joint Analysis

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**Abstract**

We report the results of the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1+6 dataset, using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for CP violation by setting limits on $\delta_{CP}$ and measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, $\Delta m^2_{32}$ ($\Delta m^2_{13}$) ($\Delta m^2_{32}$ for normal hierarchy and $\Delta m^2_{13}$ for inverted hierarchy). Though T2K has limited sensitivity to the mass hierarchy given the relatively short baseline, fits under both hypotheses are performed to show which is preferred by data. Furthermore the search for $\bar{\nu}_e$ appearance and the measurement of $\bar{\nu}_\mu$ disappearance are shown as well.

This analysis adapts the work presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] to this dataset. The Run 1-6 dataset corresponds to an integrated J-PARC neutrino beam exposure of $6.914\times10^{20}$ POT in neutrino mode and $4.011\times10^{20}$ POT in antineutrino mode.

This analysis predicts 125.914 ± 6.459 (syst) for neutrino $\mu$-like, 34.284 ± 1.755 (syst) for antineutrino $\mu$-like, 26.594 ± 1.808 (syst) for neutrino $e$-like, and 3.218 ± 0.238 (syst) for antineutrino $e$-like single ring events in Super K, given the oscillation parameters of the Asimov data set A in Tab. 12, and observes respectively 124 for neutrino $\mu$-like, 34 for antineutrino $\mu$-like, 31 for neutrino $e$-like, 3 for antineutrino $e$-like.

A joint analysis of all the four selected samples is performed in a framework of 3-flavour oscillations including matter effects in constant-density matter.

For $\mu$-like samples the templates are built as a function of the reconstructed neutrino energy, while for $e$-like the analysis templates are built as a function of the reconstructed neutrino energy and lepton angle, in order to improve the separation of different flavour/reaction components and increase the sensitivity to $\delta_{CP}$. The observed single ring events spectra are compared to the expected spectra in order to infer all the 3-flavour oscillation parameters.

As in [10], all the systematic parameters are marginalised [55] in the fit and confidence intervals are quoted for all the oscillation parameters. Since the most important result of this analysis release is the search for CP violation, the Feldman-Cousins method [54] is performed only for the parameter $\delta_{CP}$. Since Feldman-Cousins requires a lot of CPU time, it will be computed during the first review of the note and added to a later version.

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## 1. Introduction

In this technical note, we report the results of a 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1-6 dataset, using data taken in both neutrino mode ($6.914\times10^{20}$ POT), with the polarity set to forward horn current (FHC), and antineutrino ($4.011\times10^{20}$ POT) mode with polarity set reverse horn current (RHC). This analysis builds on our other analyses presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] As in [10], a hybrid frequentist-Bayesian method [55], which consists of marginalizing instead of profiling the nuisance parameters, is used.

The consistency of the data with the expectation is given by comparing the predicted reconstructed neutrino energy and outgoing lepton angle spectra of 1 ring events at Super-K to the observed spectra. The method of calculation of these predicted spectra is presented in Sec. 2. This analysis uses flux tuning based on recent NA61/SHINE 2009 data [17][46]. It also uses an indirect extrapolation from ND280 to Super-K, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and $\nu_\mu$ for different event topologies with the neutrino interaction vertex either in FGD1 or FGD2 [29].

A significant difference between this and the previous analyses is that for $e$-like samples the analysis templates are built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ and $\nu_\mu \rightarrow \nu_e$, fundamental in measuring $\delta_{CP}$. Furthermore the $\nu_e$ intrinsic beam and NC components can also be better separated from signal. Though the new variable $\theta$ increase the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For $\mu$-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

Updated nominal predictions of the reconstructed neutrino energy and lepton angle spectrum of single ring events in Super-K are presented for Run 1-6 in Sec. 2

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [52]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 101 parameters including energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state re-scattering systematics plus the oscillation parameters (nuisance parameters). In this Run 1-6 analysis, all the nuisance parameters are marginalised [55] . The nuisance parameters and their effects on the extrapolated single ring event spectra are described in Sec. 3. The analysis method is presented in detail in Sec. 4.1

Fit validation and coverage studies are presented in Sec. 5.

The main goal of this analysis is to search for CP violation in the context of the PMNS matrix and confidence intervals are set using the Feldman-Cousins method [54]. Confidence intervals of the other oscillation parameters are set using the constant $\Delta\chi^2$ method. Sensitivity studies are shown in Sec. 5.1. The results of the measurement using the Run 1-6 dataset is given in Sec. 6 of this note.

Finally, a summary is given in Sec. 7.

## 2. Predictions of single ring event spectra in Super-K

### 2.1. Methodology

In this analysis a comparison of the reconstructed energy ($E_r$) and lepton angle ($\theta$) spectrum of single ring events observed at Super-K with the predicted spectrum is used to build the test statistic to quote confidence intervals of the oscillation parameters. For $\mu$-like samples the templates are built as a function of the reconstructed neutrino energy, as in the past analyses, and a unique $\theta$ bin is used. The predicted number $N_{SK;r,\theta}$ of single ring events in the $r$-th and $\theta$-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} \beta \cdot P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode $m$ in the true energy bin t, the reconstructed energy bin $r\prime$ and lepton angle bin $\theta$. The construction of the nominal MC templates is discussed in Sec. 2.2. Both flux-tuning weights based on NA61/SHINE hadron-production data (see Sec. 2.3) and flux and cross-section tuning weights based on ND280 data (see Sec. 2.4) are applied to the nominal MC templates. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode $m$, the true energy bin $t$, the reconstructed energy bin $r\prime$, the lepton angle bin $\theta$ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 3. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins $r$ and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode $m$; application of the term $P_{m;t}$ is discussed in Sec. 2.5.

### 2.2. Construction of nominal Super-K Monte Carlo templates

The nominal Super-K single ring MC analysis templates $N^{MC}_{SK;m;r\prime,\theta;t}$ are constructed by applying either the single $e$-like ring selection cuts or the single $\mu$-like ring selection cuts to the official Super-K MC samples. The selection is exactly the same for neutrino and antineutrino beam mode. In total 4 samples are obtained:

- neutrino mode single $\mu$-like ring sample
- antineutrino mode single $\mu$-like ring sample
- neutrino mode single $e$-like ring sample
- antineutrino mode single $e$-like ring sample

Details of the MC samples are given in Sec. 2.2.1. The calculation of the normalization of each MC sample is described in Sec. 2.2.2. The official single $\mu$-like ring selection cuts are listed in Sec. 2.2.3. while the official single $e$-like ring selection cuts are listed in Sec. 2.2.4. The list of MC templates used in this analysis is shown in Sec. 2.2.5, while the true and reconstructed energy as well as lepton angle binning of the templates is discussed in Sec. 2.2.6.

#### 2.2.1. Input Monte Carlo samples

This analysis uses the official 14a Super-K MC samples. These were generated using the 13a JNUBEAM flux, NEUT 5.3.2, skdetsim v13p90 and the 14b Super-K software [33] with minor tunings. The CCQE model is reweighted from the Spectral Function to a Relative Fermi Gas model with a relativistic Random Phase Approximation on an event by event basis. The NC1$\gamma$ events are scaled up by a factor of two in accordance with improved calculations of single photon emission in NC interactions [34]. Lastly, a simplified reweighting, as a function of the outgoing pion energy, is used to scale the CCCoherent events from the Rein Sehgal model to the Berger-Sehgal model prediction. Six MC samples were generated for both neutrino and antineutrino beam mode running (horn current $\pm$250kA). Four of the samples ($\bar{\nu}_\mu$, $\nu_\mu$, $\bar{\nu}_e$ and $\nu_e$) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample. The sixth sample is $\nu_e$ interactions generated with the $\nu_\mu$ flux, and will be denoted the "oscillated $\nu_e$" sample.

| Running mode | Neutrino type | Number of events (flux averaged) per 22.5 kt fiducial per $10^{21}$ POT | Number of generated events in true fiducial volume in all available MC |
|---|---|---|---|
| Neutrino | $\bar{\nu}_\mu$ | 50.964 | 634600 |
| | $\nu_\mu$ | 1396.245 | 634881 |
| | $\bar{\nu}_e$ | 2.884 | 126728 |
| | $\nu_e$ | 26.663 | 126899 |
| | Oscillated $\bar{\nu}_e$ | 52.939 | 126984 |
| | Oscillated $\nu_e$ | 1483.994 | 126916 |
| Antineutrino | $\bar{\nu}_\mu$ | 353.789 | 633988 |
| | $\nu_\mu$ | 208.369 | 634659 |
| | $\bar{\nu}_e$ | 6.811 | 127013 |
| | $\nu_e$ | 8.941 | 127262 |
| | Oscillated $\bar{\nu}_e$ | 376.420 | 126812 |
| | Oscillated $\nu_e$ | 215.099 | 127278 |

Table 1: Normalization of input Super-K MC samples [33]

*2.2.2. Normalization of Monte Carlo samples*

The normalization (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{2}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin $dE_\nu$, per unit area $dS$ and per POT, $\sigma_{H_2O}$ is the total interaction cross section on water for the given neutrino species, I is the beam intensity in terms of POT, $N_A$ is Avogadro's number, $A$ is the mass number for water, $\rho$ is the water density and L is the neutrino path length in the water volume.

The Super-K MC samples are normalised to the Run 1-6 POT ($6.914\times10^{20}$ POT for neutrino and $4.011\times10^{20}$ POT for antineutrino mode) using the numbers in Tab. 1. Assuming that all the MC files are used, each sample is first normalised to $10^{21}$ POT by weighting it with the ratio of the number of events per 22.5 kt per $10^{21}$ POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to $6.914\times10^{20}$ POT for neutrino and $4.011\times10^{20}$ for antineutrino mode, by weighting the result respectively by 0.6914 and 0.4011 (the Run 1-6 POT in units of $10^{21}$ POT).

*2.2.3. Single $\mu$-like ring event selection cuts*

The Super-K MC templates contain events from the $\nu_\mu$, $\bar{\nu}_\mu$, $\nu_e$, $\bar{\nu}_e$, oscillated $\nu_e$ and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single $\mu$-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is $\mu$-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay $e^-$ in the event (`nmue <= 1`).

The same selection is applied to both neutrino and antineutrino mode samples.

*2.2.4. Single e-like ring event selection cuts*

The Super-K MC templates contain events from the $\nu_\mu$, $\bar{\nu}_\mu$, $\nu_e$, $\bar{\nu}_e$, oscillated $\nu_e$ and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single $e$-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts
- the reconstructed vertex is at least 2 metres away from the ID wall
- the visible energy in the event is above 30 MeV
- there is exactly 1 reconstructed ring in the event
- the reconstructed ring is $e$-like
- number of decay electrons is zero
- the reconstructed neutrino energy is below 1250 MeV
- fiTQun rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

The same selection is applied to both neutrino and antineutrino mode samples.

*2.2.5. List of Monte Carlo templates*

For each Super-K MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6].

In our previous $\bar{\nu}_\mu$-disappearance and $\bar{\nu}_e$ appearance analyses the NC interaction modes were subdivided into NC1$\pi^\pm$, NC1$\pi^0$, NC coherent and NC other. In this analysis the new mode NC1$\gamma$, which was previously included in NC other, is used. We have also added 5 CC MEC templates for $\bar{\nu}_\mu$, $\nu_\mu$, $\bar{\nu}_e$, $\nu_e$ and oscillated $\nu_e$ and 5 templates for the oscillated $\bar{\nu}_e$ MC sample. This gives a total of 50 templates:

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC MEC,
- $\bar{\nu}_\mu$ CC1$\pi$,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^0$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC 1$\gamma$,
- $\nu_\mu$ CCQE,
- $\nu_\mu$ CC MEC,
- $\nu_\mu$ CC1$\pi$,
- $\nu_\mu$ CC coherent,

- $\nu_\mu$ CC other,
- $\nu_\mu$ (and $\rightarrow \nu_e$, $\nu_\tau$) NC1$\pi^\pm$,
- $\nu_\mu$ (and $\rightarrow \nu_e$, $\nu_\tau$) NC1$\pi^0$,
- $\nu_\mu$ (and $\rightarrow \nu_e$, $\nu_\tau$) NC coherent,
- $\nu_\mu$ (and $\rightarrow \nu_e$, $\nu_\tau$) NC other,
- $\nu_\mu$ (and $\rightarrow \nu_e$, $\nu_\tau$) NC 1$\gamma$,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC MEC,
- $\bar{\nu}_e$ CC1$\pi$,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^0$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC 1$\gamma$,
- $\nu_e$ CCQE,
- $\nu_e$ CC MEC,
- $\nu_e$ CC1$\pi$,
- $\nu_e$ CC coherent,
- $\nu_e$ CC other,
- $\nu_e$ (and $\rightarrow \nu_\mu$, $\nu_\tau$) NC1$\pi^\pm$,
- $\nu_e$ (and $\rightarrow \nu_\mu$, $\nu_\tau$) NC1$\pi^0$,
- $\nu_e$ (and $\rightarrow \nu_\mu$, $\nu_\tau$) NC coherent,
- $\nu_e$ (and $\rightarrow \nu_\mu$, $\nu_\tau$) NC other,
- $\nu_e$ (and $\rightarrow \nu_\mu$, $\nu_\tau$) NC 1$\gamma$,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC MEC,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1$\pi$,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC MEC,

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 66.077 | 0.035 | 4.291 | 0.002 | 0.225 | 0.001 | 70.631 |
| CC1pi | 16.973 | 0.017 | 1.566 | 0.001 | 0.067 | 0.000 | 18.625 |
| CCcoherent | 0.326 | 0.000 | 0.080 | 0.000 | 0.001 | 0.000 | 0.408 |
| MEC | 19.038 | 0.010 | 0.845 | 0.000 | 0.050 | 0.000 | 19.944 |
| CCother | 7.735 | 0.008 | 0.494 | 0.001 | 0.002 | 0.000 | 8.240 |
| NC1pi0 | 0.537 | 0.016 | 0.017 | 0.001 | N/A | N/A | 0.572 |
| NC1piPM | 3.944 | 0.089 | 0.142 | 0.008 | N/A | N/A | 4.183 |
| NCcoherent | 0.016 | 0.000 | 0.001 | 0.000 | N/A | N/A | 0.018 |
| NCother | 2.993 | 0.121 | 0.165 | 0.012 | N/A | N/A | 3.290 |
| NC1gamma | 0.004 | 0.000 | 0.000 | 0.000 | N/A | N/A | 0.004 |
| Total | 117.643 | 0.297 | 7.602 | 0.025 | 0.345 | 0.002 | 125.914 |

Table 2: Predicted number of events in the neutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $6.914\times10^{20}$ POT. The Asimov data set A in tab. 12 is used.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 6.993 | 0.004 | 12.792 | 0.005 | 0.006 | 0.014 | 19.813 |
| CC1pi | 2.707 | 0.003 | 3.033 | 0.002 | 0.002 | 0.003 | 5.750 |
| CCcoherent | 0.054 | 0.000 | 0.169 | 0.000 | 0.000 | 0.000 | 0.223 |
| MEC | 2.503 | 0.001 | 2.053 | 0.001 | 0.002 | 0.002 | 4.562 |
| CCother | 1.280 | 0.001 | 0.885 | 0.000 | 0.000 | 0.000 | 2.168 |
| NC1pi0 | 0.060 | 0.002 | 0.050 | 0.001 | N/A | N/A | 0.114 |
| NC1piPM | 0.378 | 0.015 | 0.380 | 0.010 | N/A | N/A | 0.784 |
| NCcoherent | 0.001 | 0.000 | 0.006 | 0.000 | N/A | N/A | 0.008 |
| NCother | 0.530 | 0.025 | 0.296 | 0.012 | N/A | N/A | 0.862 |
| NC1gamma | 0.000 | 0.000 | 0.000 | 0.000 | N/A | N/A | 0.000 |
| Total | 14.508 | 0.051 | 19.664 | 0.032 | 0.010 | 0.019 | 34.284 |

Table 3: Predicted number of events in the antineutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $4.011\times10^{20}$ POT. The Asimov data set A in tab. 12 is used.

- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC1$\pi$,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC other.

In Tab. 2, 3, 4 and 5 the predicted number of events is shown for all the selected samples and each flavour and reaction mode.

*2.2.6. Binning of Monte Carlo templates*

This analyses uses an updated binning scheme for reconstructed variables. The difference with previous analyses is that for $e$-like samples the analysis templates are built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring $\delta_{CP}$. Furthermore the $\nu_e$ intrinsic beam and NC components can also be better separated from signal. Though the new variable $\theta$ increases the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For $\mu$-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

$\mu$-like templates have 84 true energy bins and 73 reconstructed energy bins. $e$-like templates have 84 true energy bins, 25 reconstructed energy bins and 15 lepton angle bins.

For all templates the 84 true energy bins of constant width are the following:

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.056 | 2.129 | 0.001 | 0.088 | 15.875 | 0.099 | 18.249 |
| CC1pi | 0.016 | 0.348 | 0.000 | 0.027 | 1.688 | 0.017 | 2.096 |
| CCcoherent | 0.000 | 0.002 | 0.000 | 0.002 | 0.015 | 0.002 | 0.021 |
| MEC | 0.008 | 0.710 | 0.000 | 0.022 | 3.901 | 0.019 | 4.659 |
| CCother | 0.001 | 0.045 | 0.000 | 0.003 | 0.029 | 0.001 | 0.080 |
| NC1pi0 | 0.524 | 0.012 | 0.017 | 0.001 | N/A | N/A | 0.553 |
| NC1piPM | 0.099 | 0.002 | 0.003 | 0.000 | N/A | N/A | 0.105 |
| NCcoherent | 0.174 | 0.004 | 0.018 | 0.001 | N/A | N/A | 0.197 |
| NCother | 0.126 | 0.005 | 0.008 | 0.001 | N/A | N/A | 0.139 |
| NC1gamma | 0.464 | 0.007 | 0.021 | 0.001 | N/A | N/A | 0.494 |
| Total | 1.469 | 3.264 | 0.069 | 0.145 | 21.508 | 0.138 | 26.594 |

Table 4: Predicted number of events in the neutrino mode $e$-like sample obtained after applying the BANFF weight for a total exposure of $6.914\times10^{20}$ POT. The Asimov data set A in tab. 12 is used.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.002 | 0.189 | 0.003 | 0.339 | 0.375 | 1.156 | 2.065 |
| CC1pi | 0.002 | 0.040 | 0.001 | 0.060 | 0.057 | 0.137 | 0.298 |
| CCcoherent | 0.000 | 0.000 | 0.000 | 0.006 | 0.000 | 0.019 | 0.026 |
| MEC | 0.001 | 0.069 | 0.000 | 0.063 | 0.114 | 0.185 | 0.433 |
| CCother | 0.001 | 0.008 | 0.000 | 0.004 | 0.004 | 0.002 | 0.019 |
| NC1pi0 | 0.045 | 0.002 | 0.049 | 0.001 | N/A | N/A | 0.098 |
| NC1piPM | 0.011 | 0.000 | 0.008 | 0.000 | N/A | N/A | 0.020 |
| NCcoherent | 0.021 | 0.001 | 0.081 | 0.001 | N/A | N/A | 0.103 |
| NCother | 0.022 | 0.001 | 0.012 | 0.000 | N/A | N/A | 0.035 |
| NC1gamma | 0.037 | 0.002 | 0.081 | 0.002 | N/A | N/A | 0.121 |
| Total | 0.141 | 0.313 | 0.235 | 0.478 | 0.551 | 1.500 | 3.218 |

Table 5: Predicted number of events in the antineutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $4.011\times10^{20}$ POT. The Asimov data set A in tab. 12 is used.

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 50-MeV bins from 0-1.25 GeV.

which is identical to the past analyses.

For $\mu$-like samples the 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,

- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

The $\theta$ binning used for $e$-like samples is

- 14 bins with constant width from 0° to 140°,
- 1 bin for the range $140° - 180°$.

that is the same as used by the $p - \theta$ analysis group [15].

For $\mu$-like samples a unique $\theta$ bin, that covers the whole range 0° - 180°, is used. The same reconstructed neutrino energy and lepton angle binning is used for the fit of the single ring reconstructed energy spectra. The predicted $e$-like 2-dimensional spectra are shown in App. B for each flavour component and compared with the 1-dimensional reconstructed energy spectra. It is clear that the components $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ are better separated by the reconstructed energy Vs $\theta$ templates.

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4]. The binning described here is unchanged with respect to the Run 1+2+3+4 joint fit analysis.

### *2.3. Applying flux tuning based on NA61/SHINE data*

In this analysis, the nominal MC templates, which are constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which is tuned on NA61/SHINE hadron production measurements [36][45]. The reweighted MC templates are denoted the *NA61/SHINE-tuned* MC templates in this document. The applied weights are obtained from [36] and are shown as a function of the true neutrino energy in Figs. 1-4. The effect of this flux tuning on the predicted spectra events is discussed in Sec. 2.6

### *2.4. Applying flux and cross-section (BANFF) tuning based on ND280 data*

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and $\nu_\mu$ ND280 samples to find the best-fit values of the flux normalisation and cross-section parameters [28]. This analysis uses these best-fit values to reweight the NA61/SHINE-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their $1\sigma$ prefit and postfit errors is given in Tab. 6, 7, 8 and 9.

The effects of the BANFF tune on the predicted spectra is discussed in Sec. 2.6.

### *2.5. Applying neutrino oscillations*

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\overset{(-)}{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), $\Delta m^2_{21}$ and $\delta_{CP}$. The $\overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_e \to \overset{(-)}{\nu}_e)$, while the oscillated $\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e)$ (the sample was generated assuming 100% of $\overset{(-)}{\nu}_\mu$ transform to $\overset{(-)}{\nu}_e$). All these oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm$^3$). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code are first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [69], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of $\overset{(-)}{\nu}_e$ and $\overset{(-)}{\nu}_\mu$ can yield $\overset{(-)}{\nu}_\tau$. In this analysis, we neglect contributions from $\nu_\tau$-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect

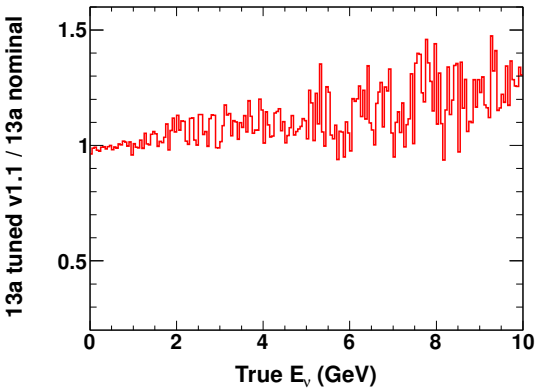



Figure 1: $\nu_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\nu_\mu$ and oscillated-$\nu_e$ MC templates [36].

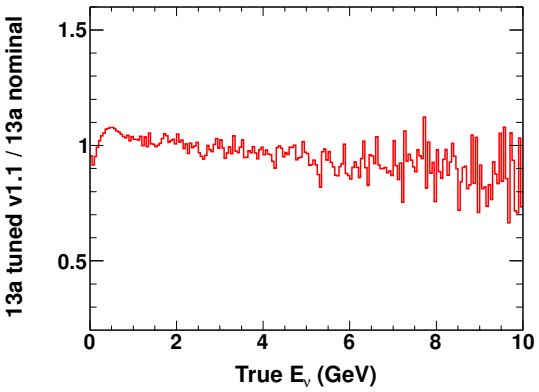



Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [36].

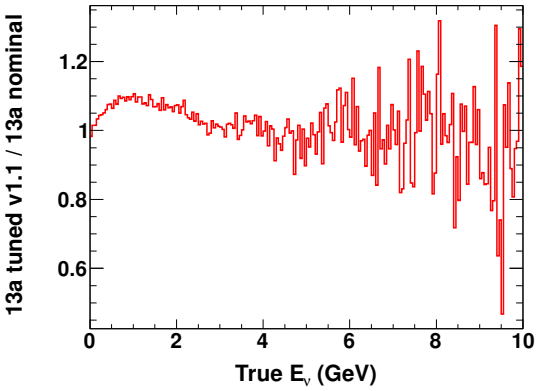



Figure 3: $\nu_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\nu_e$ MC templates [36].

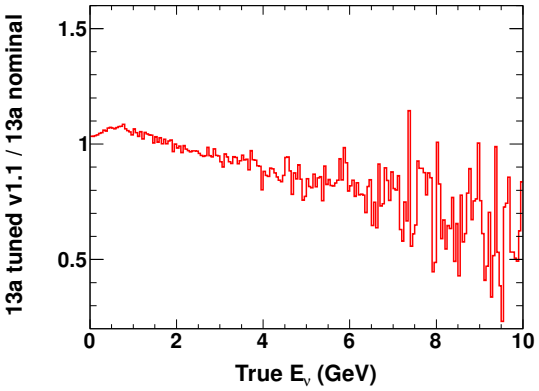



Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [36].

is negligible[1]. Accordingly, this analysis uses no $\nu_\tau$-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no $\nu_\tau$ and $\bar{\nu}_\tau$ Super-K MC samples are available).

It should be emphasized here that the $\nu_\mu$ NC MC templates for a mode $m$ are proxies for the NC MC templates for the mixture of $\nu_e$+$\nu_\mu$+$\nu_\tau$ resulting from 3-flavour $\nu_\mu$ oscillations for that mode $m$. The same applies to the $\bar{\nu}_\mu$, $\nu_e$ and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

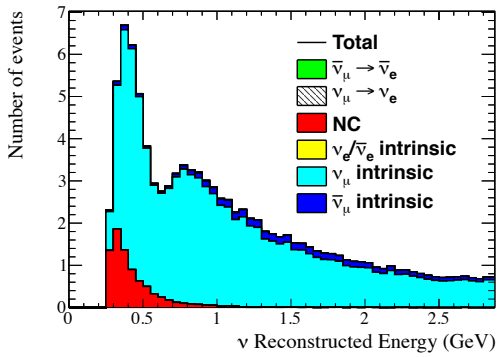
Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and $\nu_e$ samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated $\nu_e$.

### *2.6. Predictions of nominal and tuned-MC spectra*

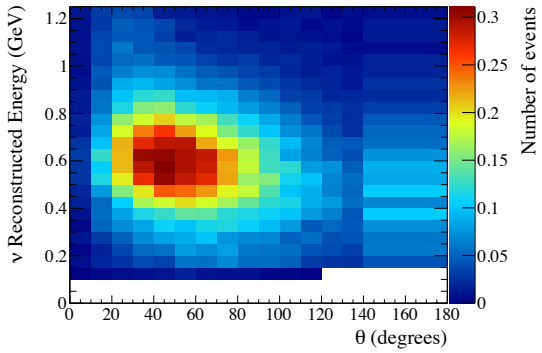
In this section, we present the expected spectra in Super-K for various oscillation scenarios. Unless explicitly stated otherwise, the oscillation parameters are set to the values shown in Tab. 12 and the normal mass hierarchy is assumed. All plots are generated for an integrated exposure of $6.914\times10^{20}$ POT for neutrino and $4.011\times10^{20}$ POT for antineutrino mode.

Predicted single ring Super-K spectra are shown in Fig. 5. By comparing neutrino and antineutrino mode $e$-like plots it is clear that $E_{reco}$ Vs $\theta$ distributions are differently populated by $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events. More details on the advantage of the 2-dimensional templates are given in App. B.

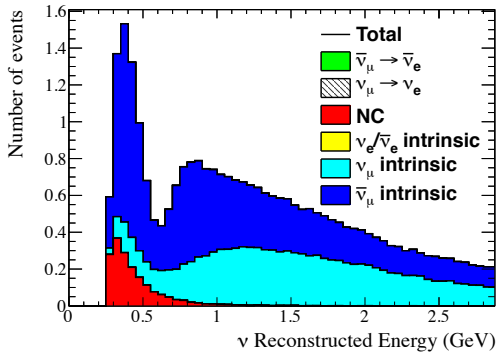
[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ $\nu_\tau$ contamination in the 1-ring $\mu$-like Super-K sample. A Super-K MC study is required to determine whether the Super-K



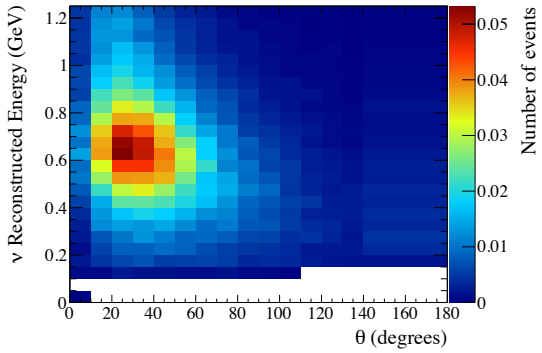

(a) Muon-like, neutrino mode




(b) Electron-like, neutrino mode




(c) Muon-like, antineutrino mode




(d) Electron-like antineutrino mode

Figure 5: Predicted spectra, muon-like distributions are function of the reconstructed neutrino energy, while the electron-like distributions are function of both the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full run 1-6 data set for an exposure of $6.914 \times 10^{20}$ POT in neutrino mode and of $4.011 \times 10^{20}$ POT in antineutrino mode. The muon-like distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), $\nu_\mu$ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra are generated with the systematic parameters measured with the ND280 data shown in Tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

efficiency assumptions made there are valid.

## 3. Systematic effects in the prediction of the single $\mu$-like ring event energy spectrum

### *3.1. Input systematic parameters*

This analysis considers 101 systematic parameters which can be grouped into 4 categories:

1. **Flux and cross-section parameters** (65 parameters).

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit. However, since ND280 is not sensitive to some cross section systematics they remain unconstrained (detailed later). Parameters fit are 11 $\nu_\mu$, 5 $\bar{\nu}_\mu$, 7 $\nu_e$ and 2 $\bar{\nu}_e$ neutrino mode (FHC) flux parameters, 5 $\nu_\mu$, 11 $\bar{\nu}_\mu$, 2 $\nu_e$ and 7 $\bar{\nu}_e$ antineutrino mode (RHC) flux parameters and 15 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and $\nu_\mu$ ND280 samples as described in Ref [28]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 6, 7, 8 and 9. The best-fit values are used to reweight the templates as described in Sec. 2.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis.

The effects of these correlated uncertainties are evaluated using the Cholesky method, that allows to throw the systematic parameters according to their multivariate distribution taking into account the existing correlations.

Some of the cross-section parameters have a complicated, often non-linear, effects on true and reconstructed neutrino energy bin normalisations; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3$\sigma$ to +3$\sigma$ in steps of 0.5$\sigma$, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7$\sigma$ to +3$\sigma$). Cubic splines are made for the following cross-section uncertainties:

- CCQE axial-mass scaling factor (MaQE)
- Fermi momentum for $^{16}$O
- Binding energy for $^{16}$O
- Nucleon to $\Delta$ transition axial form factor ($C_5^A$)
- Resonance-production axial-mass scaling factor (MaRes)
- Scale of isospin 1/2 nonresonant background
- CC other shape

The splines are made in bins of theta, true and reconstructed energy, and the binning is given in Ref. [35].

The effects of the flux and cross-section normalization uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalization for a limited range of true energy). The cross section parameters which simply normalise the overall spectrum component are:

- MEC, applied to all the MEC events (both neutrino and antineutrino)
- MEC-$\bar{\nu}$, applied to all the antineutrino MEC events
- CC coherent for $^{16}$O normalisation
- NC coherent normalisation
- NC other normalisation
- NC 1$\gamma$ normalisation
- CC $\nu_e$ normalisation
- CC $\bar{\nu}_e$ normalisation

Uncertainties due to "second class currents" are implemented as an additional uncertainty on CC $\bar{\nu}_e$ and CC $\nu_e$ parameters with a 2% anti-correlated constraint between them. The BANFF parameters are applied to the MC templates as specified in Tab. 37, 38, 39 and 40. The normalisation parameter of the NC1$\gamma$ component was not used in the previous analyses, since it was included in NC other.

In the BANFF fit a bias on the best-fit flux parameters is present due to the degeneracy between the neutrino flux and ND280 detector parameters. For this reason a covariance matrix that accounts for the bias of the flux as well as all the other parameters is summed in quadrature to the total post-fit covariance. Furthermore the best-fit values have been corrected in order to remove the bias. The description of the procedure is given in [28]. As shown in section D the impact of the bias on the systematic parameters fitted by the BANFF analysis is small compared to the other systematic uncertainties.

The predicted number of events in antineutrino mode $e$-like sample is now smaller compared to the $\bar{\nu}_e$ appearance analysis presented at the 2015 summer conferences [10], 3.218 instead of 3.729. The differences between the current and previous oscillation analyses is mainly due to:

(a) new value of $\sin^2\theta_{13}$ corresponding to the PDG 2015 release, as shown in Tab. 12, which is 0.0217 while in the previous analysis it was 0.0250. If the old PDG value is used the expected number of antineutrino mode $e$-like events becomes 3.542.

(b) Subsequently, if the cross section parameters are set to the old post-BANFF fit values, the expected number of events becomes 3.63. Indeed all the post-BANFF 2015 fit cross section parameters are higher compared to the 2016 BANFF fit result. The effect of the flux parameters is smaller, indeed if they are set to BANFF 2015 result, the expected number of events becomes 3.66.

(c) Finally, after all the changes listed above, the rescaling of CC coherent and NC1$\gamma$ components (see 2.2.1) is not applied, the agreement between the 2015 and 2016 analysis is obtained. The first one increases the number of events up to 3.772 while the latter one reduces down to 3.711, which is approximately the number of events expected in the 2015 analysis.

| Parameter | Description | Best fit | $1\sigma$ pre/postfit fractional error |
|---|---|---|---|
| 0 | FHC $\nu_\mu$ flux normalisation, E = 0.0 - 0.4 GeV | 1.128 | 0.099 / 0.064 |
| 1 | FHC $\nu_\mu$ flux normalisation, E = 0.4 - 0.5 GeV | 1.156 | 0.103 / 0.061 |
| 2 | FHC $\nu_\mu$ flux normalisation, E = 0.5 - 0.6 GeV | 1.148 | 0.096 / 0.051 |
| 3 | FHC $\nu_\mu$ flux normalisation, E = 0.6 - 0.7 GeV | 1.128 | 0.087 / 0.043 |
| 4 | FHC $\nu_\mu$ flux normalisation, E = 0.7 - 1.0 GeV | 1.104 | 0.113 / 0.046 |
| 5 | FHC $\nu_\mu$ flux normalisation, E = 1.0 - 1.5 GeV | 1.100 | 0.092 / 0.045 |
| 6 | FHC $\nu_\mu$ flux normalisation, E = 1.5 - 2.5 GeV | 1.127 | 0.070 / 0.044 |
| 7 | FHC $\nu_\mu$ flux normalisation, E = 2.5 - 3.5 GeV | 1.124 | 0.074 / 0.048 |
| 8 | FHC $\nu_\mu$ flux normalisation, E = 3.5 - 5.0 GeV | 1.121 | 0.087 / 0.049 |
| 9 | FHC $\nu_\mu$ flux normalisation, E = 5.0 - 7.0 GeV | 1.075 | 0.098 / 0.053 |
| 10 | FHC $\nu_\mu$ flux normalisation, E = 7.0 - 30.0 GeV | 1.064 | 0.114 / 0.065 |
| 11 | FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.7 GeV | 1.100 | 0.103 / 0.081 |
| 12 | FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV | 1.112 | 0.079 / 0.048 |
| 13 | FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV | 1.111 | 0.084 / 0.060 |
| 14 | FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV | 1.116 | 0.086 / 0.070 |
| 15 | FHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 30.0 GeV | 1.162 | 0.086 / 0.069 |
| 16 | FHC $\nu_e$ flux normalisation, E = 0.0 - 0.5 GeV | 1.134 | 0.090 / 0.052 |
| 17 | FHC $\nu_e$ flux normalisation, E = 0.5 - 0.7 GeV | 1.135 | 0.090 / 0.049 |
| 18 | FHC $\nu_e$ flux normalisation, E = 0.7 - 0.8 GeV | 1.135 | 0.086 / 0.047 |
| 19 | FHC $\nu_e$ flux normalisation, E = 0.8 - 1.5 GeV | 1.119 | 0.081 / 0.043 |
| 20 | FHC $\nu_e$ flux normalisation, E = 1.5 - 2.5 GeV | 1.115 | 0.079 / 0.046 |
| 21 | FHC $\nu_e$ flux normalisation, E = 2.5 - 4.0 GeV | 1.111 | 0.084 / 0.050 |
| 22 | FHC $\nu_e$ flux normalisation, E = 4.0 - 30.0 GeV | 1.118 | 0.094 / 0.067 |
| 23 | FHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 2.5 GeV | 1.121 | 0.074 / 0.057 |
| 24 | FHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 30.0 GeV | 1.153 | 0.128 / 0.117 |

Table 6: Summary of neutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

| Parameter | Description | Best fit | $1\sigma$ fractional error |
|---|---|---|---|
| 25 | RHC $\nu_\mu$ flux normalisation, E = 0.0 - 0.7 GeV | 1.098 | 0.094 / 0.072 |
| 26 | RHC $\nu_\mu$ flux normalisation, E = 0.7 - 1.0 GeV | 1.121 | 0.079 / 0.052 |
| 27 | RHC $\nu_\mu$ flux normalisation, E = 1.0 - 1.5 GeV | 1.130 | 0.077 / 0.048 |
| 28 | RHC $\nu_\mu$ flux normalisation, E = 1.5 - 2.5 GeV | 1.155 | 0.081 / 0.054 |
| 29 | RHC $\nu_\mu$ flux normalisation, E = 2.5 - 30.0 GeV | 1.111 | 0.080 / 0.055 |
| 30 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV | 1.118 | 0.104 / 0.071 |
| 31 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV | 1.127 | 0.102 / 0.060 |
| 32 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV | 1.117 | 0.096 / 0.052 |
| 33 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV | 1.121 | 0.085 / 0.044 |
| 34 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV | 1.155 | 0.125 / 0.066 |
| 35 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV | 1.132 | 0.105 / 0.057 |
| 36 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV | 1.139 | 0.080 / 0.053 |
| 37 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV | 1.141 | 0.074 / 0.054 |
| 38 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV | 1.151 | 0.094 / 0.071 |
| 39 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV | 1.133 | 0.093 / 0.070 |
| 40 | RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV | 1.082 | 0.130 / 0.110 |
| 41 | RHC $\nu_e$ flux normalisation, E = 0.0 - 2.5 GeV | 1.117 | 0.069 / 0.051 |
| 42 | RHC $\nu_e$ flux normalisation, E = 2.5 - 30.0 GeV | 1.112 | 0.085 / 0.071 |
| 43 | RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV | 1.126 | 0.095 / 0.058 |
| 44 | RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV | 1.127 | 0.091 / 0.051 |
| 45 | RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV | 1.133 | 0.091 / 0.052 |
| 46 | RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV | 1.132 | 0.084 / 0.046 |
| 47 | RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV | 1.125 | 0.080 / 0.056 |
| 48 | RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV | 1.119 | 0.089 / 0.071 |
| 49 | RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV | 1.166 | 0.156 / 0.141 |

Table 7: Summary of antineutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

| Parameter | Description | Best fit | $1\sigma$ pre/postfit fractional error |
|---|---|---|---|
| 50 | Meson exchange current normalisation for $^{16}$O | 1.546 | - / 0.343 |
| 51 | $C_5^A$ nucleon to $\Delta$ transition axial form factor | 0.789 | 0.119 / 0.062 |
| 52 | Scale of isospin 1/2 nonresonant background | 1.042 | 0.154 / 0.134 |
| 53 | CCQE axial-mass scaling factor | 0.931 | - / 0.028 |
| 54 | Resonance-production axial-mass scaling factor | 0.889 | 0.158 / 0.041 |
| 55 | BANFF; Second current class axial | 1.000 | 1.000 / 1.000 |
| 56 | BANFF; Second current class vector | 1.000 | 1.000 / 1.000 |
| 57 | Fermi momentum for $^{16}$O | 1.041 | - / 0.105 |
| 58 | CC other shape | -0.022 | 0.400 / 0.208 |
| 59 | Binding energy for $^{16}$O | 0.882 | 0.333 / 0.282 |
| 60 | CC coherent for $^{16}$O normalisation | 0.858 | 0.300 / 0.228 |
| 61 | NC coherent normalisation | 0.931 | 0.300 / 0.298 |
| 62 | NC other normalisation | 1.000 | 0.300 / 0.300 |
| 63 | CC $\nu_e$ normalisation | 1.000 | 0.028 / 0.028 |
| 64 | NC $1\gamma$ normalisation | 1.000 | 1.000 / 1.000 |
| 65 | CC $\bar{\nu}_e$ normalisation | 1.000 | 0.028 / 0.028 |
| 66 | Antineutrino Meson exchange current normalisation for $^{16}$O | 0.578 | - / 0.177 |

Table 8: Summary of cross section systematics included in the *VALOR* joint fit analysis. Parameters with no prefit error were unconstrained. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

| Parameter | Description | $1\sigma$ fractional error |
|---|---|---|
| 0 | SKDet + FSI/SI 0; $E_{reco}$ 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu) | 0.018 |
| 1 | SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu) | 0.017 |
| 2 | SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu) | 0.018 |
| 3 | SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu) | 0.077 |
| 4 | SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sig$\nu_e$ CC (1Rmu) | 1.004 |
| 5 | SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu) | 0.598 |
| 6 | SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc $\nu_e$ CC (1Re) | 0.100 |
| 7 | SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc $\nu_e$ CC (1Re) | 0.030 |
| 8 | SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc $\nu_e$ CC (1Re) | 0.033 |
| 9 | SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re) | 1.225 |
| 10 | SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re) | 1.166 |
| 11 | SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re) | 1.028 |
| 12 | SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re) | 0.063 |
| 13 | SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re) | 0.040 |
| 14 | SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re) | 0.061 |
| 15 | SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re) | 0.488 |
| 16 | SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re) | 0.194 |
| 17 | SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re) | 0.590 |
| 18 | SKDet + FSI/SI 0; Ereco 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC | 0.018 |
| 19 | SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC | 0.016 |
| 20 | SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$CCQE (1Rmu); RHC | 0.016 |
| 21 | SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu); RHC | 0.093 |
| 22 | SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sig$\nu_e$ CC (1Rmu); RHC | 1.004 |
| 23 | SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu); RHC | 0.597 |
| 24 | SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc $\nu_e$ CC (1Re); RHC | 0.060 |
| 25 | SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc $\nu_e$ CC (1Re); RHC | 0.027 |
| 26 | SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc $\nu_e$ CC (1Re); RHC | 0.056 |
| 27 | SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC | 1.244 |
| 28 | SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC | 1.123 |
| 29 | SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC | 0.904 |
| 30 | SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC | 0.044 |
| 31 | SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC | 0.032 |
| 32 | SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC | 0.068 |
| 33 | SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re); RHC | 0.482 |
| 34 | SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re); RHC | 0.171 |
| 35 | SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re); RHC | 0.538 |

Table 9: Summary of SK detector + FSI +SI + PN systematics included in the *VALOR* joint fit analysis.

| Parameter | Sample | Category | Energy range (GeV) |
|---|---|---|---|
| 0 \| 18 | electron-like | oscillated $\nu_e/\bar{\nu}_e$ CC | $E_{reco} < 0.35$ |
| 1 \| 19 | electron-like | oscillated $\nu_e/\bar{\nu}_e$ CC | $0.35 < E_{reco} < 0.80$ |
| 2 \| 20 | electron-like | oscillated $\nu_e/\bar{\nu}_e$ CC | $0.80 < E_{reco} < 1.25$ |
| 3 \| 21 | electron-like | $\nu_\mu/\bar{\nu}_\mu$ CC | $E_{reco} < 0.35$ |
| 4 \| 22 | electron-like | $\nu_\mu/\bar{\nu}_\mu$ CC | $0.35 < E_{reco} < 0.80$ |
| 5 \| 23 | electron-like | $\nu_\mu/\bar{\nu}_\mu$ CC | $0.80 < E_{reco} < 1.25$ |
| 6 \| 24 | electron-like | $\nu_e/\bar{\nu}_e$ CC | $E_{reco} < 0.35$ |
| 7 \| 25 | electron-like | $\nu_e/\bar{\nu}_e$ CC | $0.35 < E_{reco} < 0.80$ |
| 8 \| 26 | electron-like | $\nu_e/\bar{\nu}_e$ CC | $0.80 < E_{reco} < 1.25$ |
| 9 \| 27 | electron-like | all NC | $E_{reco} < 0.35$ |
| 10 \| 28 | electron-like | all NC | $0.35 < E_{reco} < 0.80$ |
| 11 \| 29 | electron-like | all NC | $0.80 < E_{reco} < 1.25$ |
| 12 \| 30 | muon-like | $\nu_\mu/\bar{\nu}_\mu$ CCQE | $E_{reco} < 0.40$ |
| 13 \| 31 | muon-like | $\nu_\mu/\bar{\nu}_\mu$ CCQE | $0.40 < E_{reco} < 1.10$ |
| 14 \| 32 | muon-like | $\nu_\mu/\bar{\nu}_\mu$ CCQE | $1.10 < E_{reco} < 30.0$ |
| 15 \| 33 | muon-like | all $\nu_\mu/\bar{\nu}_\mu$ CCnQE | $E_{reco} < 30.0$ |
| 16 \| 34 | muon-like | $\nu_e/\bar{\nu}_e$ CC | $E_{reco} < 30.0$ |
| 17 \| 35 | muon-like | all NC | $E_{reco} < 30.0$ |

Table 10: Binning of Super-K selection efficiency systematic parameters in covariance matrices for both electron- and muon-like samples for each category and reconstructed neutrino energy range. The same parametrization is used for both neutrino and antineutrino mode samples. The parameter number is shown for both neutrino (0-17) and antineutrino (18-35) mode.

2. **Super-K efficiencies & effects of intranuclear and secondary re-interactions** (36 parameters).

The Super-K detector uncertainties include the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using $\nu_\mu$ CCQE, $\nu_\mu$ CCnQE, NC, and $\nu_e$ CC-enriched control samples. The NC PID error is evaluated using a NC-enriched control sample, which does not use PID information. Each Monte Carlo control sample is fitted to the data using the efficiency as a fit parameter, and the uncertainties in the efficiencies are estimated by adding the covariance matrix containing the differences between the best-fit and nominal efficiencies to the covariance matrix containing the statistical errors in the fitted efficiencies [30] [31].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [20]. The NIWG group produced 16 parameter sets that span the $1\sigma$ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [20]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the Super-K detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [23]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single $e$-like or $\mu$-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [19]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [20]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the Super-K detector efficiency and FSI+SI systematics together, and add their covari-

| Parameter(s) | Prior | Range |
|---|---|---|
| $\sin^2\theta_{23}$ | uniform | $[0.3; 0.7]$ |
| $\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors | gauss | $0.085 \pm 0.005$ |
| $\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) T2K only | uniform | $[0; 0.4]$ |
| $\sin^2 2\theta_{12}$ | gauss | $0.846 \pm 0.021$ |
| $\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH) | uniform | $[2; 3] \times 10^{-3}$ eV$^2$/c$^4$ |
| $\Delta m^2_{21}$ | gauss | $(7.53 \pm 0.18) \times 10^{-5}$ eV$^2$/c$^4$ |
| $\delta_{CP}$ | uniform | $[-\pi; +\pi]$ |
| Mass Hierarchy | fixed | NH or IH |

Table 11: Treatment of the oscillation parameters in the $\nu/\bar{\nu}$ joint analysis. All the gaussian priors are from [66]. The parameter $\sin^2 2\theta_{13}$ can be constrained with the measurement of reactor experiments using the gaussian prior, otherwise a uniform prior is used and the measurement relies only on the T2K data. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and $\Delta m^2_{21}$ are from [66], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH.

ance matrices linearly. Each of these covariance matrices have 36 correlated parameters, corresponding to different modes and reconstructed energy bins, as shown in Tab. 10. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **Super-K energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the Super-K reconstructed energy scale. This uncertainty is estimated to be 2.4% [32]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in Tab. 37, 38, 39 and 40 and is common to all the super-K samples.

4. **Oscillation parameters and mass hierarchy** (7 parameters).

The 3-flavour oscillation probability is used. All the nuisance oscillation parameters, i.e. those parameters which are not shown in a given contour, are marginalised except the mass hierarchy which is fixed to either normal or inverted hierarchy. The treatment of the oscillation parameters in the analysis is shown in Tab. 11 while the nominal values are shown in Tab. 12. For $\sin^2\theta_{13}$ two different priors can be used, depending on whether the reactor constraint is applied, denoted as "reactors", or not, denoted as "T2K only". In the first case a gaussian prior given by the combined measurement of reactor experiments is used, while in the latter case a flat prior which covers a large range of values is adopted. With respect to the older global fit [65], the value of $\sin^2\theta_{13}$ has been reduced of about 10% and the error reduced as well. Since an excess of events was seen in the T2K Run 1-4 neutrino mode $e$-like sample and not much data are collected in neutrino mode during the Run 5+6, this change is expected to have an impact on the search for CP violation, making the confidence intervals tighter.

| Parameter(s) | Asimov A | Asimov B |
|---|---|---|
| $\sin^2\theta_{23}$ | 0.528 | 0.45 |
| $\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors | 0.085 (0.0217) | 0.085 (0.0217) |
| $\sin^2 2\theta_{12}$ | 0.846 | 0.846 |
| $\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH) | $2.509 \times 10^{-3}$ eV$^2$/c$^4$ | $2.509 \times 10^{-3}$ eV$^2$/c$^4$ |
| $\Delta m^2_{21}$ | $7.53 \times 10^{-5}$ eV$^2$/c$^4$ | $7.53 \times 10^{-5}$ eV$^2$/c$^4$ |
| $\delta_{CP}$ | -1.601 | 0 |
| Mass Hierarchy | Normal | Normal |

Table 12: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and $\Delta m^2_{21}$ are from [66], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH. The only difference between the Asimov data sets A and B is given by $\delta_{CP}$, set to 0, and $\sin^2\theta_{23}$ which corresponds to the opposite octant.

### *3.2. Evaluation of the effects of systematic parameters on the predicted single e-like ring reconstructed energy spectrum*

In this section, we present the effects of the systematics described in Sec. 3.1 on the predicted number of single ring events and the predicted spectra in Super-K. All results are generated for an integrated exposure of $6.914\times10^{20}$ POT for neutrino and $4.011\times10^{20}$ POT for antineutrino mode and the BANFF-tuned MC templates are used.

In Tab. 13, 14, 15 and 16 the effect of each single systematic parameter obtained by varying its value by $\pm1\sigma$ is shown.

| Parameter | Nominal | BANFF tuned | BANFF tuned $+1\sigma$ | | BANFF tuned $-1\sigma$ | |
|---|---|---|---|---|---|---|
| $f^{banff}_{0;t,r}$ | 120.349 | 125.914 | 126.429 | 0.40884 % | 125.399 | -0.40884 % |
| $f^{banff}_{1;t,r}$ | 120.349 | 125.914 | 126.409 | 0.39338 % | 125.418 | -0.39338 % |
| $f^{banff}_{2;t,r}$ | 120.349 | 125.914 | 126.033 | 0.094558 % | 125.795 | -0.094558 % |
| $f^{banff}_{3;t,r}$ | 120.349 | 125.914 | 126.012 | 0.078072 % | 125.815 | -0.078072 % |
| $f^{banff}_{4;t,r}$ | 120.349 | 125.914 | 126.834 | 0.73082 % | 124.994 | -0.73082 % |
| $f^{banff}_{5;t,r}$ | 120.349 | 125.914 | 126.771 | 0.68095 % | 125.056 | -0.68095 % |
| $f^{banff}_{6;t,r}$ | 120.349 | 125.914 | 126.733 | 0.65073 % | 125.094 | -0.65073 % |
| $f^{banff}_{7;t,r}$ | 120.349 | 125.914 | 126.452 | 0.42768 % | 125.375 | -0.42768 % |
| $f^{banff}_{8;t,r}$ | 120.349 | 125.914 | 126.471 | 0.44277 % | 125.356 | -0.44277 % |
| $f^{banff}_{9;t,r}$ | 120.349 | 125.914 | 126.126 | 0.16892 % | 125.701 | -0.16892 % |
| $f^{banff}_{10;t,r}$ | 120.349 | 125.914 | 125.96 | 0.036964 % | 125.867 | -0.036964 % |
| $f^{banff}_{11;t,r}$ | 120.349 | 125.914 | 125.941 | 0.021242 % | 125.887 | -0.021242 % |
| $f^{banff}_{12;t,r}$ | 120.349 | 125.914 | 125.934 | 0.015671 % | 125.894 | -0.015671 % |
| $f^{banff}_{13;t,r}$ | 120.349 | 125.914 | 126.002 | 0.069974 % | 125.826 | -0.069974 % |
| $f^{banff}_{14;t,r}$ | 120.349 | 125.914 | 126.08 | 0.13209 % | 125.747 | -0.13209 % |
| $f^{banff}_{15;t,r}$ | 120.349 | 125.914 | 126.063 | 0.1185 % | 125.765 | -0.1185 % |
| $f^{banff}_{16;t,r}$ | 120.349 | 125.914 | 125.914 | 0.00034323 % | 125.913 | -0.00034323 % |
| $f^{banff}_{17;t,r}$ | 120.349 | 125.914 | 125.915 | 0.00058236 % | 125.913 | -0.00058236 % |
| $f^{banff}_{18;t,r}$ | 120.349 | 125.914 | 125.914 | 0.00041762 % | 125.913 | -0.00041762 % |
| $f^{banff}_{19;t,r}$ | 120.349 | 125.914 | 125.916 | 0.0021262 % | 125.911 | -0.0021262 % |
| $f^{banff}_{20;t,r}$ | 120.349 | 125.914 | 125.917 | 0.0028071 % | 125.91 | -0.0028071 % |
| $f^{banff}_{21;t,r}$ | 120.349 | 125.914 | 125.917 | 0.0024707 % | 125.911 | -0.0024707 % |
| $f^{banff}_{22;t,r}$ | 120.349 | 125.914 | 125.916 | 0.0015978 % | 125.912 | -0.0015978 % |
| $f^{banff}_{23;t,r}$ | 120.349 | 125.914 | 125.914 | 0.00041569 % | 125.913 | -0.00041569 % |
| $f^{banff}_{24;t,r}$ | 120.349 | 125.914 | 125.915 | 0.0011942 % | 125.912 | -0.0011942 % |
| $f^{banff}_{Norm_{MEC}}$ | 120.349 | 125.914 | 130.333 | 3.5095 % | 121.495 | -3.5095 % |
| $f^{banff}_{C_5^A}$ | 120.349 | 125.914 | 127.621 | 1.3562 % | 124.311 | -1.2726 % |
| $f^{banff}_{Bg_{RES}}$ | 120.349 | 125.914 | 126.985 | 0.85109 % | 124.971 | -0.74846 % |
| $f^{banff}_{M_A^{QE}}$ | 120.349 | 125.914 | 127.832 | 1.5237 % | 123.975 | -1.5395 % |
| $f^{banff}_{M_A^{RES}}$ | 120.349 | 125.914 | 127.008 | 0.86933 % | 124.837 | -0.85554 % |
| $f^{banff}_{p_f}$ | 120.349 | 125.914 | 123.763 | -1.7078 % | 127.899 | 1.5763 % |
| $f^{banff}_{Shape_{CCoth}}$ | 120.349 | 125.914 | 126.435 | 0.41379 % | 125.393 | -0.41379 % |
| $f^{banff}_{E_B}$ | 120.349 | 125.914 | 125.899 | -0.01212 % | 125.93 | 0.012715 % |
| $f^{banff}_{Norm_{CCcoh}}$ | 120.349 | 125.914 | 126.022 | 0.086181 % | 125.805 | -0.086181 % |
| $f^{banff}_{Norm_{NCcoh}}$ | 120.349 | 125.914 | 125.919 | 0.0044519 % | 125.908 | -0.0044519 % |
| $f^{banff}_{Norm_{NCother}}$ | 120.349 | 125.914 | 126.901 | 0.78393 % | 124.927 | -0.78393 % |
| $f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$ | 120.349 | 125.914 | 125.926 | 0.0093354 % | 125.902 | -0.0093354 % |
| $f^{banff}_{Norm_{NC1\gamma}}$ | 120.349 | 125.914 | 125.918 | 0.0029615 % | 125.91 | -0.0029615 % |
| $f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$ | 120.349 | 125.914 | 125.914 | 0.00011837 % | 125.914 | -0.00011837 % |
| $f^{banff}_{Norm_{MECBar}}$ | 120.349 | 125.914 | 126.172 | 0.20522 % | 125.655 | -0.20522 % |
| $f^{SK}_{E;r}$ | 120.349 | 125.914 | 125.914 | -4.5145e-14 % | 125.914 | -7.9003e-14 % |
| $f^{SK+FSI}_{0;t,r}$ | 120.349 | 125.914 | 126.074 | 0.1271 % | 125.754 | -0.1271 % |
| $f^{SK+FSI}_{1;t,r}$ | 120.349 | 125.914 | 126.556 | 0.5104 % | 125.271 | -0.5104 % |
| $f^{SK+FSI}_{2;t,r}$ | 120.349 | 125.914 | 126.675 | 0.60427 % | 125.153 | -0.60427 % |
| $f^{SK+FSI}_{3;t,r}$ | 120.349 | 125.914 | 128.014 | 1.6677 % | 123.814 | -1.6677 % |
| $f^{SK+FSI}_{4;t,r}$ | 120.349 | 125.914 | 126.336 | 0.33551 % | 125.491 | -0.33551 % |
| $f^{SK+FSI}_{5;t,r}$ | 120.349 | 125.914 | 130.738 | 3.8311 % | 121.09 | -3.8311 % |

Table 13: $\pm1\sigma$ effect of the systematic parameters on the total number of $\mu$-like Super-K events with $6.914\times10^{20}$ POT for neutrino mode.

| Parameter | Nominal | BANFF tuned | BANFF tuned $+1\sigma$ | | BANFF tuned $-1\sigma$ | |
|---|---|---|---|---|---|---|
| $f^{banff}_{0;t,r}$ | 24.2563 | 26.5935 | 26.6217 | 0.10583 % | 26.5654 | -0.10583 % |
| $f^{banff}_{1;t,r}$ | 24.2563 | 26.5935 | 26.7186 | 0.47018 % | 26.4685 | -0.47018 % |
| $f^{banff}_{2;t,r}$ | 24.2563 | 26.5935 | 26.8418 | 0.93345 % | 26.3453 | -0.93345 % |
| $f^{banff}_{3;t,r}$ | 24.2563 | 26.5935 | 26.8365 | 0.91353 % | 26.3506 | -0.91353 % |
| $f^{banff}_{4;t,r}$ | 24.2563 | 26.5935 | 26.8768 | 1.0652 % | 26.3102 | -1.0652 % |
| $f^{banff}_{5;t,r}$ | 24.2563 | 26.5935 | 26.6316 | 0.14302 % | 26.5555 | -0.14302 % |
| $f^{banff}_{6;t,r}$ | 24.2563 | 26.5935 | 26.6004 | 0.025826 % | 26.5867 | -0.025826 % |
| $f^{banff}_{7;t,r}$ | 24.2563 | 26.5935 | 26.5978 | 0.015909 % | 26.5893 | -0.015909 % |
| $f^{banff}_{8;t,r}$ | 24.2563 | 26.5935 | 26.5992 | 0.021436 % | 26.5878 | -0.021436 % |
| $f^{banff}_{9;t,r}$ | 24.2563 | 26.5935 | 26.5962 | 0.010031 % | 26.5909 | -0.010031 % |
| $f^{banff}_{10;t,r}$ | 24.2563 | 26.5935 | 26.595 | 0.0054078 % | 26.5921 | -0.0054078 % |
| $f^{banff}_{11;t,r}$ | 24.2563 | 26.5935 | 26.5975 | 0.014836 % | 26.5896 | -0.014836 % |
| $f^{banff}_{12;t,r}$ | 24.2563 | 26.5935 | 26.5963 | 0.010428 % | 26.5907 | -0.010428 % |
| $f^{banff}_{13;t,r}$ | 24.2563 | 26.5935 | 26.596 | 0.0093637 % | 26.591 | -0.0093637 % |
| $f^{banff}_{14;t,r}$ | 24.2563 | 26.5935 | 26.595 | 0.00542 % | 26.5921 | -0.00542 % |
| $f^{banff}_{15;t,r}$ | 24.2563 | 26.5935 | 26.5948 | 0.0046444 % | 26.5923 | -0.0046444 % |
| $f^{banff}_{16;t,r}$ | 24.2563 | 26.5935 | 26.6185 | 0.093913 % | 26.5685 | -0.093913 % |
| $f^{banff}_{17;t,r}$ | 24.2563 | 26.5935 | 26.628 | 0.12954 % | 26.5591 | -0.12954 % |
| $f^{banff}_{18;t,r}$ | 24.2563 | 26.5935 | 26.6109 | 0.065228 % | 26.5762 | -0.065228 % |
| $f^{banff}_{19;t,r}$ | 24.2563 | 26.5935 | 26.6464 | 0.19869 % | 26.5407 | -0.19869 % |
| $f^{banff}_{20;t,r}$ | 24.2563 | 26.5935 | 26.598 | 0.016936 % | 26.589 | -0.016936 % |
| $f^{banff}_{21;t,r}$ | 24.2563 | 26.5935 | 26.594 | 0.0017641 % | 26.5931 | -0.0017641 % |
| $f^{banff}_{22;t,r}$ | 24.2563 | 26.5935 | 26.5937 | 0.00071333 % | 26.5933 | -0.00071333 % |
| $f^{banff}_{23;t,r}$ | 24.2563 | 26.5935 | 26.6008 | 0.027288 % | 26.5863 | -0.027288 % |
| $f^{banff}_{24;t,r}$ | 24.2563 | 26.5935 | 26.5937 | 0.00079831 % | 26.5933 | -0.00079831 % |
| $f^{banff}_{Norm_{MEC}}$ | 24.2563 | 26.5935 | 27.6259 | 3.882 % | 25.5612 | -3.882 % |
| $f^{banff}_{C^A_5}$ | 24.2563 | 26.5935 | 26.8014 | 0.78157 % | 26.3972 | -0.73812 % |
| $f^{banff}_{Bg_{RES}}$ | 24.2563 | 26.5935 | 26.6753 | 0.30745 % | 26.5216 | -0.27037 % |
| $f^{banff}_{M^{QE}_A}$ | 24.2563 | 26.5935 | 27.0363 | 1.6648 % | 26.1369 | -1.7172 % |
| $f^{banff}_{M^{RES}_A}$ | 24.2563 | 26.5935 | 26.6925 | 0.37237 % | 26.4929 | -0.37839 % |
| $f^{banff}_{p_f}$ | 24.2563 | 26.5935 | 25.9799 | -2.3076 % | 27.1501 | 2.093 % |
| $f^{banff}_{Shape_{CCoth}}$ | 24.2563 | 26.5935 | 26.6063 | 0.048094 % | 26.5807 | -0.048083 % |
| $f^{banff}_{E_B}$ | 24.2563 | 26.5935 | 26.5533 | -0.15134 % | 26.626 | 0.12201 % |
| $f^{banff}_{Norm_{CCcoh}}$ | 24.2563 | 26.5935 | 26.5991 | 0.021135 % | 26.5879 | -0.021135 % |
| $f^{banff}_{Norm_{NCcoh}}$ | 24.2563 | 26.5935 | 26.6567 | 0.23764 % | 26.5303 | -0.23764 % |
| $f^{banff}_{Norm_{NCother}}$ | 24.2563 | 26.5935 | 26.6352 | 0.15691 % | 26.5518 | -0.15691 % |
| $f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$ | 24.2563 | 26.5935 | 27.2933 | 2.6315 % | 25.8937 | -2.6315 % |
| $f^{banff}_{Norm_{NC1\gamma}}$ | 24.2563 | 26.5935 | 27.0873 | 1.8568 % | 26.0997 | -1.8568 % |
| $f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$ | 24.2563 | 26.5935 | 26.6014 | 0.029817 % | 26.5856 | -0.029817 % |
| $f^{banff}_{Norm_{MECBar}}$ | 24.2563 | 26.5935 | 26.6061 | 0.047159 % | 26.581 | -0.047159 % |
| $f^{SK}_{E;r}$ | 24.2563 | 26.5935 | 26.5935 | 1.3359e-14 % | 26.5935 | -1.3359e-14 % |
| $f^{SK+FSI}_{6;t,r}$ | 24.2563 | 26.5935 | 26.8445 | 0.94359 % | 26.3426 | -0.94359 % |
| $f^{SK+FSI}_{7;t,r}$ | 24.2563 | 26.5935 | 27.0622 | 1.7624 % | 26.1248 | -1.7624 % |
| $f^{SK+FSI}_{8;t,r}$ | 24.2563 | 26.5935 | 26.7121 | 0.44605 % | 26.4749 | -0.44605 % |
| $f^{SK+FSI}_{9;t,r}$ | 24.2563 | 26.5935 | 26.6525 | 0.22178 % | 26.5345 | -0.22178 % |
| $f^{SK+FSI}_{10;t,r}$ | 24.2563 | 26.5935 | 26.6244 | 0.116 % | 26.5627 | -0.116 % |
| $f^{SK+FSI}_{11;t,r}$ | 24.2563 | 26.5935 | 26.6021 | 0.0321 % | 26.585 | -0.0321 % |
| $f^{SK+FSI}_{12;t,r}$ | 24.2563 | 26.5935 | 26.6184 | 0.093385 % | 26.5687 | -0.093385 % |
| $f^{SK+FSI}_{13;t,r}$ | 24.2563 | 26.5935 | 26.6657 | 0.27133 % | 26.5214 | -0.27133 % |
| $f^{SK+FSI}_{14;t,r}$ | 24.2563 | 26.5935 | 26.6662 | 0.27323 % | 26.5209 | -0.27323 % |
| $f^{SK+FSI}_{15;t,r}$ | 24.2563 | 26.5935 | 26.8604 | 1.0036 % | 26.3266 | -1.0036 % |
| $f^{SK+FSI}_{16;t,r}$ | 24.2563 | 26.5935 | 26.7358 | 0.53512 % | 26.4512 | -0.53512 % |
| $f^{SK+FSI}_{17;t,r}$ | 24.2563 | 26.5935 | 26.7166 | 0.46264 % | 26.4705 | -0.46264 % |

Table 14: $\pm1\sigma$ effect of the systematic parameters on the total number of $e$-like Super-K events with $6.914\times10^{20}$ POT for neutrino mode.

| Parameter | Nominal | BANFF tuned | BANFF tuned $+1\sigma$ | | BANFF tuned $-1\sigma$ | |
|---|---|---|---|---|---|---|
| $f^{banff}_{0;t,r}RHC$ | 33.354 | 34.2841 | 34.3212 | 0.1084 % | 34.2469 | -0.1084 % |
| $f^{banff}_{1;t,r}RHC$ | 33.354 | 34.2841 | 34.3316 | 0.13851 % | 34.2366 | -0.13851 % |
| $f^{banff}_{2;t,r}RHC$ | 33.354 | 34.2841 | 34.4224 | 0.4034 % | 34.1458 | -0.4034 % |
| $f^{banff}_{3;t,r}RHC$ | 33.354 | 34.2841 | 34.5137 | 0.66983 % | 34.0544 | -0.66983 % |
| $f^{banff}_{4;t,r}RHC$ | 33.354 | 34.2841 | 34.5214 | 0.69216 % | 34.0468 | -0.69216 % |
| $f^{banff}_{5;t,r}RHC$ | 33.354 | 34.2841 | 34.4278 | 0.41934 % | 34.1403 | -0.41934 % |
| $f^{banff}_{6;t,r}RHC$ | 33.354 | 34.2841 | 34.3655 | 0.23759 % | 34.2026 | -0.23759 % |
| $f^{banff}_{7;t,r}RHC$ | 33.354 | 34.2841 | 34.301 | 0.049526 % | 34.2671 | -0.049526 % |
| $f^{banff}_{8;t,r}RHC$ | 33.354 | 34.2841 | 34.2964 | 0.036085 % | 34.2717 | -0.036085 % |
| $f^{banff}_{9;t,r}RHC$ | 33.354 | 34.2841 | 34.4627 | 0.52119 % | 34.1054 | -0.52119 % |
| $f^{banff}_{10;t,r}RHC$ | 33.354 | 34.2841 | 34.4614 | 0.51727 % | 34.1067 | -0.51727 % |
| $f^{banff}_{11;t,r}RHC$ | 33.354 | 34.2841 | 34.4688 | 0.53874 % | 34.0994 | -0.53874 % |
| $f^{banff}_{12;t,r}RHC$ | 33.354 | 34.2841 | 34.3882 | 0.30381 % | 34.1799 | -0.30381 % |
| $f^{banff}_{13;t,r}RHC$ | 33.354 | 34.2841 | 34.3968 | 0.32868 % | 34.1714 | -0.32868 % |
| $f^{banff}_{14;t,r}RHC$ | 33.354 | 34.2841 | 34.3136 | 0.086075 % | 34.2546 | -0.086075 % |
| $f^{banff}_{15;t,r}RHC$ | 33.354 | 34.2841 | 34.2898 | 0.016631 % | 34.2784 | -0.016631 % |
| $f^{banff}_{16;t,r}RHC$ | 33.354 | 34.2841 | 34.2853 | 0.0035678 % | 34.2828 | -0.0035678 % |
| $f^{banff}_{17;t,r}RHC$ | 33.354 | 34.2841 | 34.2856 | 0.0045846 % | 34.2825 | -0.0045846 % |
| $f^{banff}_{18;t,r}RHC$ | 33.354 | 34.2841 | 34.2841 | 0.00019776 % | 34.284 | -0.00019776 % |
| $f^{banff}_{19;t,r}RHC$ | 33.354 | 34.2841 | 34.2841 | 0.00023211 % | 34.284 | -0.00023211 % |
| $f^{banff}_{20;t,r}RHC$ | 33.354 | 34.2841 | 34.2841 | 0.00018231 % | 34.284 | -0.00018231 % |
| $f^{banff}_{21;t,r}RHC$ | 33.354 | 34.2841 | 34.2843 | 0.0007317 % | 34.2838 | -0.0007317 % |
| $f^{banff}_{22;t,r}RHC$ | 33.354 | 34.2841 | 34.2845 | 0.0011599 % | 34.2837 | -0.0011599 % |
| $f^{banff}_{23;t,r}RHC$ | 33.354 | 34.2841 | 34.2846 | 0.0015682 % | 34.2835 | -0.0015682 % |
| $f^{banff}_{24;t,r}RHC$ | 33.354 | 34.2841 | 34.2846 | 0.0016844 % | 34.2835 | -0.0016844 % |
| $f^{banff}_{Norm_{MEC}}$ | 33.354 | 34.2841 | 35.2949 | 2.9484 % | 33.2732 | -2.9484 % |
| $f^{banff}_{C_5^A}$ | 33.354 | 34.2841 | 34.7665 | 1.4073 % | 33.8496 | -1.2673 % |
| $f^{banff}_{Bg_{RES}}$ | 33.354 | 34.2841 | 34.6114 | 0.95474 % | 33.9962 | -0.8396 % |
| $f^{banff}_{M_A^{QE}}$ | 33.354 | 34.2841 | 34.7782 | 1.4412 % | 33.7964 | -1.4226 % |
| $f^{banff}_{M_A^{RES}}$ | 33.354 | 34.2841 | 34.557 | 0.79619 % | 34.0229 | -0.76184 % |
| $f^{banff}_{p_f}$ | 33.354 | 34.2841 | 33.7369 | -1.596 % | 34.8188 | 1.5598 % |
| $f^{banff}_{Shape_{CCoth}}$ | 33.354 | 34.2841 | 34.4278 | 0.41939 % | 34.1403 | -0.41939 % |
| $f^{banff}_{E_B}$ | 33.354 | 34.2841 | 34.2825 | -0.0045403 % | 34.2856 | 0.0043363 % |
| $f^{banff}_{Norm_{CCcoh}}$ | 33.354 | 34.2841 | 34.3434 | 0.17311 % | 34.2247 | -0.17311 % |
| $f^{banff}_{Norm_{NCcoh}}$ | 33.354 | 34.2841 | 34.2866 | 0.0073745 % | 34.2815 | -0.0073745 % |
| $f^{banff}_{Norm_{NCother}}$ | 33.354 | 34.2841 | 34.5427 | 0.75445 % | 34.0254 | -0.75445 % |
| $f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$ | 33.354 | 34.2841 | 34.2846 | 0.0015892 % | 34.2835 | -0.0015892 % |
| $f^{banff}_{Norm_{NC1\gamma}}$ | 33.354 | 34.2841 | 34.2844 | 0.00099164 % | 34.2837 | -0.00099164 % |
| $f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$ | 33.354 | 34.2841 | 34.2848 | 0.0022231 % | 34.2833 | -0.0022231 % |
| $f^{banff}_{Norm_{MECBar}}$ | 33.354 | 34.2841 | 34.9123 | 1.8323 % | 33.6559 | -1.8323 % |
| $f^{SK}_{E;r}$ | 33.354 | 34.2841 | 34.2841 | -2.0725e-14 % | 34.2841 | -2.0725e-14 % |
| $f^{SK+FSI}_{0;t,r}RHC$ | 33.354 | 34.2841 | 34.3274 | 0.12641 % | 34.2407 | -0.12641 % |
| $f^{SK+FSI}_{1;t,r}RHC$ | 33.354 | 34.2841 | 34.4079 | 0.36115 % | 34.1602 | -0.36115 % |
| $f^{SK+FSI}_{2;t,r}RHC$ | 33.354 | 34.2841 | 34.5168 | 0.67883 % | 34.0513 | -0.67883 % |
| $f^{SK+FSI}_{3;t,r}RHC$ | 33.354 | 34.2841 | 35.0403 | 2.2058 % | 33.5278 | -2.2058 % |
| $f^{SK+FSI}_{4;t,r}RHC$ | 33.354 | 34.2841 | 34.3304 | 0.13529 % | 34.2377 | -0.13529 % |
| $f^{SK+FSI}_{5;t,r}RHC$ | 33.354 | 34.2841 | 35.3403 | 3.0809 % | 33.2278 | -3.0809 % |

Table 15: $\pm1\sigma$ effect of the systematic parameters on the total number of $\mu$-like Super-K events with $4.011\times10^{20}$ POT for antineutrino mode.

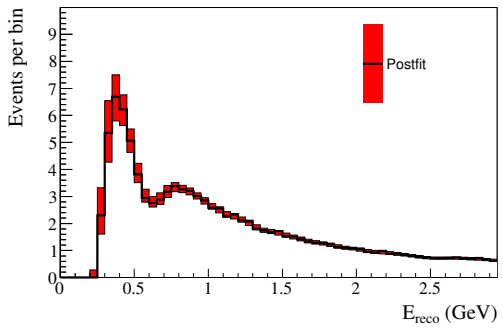
| Parameter | Nominal | BANFF tuned | BANFF tuned $+1\sigma$ | | BANFF tuned $-1\sigma$ | |
|---|---|---|---|---|---|---|
| $f^{banff}_{0;t,r}RHC$ | 3.11707 | 3.21831 | 3.23425 | 0.49537 % | 3.20237 | -0.49537 % |
| $f^{banff}_{1;t,r}RHC$ | 3.11707 | 3.21831 | 3.22891 | 0.32934 % | 3.20771 | -0.32934 % |
| $f^{banff}_{2;t,r}RHC$ | 3.11707 | 3.21831 | 3.22364 | 0.16581 % | 3.21297 | -0.16581 % |
| $f^{banff}_{3;t,r}RHC$ | 3.11707 | 3.21831 | 3.22045 | 0.066577 % | 3.21617 | -0.066577 % |
| $f^{banff}_{4;t,r}RHC$ | 3.11707 | 3.21831 | 3.22085 | 0.078879 % | 3.21577 | -0.078879 % |
| $f^{banff}_{5;t,r}RHC$ | 3.11707 | 3.21831 | 3.22043 | 0.065975 % | 3.21619 | -0.065975 % |
| $f^{banff}_{6;t,r}RHC$ | 3.11707 | 3.21831 | 3.22509 | 0.21083 % | 3.21152 | -0.21083 % |
| $f^{banff}_{7;t,r}RHC$ | 3.11707 | 3.21831 | 3.23439 | 0.49965 % | 3.20223 | -0.49965 % |
| $f^{banff}_{8;t,r}RHC$ | 3.11707 | 3.21831 | 3.2365 | 0.56517 % | 3.20012 | -0.56517 % |
| $f^{banff}_{9;t,r}RHC$ | 3.11707 | 3.21831 | 3.25195 | 1.0453 % | 3.18467 | -1.0453 % |
| $f^{banff}_{10;t,r}RHC$ | 3.11707 | 3.21831 | 3.2239 | 0.17371 % | 3.21272 | -0.17371 % |
| $f^{banff}_{11;t,r}RHC$ | 3.11707 | 3.21831 | 3.21966 | 0.041857 % | 3.21696 | -0.041857 % |
| $f^{banff}_{12;t,r}RHC$ | 3.11707 | 3.21831 | 3.21896 | 0.020249 % | 3.21766 | -0.020249 % |
| $f^{banff}_{13;t,r}RHC$ | 3.11707 | 3.21831 | 3.21907 | 0.023667 % | 3.21755 | -0.023667 % |
| $f^{banff}_{14;t,r}RHC$ | 3.11707 | 3.21831 | 3.21857 | 0.0081707 % | 3.21805 | -0.0081707 % |
| $f^{banff}_{15;t,r}RHC$ | 3.11707 | 3.21831 | 3.21855 | 0.0074537 % | 3.21807 | -0.0074537 % |
| $f^{banff}_{16;t,r}RHC$ | 3.11707 | 3.21831 | 3.23255 | 0.44247 % | 3.20407 | -0.44247 % |
| $f^{banff}_{17;t,r}RHC$ | 3.11707 | 3.21831 | 3.21849 | 0.0055449 % | 3.21813 | -0.0055449 % |
| $f^{banff}_{18;t,r}RHC$ | 3.11707 | 3.21831 | 3.22275 | 0.13805 % | 3.21387 | -0.13805 % |
| $f^{banff}_{19;t,r}RHC$ | 3.11707 | 3.21831 | 3.22312 | 0.14949 % | 3.2135 | -0.14949 % |
| $f^{banff}_{20;t,r}RHC$ | 3.11707 | 3.21831 | 3.22095 | 0.081949 % | 3.21567 | -0.081949 % |
| $f^{banff}_{21;t,r}RHC$ | 3.11707 | 3.21831 | 3.22701 | 0.2705 % | 3.2096 | -0.2705 % |
| $f^{banff}_{22;t,r}RHC$ | 3.11707 | 3.21831 | 3.21896 | 0.020273 % | 3.21766 | -0.020273 % |
| $f^{banff}_{23;t,r}RHC$ | 3.11707 | 3.21831 | 3.21839 | 0.002446 % | 3.21823 | -0.002446 % |
| $f^{banff}_{24;t,r}RHC$ | 3.11707 | 3.21831 | 3.21838 | 0.002219 % | 3.21824 | -0.002219 % |
| $f^{banff}_{Norm_{MEC}}$ | 3.11707 | 3.21831 | 3.31425 | 2.9812 % | 3.12237 | -2.9812 % |
| $f^{banff}_{C^A_5}$ | 3.11707 | 3.21831 | 3.25166 | 1.0364 % | 3.19031 | -0.86988 % |
| $f^{banff}_{Bg_{RES}}$ | 3.11707 | 3.21831 | 3.23459 | 0.5058 % | 3.20399 | -0.4448 % |
| $f^{banff}_{M^{QE}_A}$ | 3.11707 | 3.21831 | 3.26276 | 1.3812 % | 3.17411 | -1.3732 % |
| $f^{banff}_{M^{RES}_A}$ | 3.11707 | 3.21831 | 3.22955 | 0.34945 % | 3.20849 | -0.30505 % |
| $f^{banff}_{p_f}$ | 3.11707 | 3.21831 | 3.14236 | -2.3599 % | 3.29191 | 2.2869 % |
| $f^{banff}_{Shape_{CCoth}}$ | 3.11707 | 3.21831 | 3.22119 | 0.089515 % | 3.21543 | -0.089518 % |
| $f^{banff}_{E_B}$ | 3.11707 | 3.21831 | 3.21749 | -0.025439 % | 3.21891 | 0.018569 % |
| $f^{banff}_{Norm_{CCcoh}}$ | 3.11707 | 3.21831 | 3.22518 | 0.21355 % | 3.21144 | -0.21355 % |
| $f^{banff}_{Norm_{NCcoh}}$ | 3.11707 | 3.21831 | 3.2514 | 1.0282 % | 3.18522 | -1.0282 % |
| $f^{banff}_{Norm_{NCother}}$ | 3.11707 | 3.21831 | 3.22886 | 0.32779 % | 3.20776 | -0.32779 % |
| $f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$ | 3.11707 | 3.21831 | 3.24258 | 0.75412 % | 3.19404 | -0.75412 % |
| $f^{banff}_{Norm_{NC1\gamma}}$ | 3.11707 | 3.21831 | 3.33932 | 3.76 % | 3.0973 | -3.76 % |
| $f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$ | 3.11707 | 3.21831 | 3.27411 | 1.734 % | 3.1625 | -1.734 % |
| $f^{banff}_{Norm_{MECBar}}$ | 3.11707 | 3.21831 | 3.2943 | 2.3612 % | 3.14232 | -2.3612 % |
| $f^{SK}_{E;r}$ | 3.11707 | 3.21831 | 3.21831 | -1.3799e-14 % | 3.21831 | -4.1397e-14 % |
| $f^{SK+FSI}_{6;t,r}RHC$ | 3.11707 | 3.21831 | 3.22805 | 0.30255 % | 3.20857 | -0.30255 % |
| $f^{SK+FSI}_{7;t,r}RHC$ | 3.11707 | 3.21831 | 3.25713 | 1.2064 % | 3.17948 | -1.2064 % |
| $f^{SK+FSI}_{8;t,r}RHC$ | 3.11707 | 3.21831 | 3.24365 | 0.78746 % | 3.19297 | -0.78746 % |
| $f^{SK+FSI}_{9;t,r}RHC$ | 3.11707 | 3.21831 | 3.22378 | 0.16988 % | 3.21284 | -0.16988 % |
| $f^{SK+FSI}_{10;t,r}RHC$ | 3.11707 | 3.21831 | 3.2219 | 0.11164 % | 3.21472 | -0.11164 % |
| $f^{SK+FSI}_{11;t,r}RHC$ | 3.11707 | 3.21831 | 3.22052 | 0.068625 % | 3.2161 | -0.068625 % |
| $f^{SK+FSI}_{12;t,r}RHC$ | 3.11707 | 3.21831 | 3.22172 | 0.10612 % | 3.21489 | -0.10612 % |
| $f^{SK+FSI}_{13;t,r}RHC$ | 3.11707 | 3.21831 | 3.23071 | 0.38524 % | 3.20591 | -0.38524 % |
| $f^{SK+FSI}_{14;t,r}RHC$ | 3.11707 | 3.21831 | 3.23982 | 0.66827 % | 3.1968 | -0.66827 % |
| $f^{SK+FSI}_{15;t,r}RHC$ | 3.11707 | 3.21831 | 3.28217 | 1.9844 % | 3.15444 | -1.9844 % |
| $f^{SK+FSI}_{16;t,r}RHC$ | 3.11707 | 3.21831 | 3.25076 | 1.0083 % | 3.18586 | -1.0083 % |
| $f^{SK+FSI}_{17;t,r}RHC$ | 3.11707 | 3.21831 | 3.2479 | 0.91957 % | 3.18871 | -0.91957 % |

Table 16: $\pm1\sigma$ effect of the systematic parameters on the total number of $e$-like Super-K events with $4.011\times10^{20}$ POT for antineutrino mode.

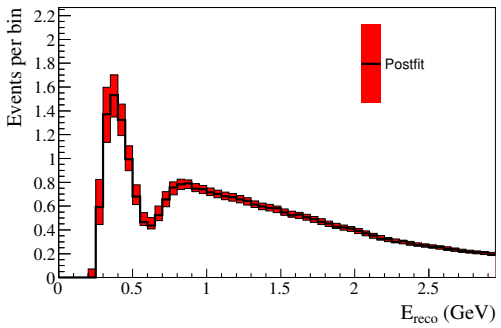
The effects of all the combined systematic uncertainties on the reconstructed energy spectrum of single ring events is shown in Fig. 6 and 7 for the typical oscillation scenario with parameters of Asimov datasets A. In App. D the same plots are shown for different oscillation scenarios. These plots show the total error envelope for the reconstructed energy spectra; this is calculated as the $\pm 1\sigma$ spread with respect to the median of bin contents from 10k toy MC experiments generated with randomized systematic parameters. All systematic parameters are considered and their correlations are taken into account.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 17, 18, 19 and 20 with the typical parameter values given in Tab. 12. For the measurement of $\delta_{CP}$ it important to understand the effect of the systematic uncertainties on the ratio of the number of events between neutrino and antineutrino mode samples, which is the consequence of correlations between systematic parameters. If correlations do not reproduce the physics correctly, a fake asymmetry between neutrino and antineutrino oscillations can be seen. On the other hand if correlations exist but are not taken into account, the measurement of $\delta_{CP}$ could be biased. Depending on the value of $\delta_{CP}$, correlations or anti-correlations between parameters are more conservative. If true $\delta_{CP} = \pm\pi/2$ the most conservative case is given by anti-correlation in the $e$-like samples, since the oscillation asymmetry could be hidden by the systematic uncertainties. If true $\delta_{CP} \sim 0$ uncorrelated parameters reduce the sensitivity to $\delta_{CP}$. The effect of the systematic parameters on the ratio between neutrino and antineutrino mode events are shown in Tab. 21. The most critical systematic parameters are $\sigma_{\nu_e}$ and $\sigma_{\bar{\nu}_e}$. In fact they are partially anti-correlated in this analysis and are applied respectively to $\nu_e$ and $\bar{\nu}_e$ events.

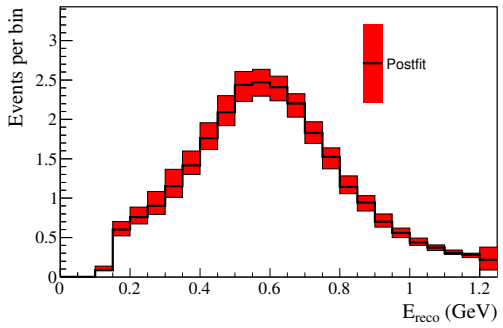
For the systematic effect study, in order to reduce the CPU time, the toys MC were produced using 1-dimensional templates $E_{reco}$ $e$-like templates, instead of the 2-dimensional $E_{reco}$ Vs $\theta$, used in the analysis. However it has been checked that using 1- or 2- dimensional templates does not make any difference in the drawing of the error envelopes and the effect of the systematic uncertainties on the total number of events.



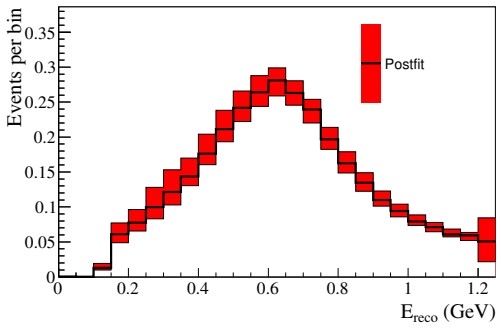

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

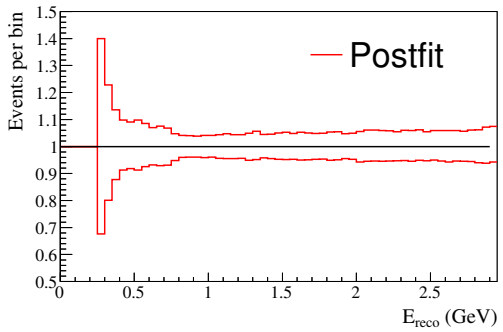



(c) Electron-like, neutrino mode

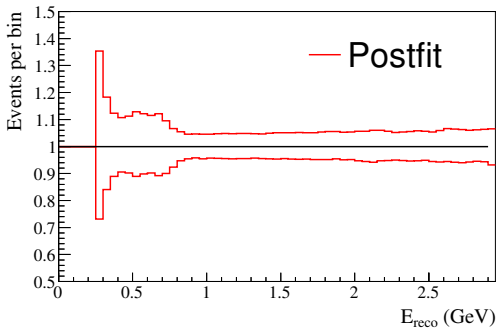



(d) Electron-like, antineutrino mode

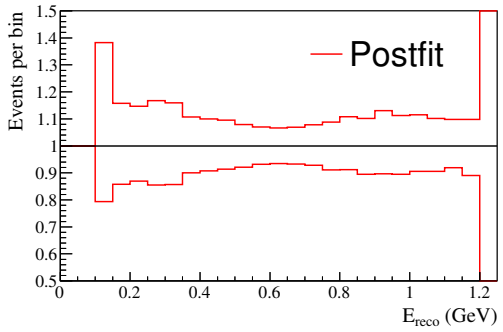
Figure 6: Error envelopes for the energy spectrum as a result of all the systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



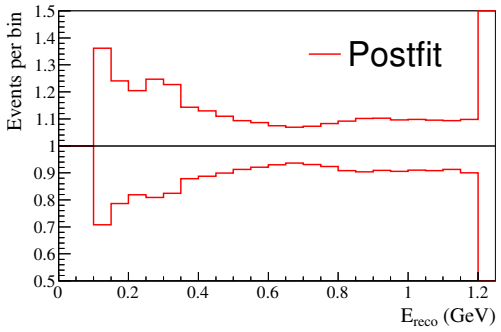

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode



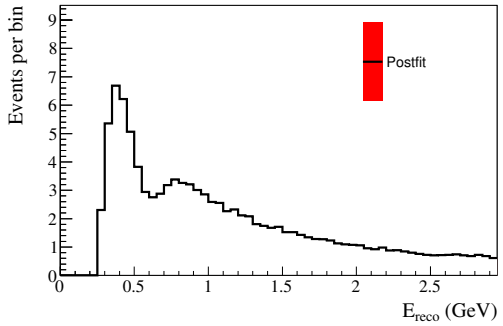

(c) Electron-like, neutrino mode



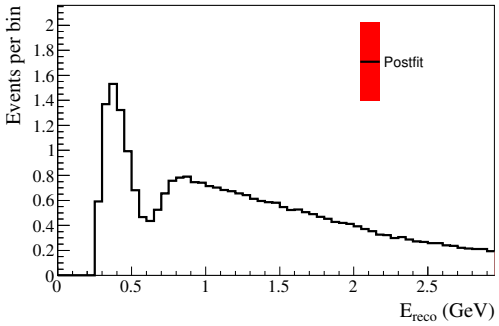

(d) Electron-like, antineutrino mode

Figure 7: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.

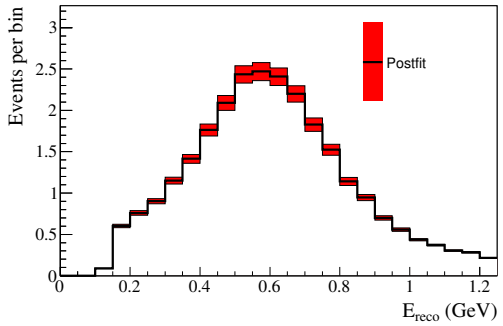
In Fig. 8, 9, 10, 11, 12 the error envelopes of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.



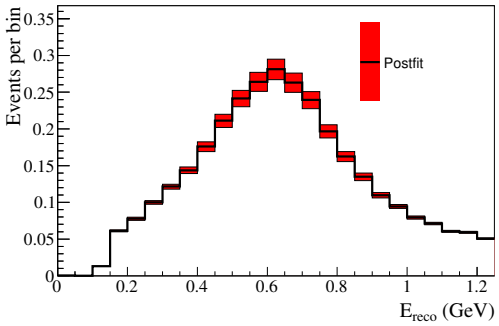

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

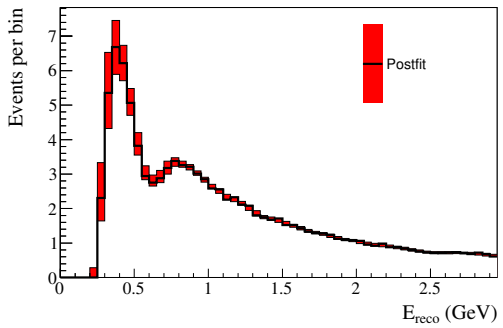



(c) Electron-like, neutrino mode

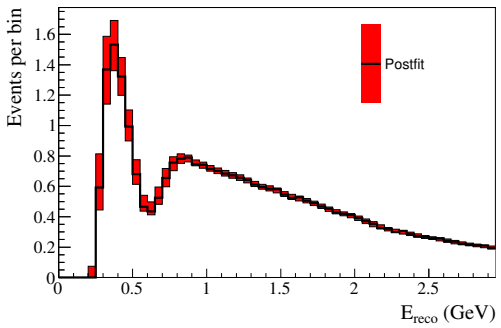



(d) Electron-like, antineutrino mode

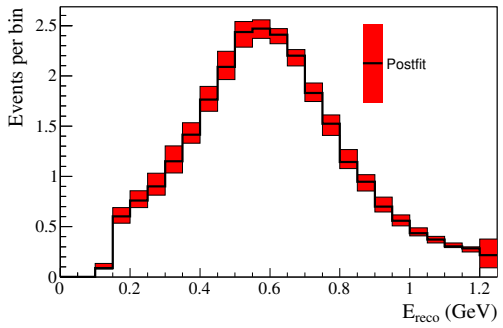
Figure 8: Error envelopes for the energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



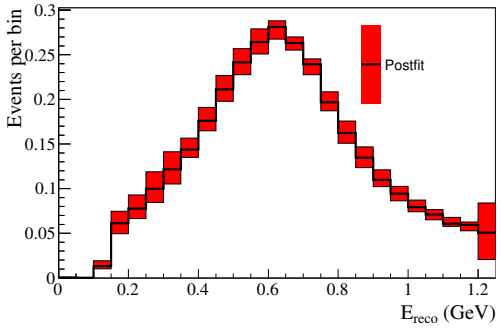

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

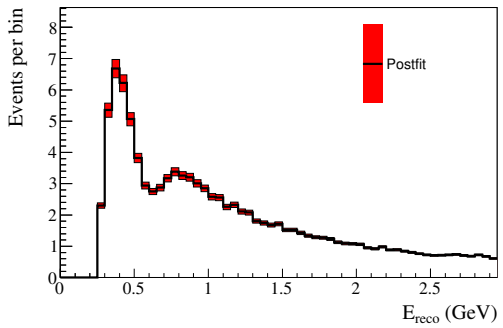



(c) Electron-like, neutrino mode

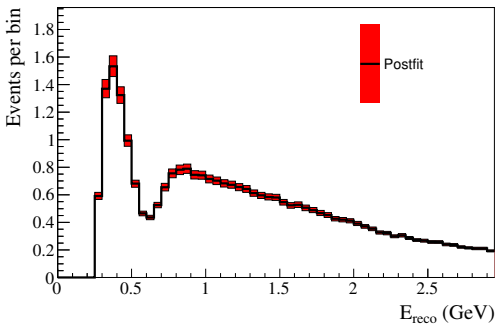



(d) Electron-like, antineutrino mode

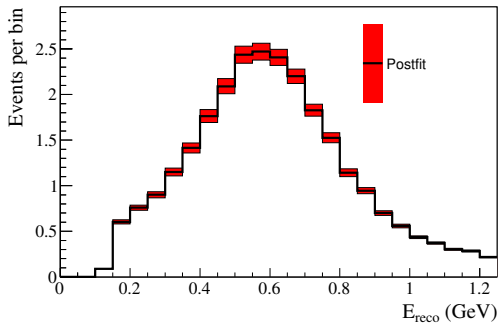
Figure 9: Error envelopes for the energy spectrum as a result of the super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



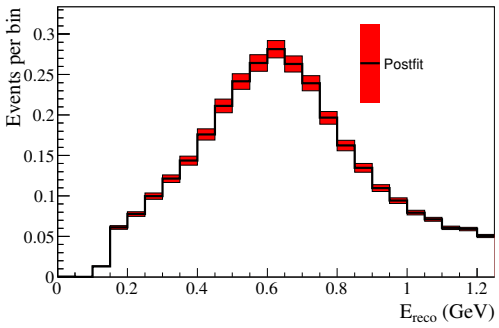

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

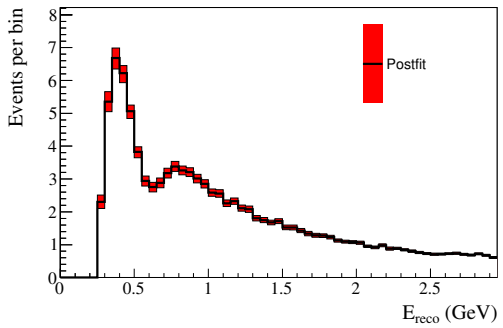



(c) Electron-like, neutrino mode

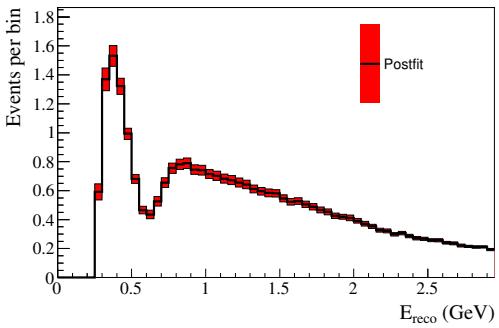



(d) Electron-like, antineutrino mode

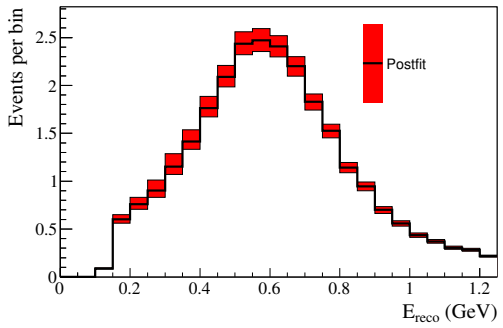
Figure 10: Error envelopes for the energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



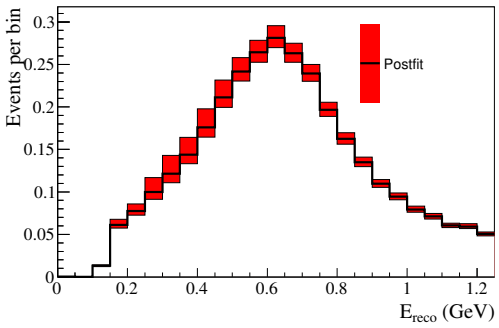

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

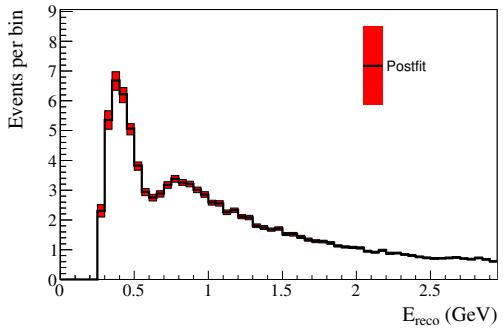



(c) Electron-like, neutrino mode

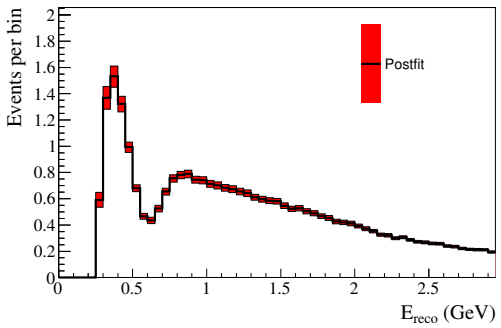



(d) Electron-like, antineutrino mode

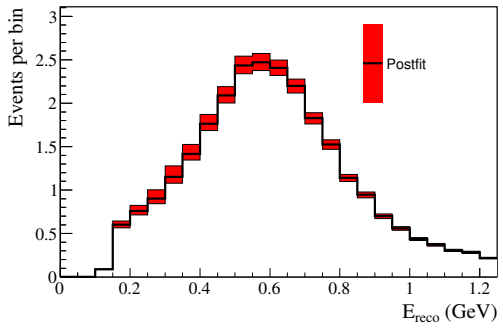
Figure 11: Error envelopes for the energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



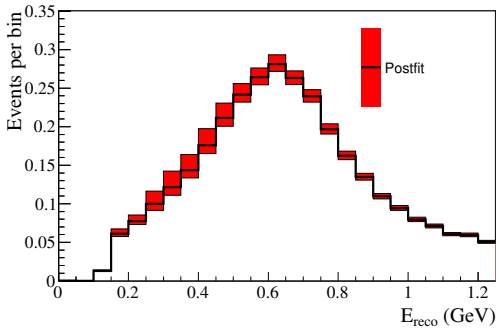

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode



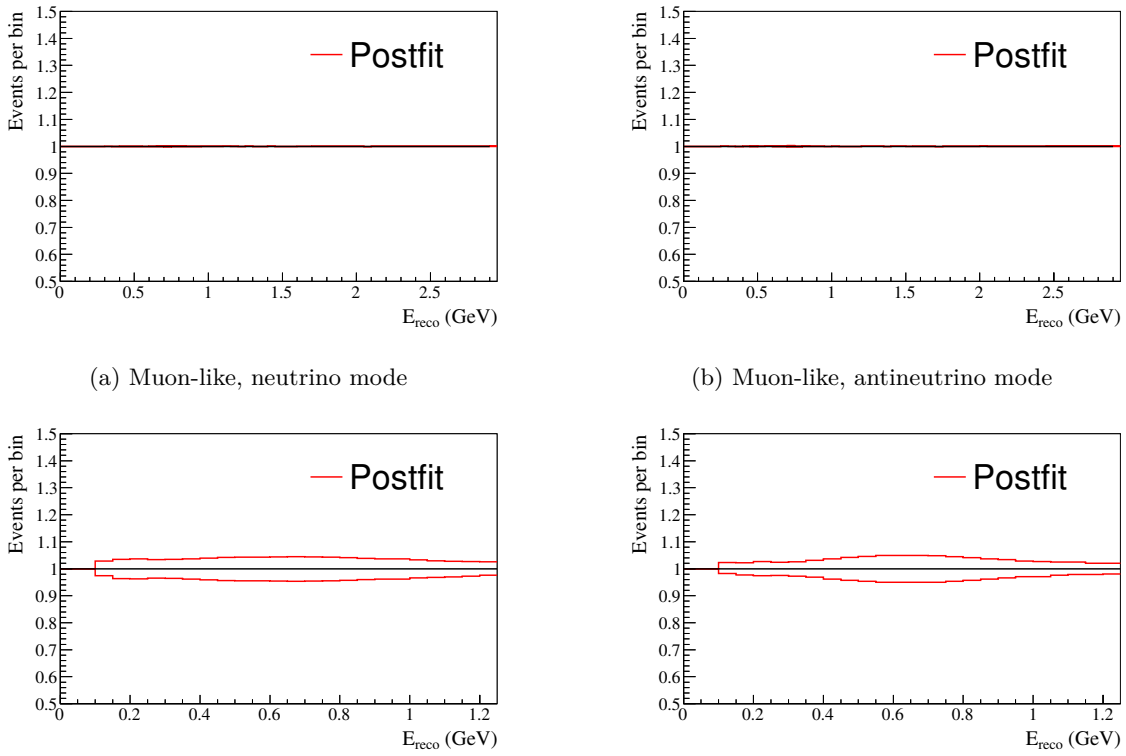

(c) Electron-like, neutrino mode




(d) Electron-like, antineutrino mode

Figure 12: Error envelopes for the energy spectrum as a result of the flux plus cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.

In Fig. 13, 14, 15, 16, 17 the ratio error envelopes of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.



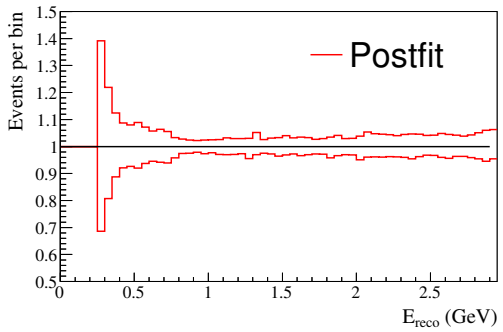

(a) Muon-like, neutrino mode

(b) Muon-like, antineutrino mode

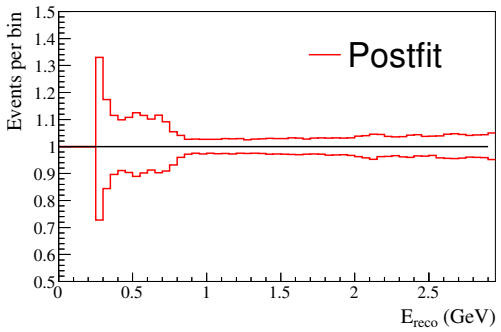
(c) Electron-like, neutrino mode

(d) Electron-like, antineutrino mode

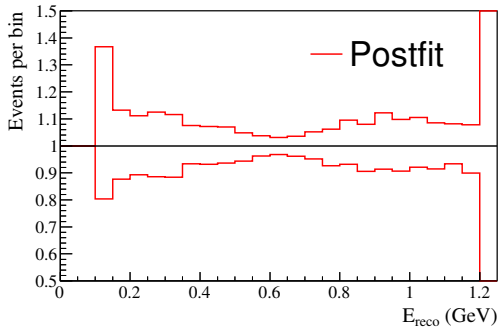
Figure 13: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



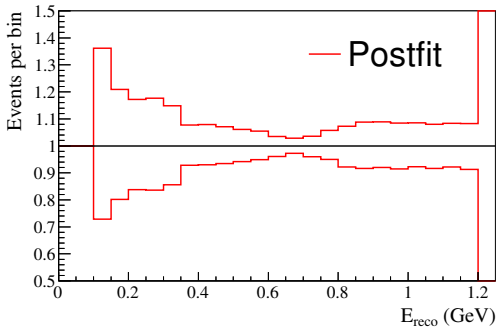

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

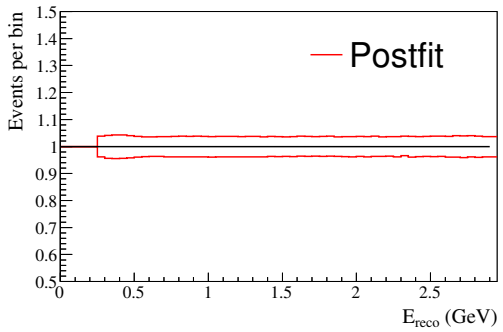



(c) Electron-like, neutrino mode

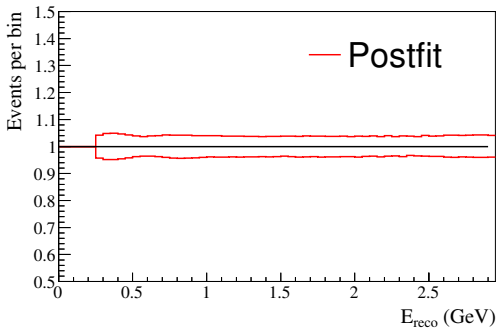



(d) Electron-like, antineutrino mode

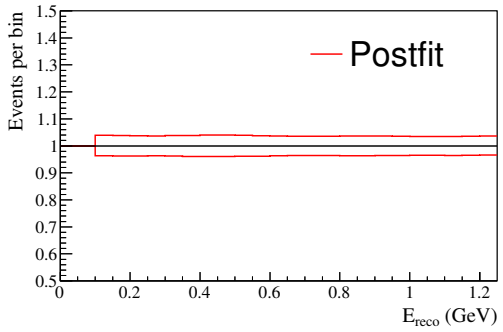
Figure 14: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



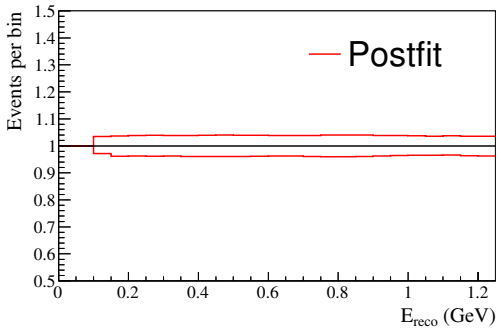

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

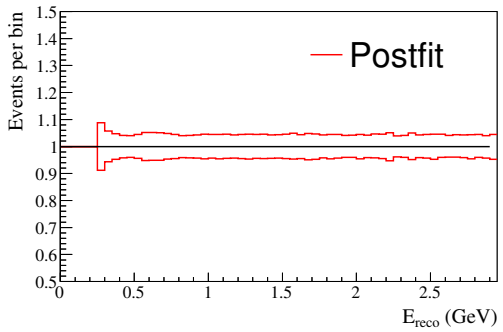



(c) Electron-like, neutrino mode

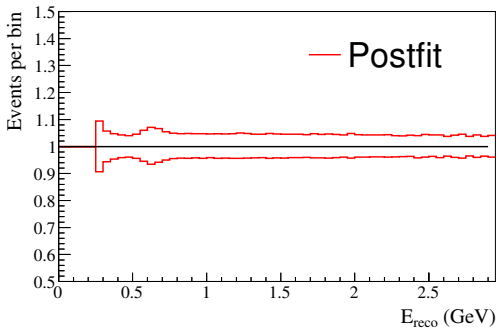



(d) Electron-like, antineutrino mode

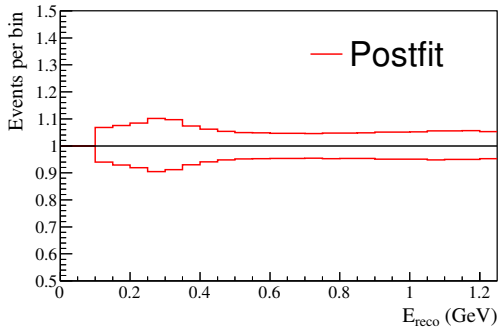
Figure 15: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



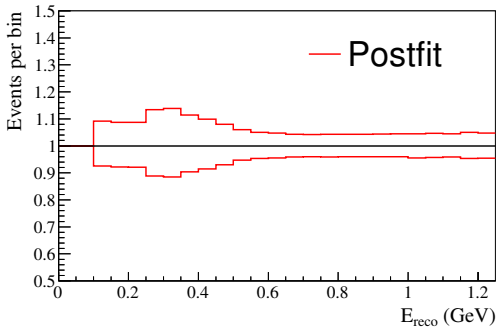

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode

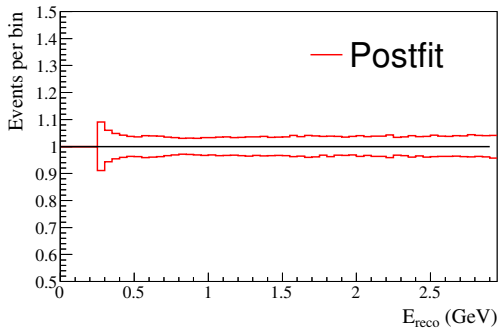



(c) Electron-like, neutrino mode

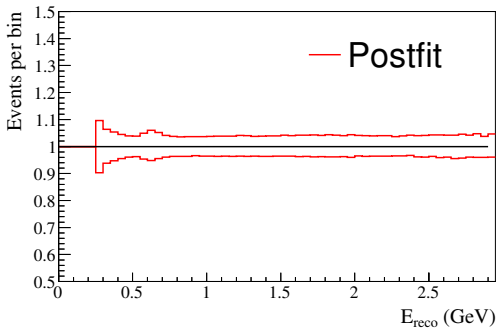



(d) Electron-like, antineutrino mode

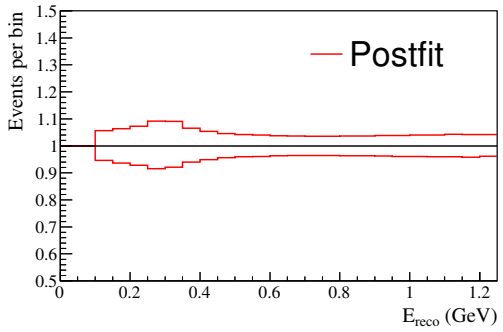
Figure 16: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.



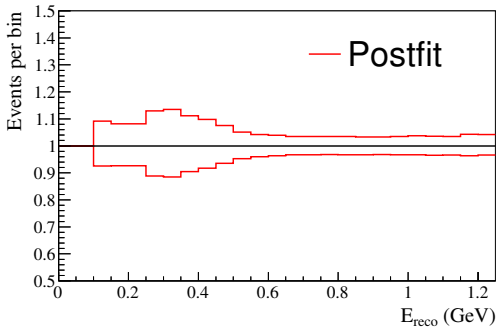

(a) Muon-like, neutrino mode




(b) Muon-like, antineutrino mode




(c) Electron-like, neutrino mode




(d) Electron-like, antineutrino mode

Figure 17: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the flux + cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents is calculated. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. BANFF tuned MC templates are used. All systematic parameters are considered and their correlations are taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 4.15% |
| Flux | 3.59% |
| XSec Corr Tot | 4.02% |
| MEC (corr) | 3.54% |
| MECbar (corr) | 0.20% |
| NC other (uncorr) | 0.77% |
| NC 1gamma (uncorr) | 0.00% |
| XSec nue/numu | 0.01% |
| XSec Tot | 4.10% |
| Flux+XSec (ND280 constrained) | 2.82% |
| Flux+XSec | 2.92% |
| Flux+XSec+SKDet+FSI+SI | 5.09% |
| Oscillations | 0.03% |
| All | 5.13% |
| All (pre-fit) | 11.9% |

Table 17: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 3.50% |
| Flux | 3.67% |
| XSec Corr Tot | 4.25% |
| MEC (corr) | 3.85% |
| MECbar (corr) | 0.05% |
| NC other (uncorr) | 0.16% |
| NC 1gamma (uncorr) | 1.47% |
| XSec nue/numu | 2.60% |
| XSec Tot | 5.20% |
| Flux+XSec (ND280 constrained) | 2.84% |
| Flux+XSec | 4.13% |
| Flux+XSec+SKDet+FSI+SI | 5.53% |
| Oscillations | 4.16% |
| All | 6.80% |
| All (pre-fit) | 12.6% |

Table 18: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 3.94% |
| Flux | 3.66% |
| XSec Corr Tot | 4.08% |
| MEC (corr) | 2.96% |
| MECbar (corr) | 1.82% |
| NC other (uncorr) | 0.75% |
| NC 1gamma (uncorr) | 0.00% |
| XSec nue/numu | 0.00% |
| XSec Tot | 4.14% |
| Flux+XSec (ND280 constrained) | 3.27% |
| Flux+XSec | 3.36% |
| Flux+XSec+SKDet+FSI+SI | 5.18% |
| Oscillations | 0.03% |
| All | 5.12% |
| All (pre-fit) | 12.7% |

Table 19: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 3.97% |
| Flux | 3.78% |
| XSec Corr Tot | 4.30% |
| MEC (corr) | 3.00% |
| MECbar (corr) | 2.36% |
| NC other (uncorr) | 0.33% |
| NC 1gamma (uncorr) | 2.93% |
| XSec nue/numu | 1.51% |
| XSec Tot | 5.43% |
| Flux+XSec (ND280 constrained) | 3.19% |
| Flux+XSec | 4.61% |
| Flux+XSec+SKDet+FSI+SI | 6.21% |
| Oscillations | 4.00% |
| All | 7.41% |
| All (pre-fit) | 14.3% |

Table 20: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| Source of uncertainty | $\mu$-like RMS/Mean | e-like RMS/Mean |
|---|---|---|
| SKDet | 0.7% | 1.6% |
| FSI+SI | 2.6% | 3.5% |
| SKDet+FSI+SI | 2.6% | 3.8% |
| Flux | 1.7% | 1.7% |
| XSec NC other (uncorr) | 0.0% | 0.2% |
| XSec NC 1$\gamma$ (uncorr) | 0.0% | 1.5% |
| XSec $\nu_e$ / $\nu_\mu$ (uncorr) | 0.0% | 3.1% |
| XSec Tot | 1.7% | 4.2% |
| Flux+XSec | 1.8% | 4.1% |
| Flux+XSec+SDet+FSI+SI | 3.2% | 5.7% |
| Oscillations | 0.0% | 0.8% |
| All | 3.2% | 5.8% |

Table 21: Effect of $1\sigma$ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

## 4. Analysis procedure

### *4.1. Fit method*

The goal of this analysis is to measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, $\Delta m^2_{32}$ and search for CP violation by setting limits on $\delta_{CP}$ for a given hypothesis of mass hierarchy, which is fixed to either normal or inverted. All the systematics as well as oscillation parameters, which are nuisance, are marginalised. The treatment of the oscillation parameters is shown in Tab. 12. The parameters of interest will be denoted with the vector $\vec{\theta}$, since more than 1 parameter at a time can be measured. $\vec{\theta}$ can include the combination of all the oscillation parameters listed above.

The *VALOR* analysis uses a binned likelihood-ratio method where the estimation of $\vec{\theta}$ is obtained by comparing the observed and predicted Super-K $E_{reco}$ Vs $\theta_l$ spectra for 1-ring events.

Let $N$ be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the Super-K reconstructed energy spectrum prediction (e.g. parameters that control the beam, neutrino interaction simulations, detector response and oscillations).

The log-likelihood function used in the analysis is

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) \tag{3}$$

where $n_i^{obs}$ is the observed number of events in the $i^{th}$ bin and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The log-likelihood function obtained by adding a gaussian penatly term can be written as

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \tag{4}$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematic parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, $n_i^{obs}$ is the observed number of events in the $i^{th}$ bin, and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The total likelihood is given by the product of the likelihood terms of each selected sample,

$$\lambda(\vec{\theta};\mathbf{a}) = \prod_{s=1}^{N_{samples}}\lambda_s(\vec{\theta};\mathbf{a}) \tag{5}$$

where $N_{samples}$ is the number of selected samples used in the analysis, or by the sum of $-2\ln\lambda(\vec{\theta},\mathbf{a})$ from all the super-K samples

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s(\vec{\theta},\mathbf{a}) \tag{6}$$

The advantage of the likelihood ratio method over the extended maximum-likelihood method is that, in the large-sample limit, the quantity $-2\ln\lambda(\vec{\theta};\mathbf{a})$ in Eq. 3 has a $\chi^2$ distribution and it can therefore be used as a goodness-of-fit test.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

### *4.2. Marginalization*

In the previous analyses, except [10], a fully frequentist approach, in which all nuisance parameters were profiled, was used. Now a hybrid Bayesian-frequentist method is used. The nuisance parameters are marginalised over while the parameters of interest are minimized with respect to the likelihood distribution. The marginalization of the nuisance parameters give a so-called marginal likelihood that depends on only on a vector of parameters of interest, $\vec{\theta}$.

The marginal likelihood is derived using the following equation:

$$\lambda_{marg}(\vec{\theta}) = \int_A \lambda(\vec{\theta};\mathbf{a})\pi(\mathbf{a})d\mathbf{a} = \frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i) \tag{7}$$

where $\mathbf{a}$ is the set of nuisance parameters, $\pi(\mathbf{a})$ is the prior of the nuisance parameters. $n$ is the number of toys MC produced for the marginalization and $\mathbf{a_i}$ is the single set of nuisance parameters extracted in the "i-th" toy MC. Many sets of $\mathbf{a}$ ($n = 10k$) should be extracted in order to effectively sample the space of the nuisance parameters. The number of extractions depends on the number of nuisance parameters as well as the precision that is required by the measurement.

We derive the marginal log-likelihood, which we will be denoted as $\chi^2_{marg}$ from now on since it follows a $\chi^2$ distribution when the Poisson statistic converges to the Gaussian:

$$\chi^2_{marg}(\vec{\theta}) = -2\ln\lambda_{marg}(\vec{\theta}) = -2\ln\left(\frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i)\right) \tag{8}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the $n$ throws, not a weighted average of the log-likelihood. During the marginalization process, random values of the nuisance parameters are drawn according to the prior constraints which come from fits to near detector data and control samples at Super-K. Therefore, each throw is already weighted based on the prior information, and as a result the nuisance parameter values are not included in the calculation of the likelihood. For this reason Eq. 3, which doesn't include the penalty term, is used.

Sometimes the free parameters, which are not marginalised, are constrained with a penalty term. In this case a modified version of Eq. 8 is used:

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + (\vec{\theta} - \vec{\theta_0})^T V^{-1}(\vec{\theta} - \vec{\theta_0}) \tag{9}$$

where $\vec{\theta_0}$ is the vector of nominal values of the parameters of interest, $\vec{\theta}$ is the vector of its fitted values and $V$ is the covariance matrix which relates all the free parameters. If the constrained parameter $\theta_{constr}$ is not correlated with the other parameters of interest, $V$ is reduced to the uncertainty $\Delta\theta^{constr}$, and Eq. 9 becomes

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + \frac{(\theta^{constr} - \theta_0^{constr})^2}{(\Delta\theta^{constr})^2} \tag{10}$$

This is the case of $\sin^2\theta_{13}$ when it is measured, i.e. it is parameter of interest, but at the same time it is constrained with the measurement of the reactor experiments. When a penalty term is applied to a parameter of interest it will be specified.

The toys MC used for the marginalization of the nuisance parameters are performed exactly in the same way as described in sec. 3.1, taking into account the correlations, but without statistical fluctuations. First $n = 10k$ distributions of $E_{reco}$ Vs $\theta_l$ for $e$-like and $E_{reco}$ only for $\mu$-like samples are produced using the extracted systematic parameters for $1\times10^{20}$ POT so that they may be scaled appropriately when comparing to datasets of differing POT. Finally they are stored in output files which can be used for whatever needed fit. A fit on a single dataset is performed as following:

- all the $n$ toy MC distributions are read from the files,
- each distribution is normalised to the number of POT of the data set,
- for each toy distribution Eq. 3 is calculated and $\lambda(\vec{\theta},\mathbf{a}) = e^{-(-2\ln\lambda(\vec{\theta},\mathbf{a}))/2}$ is obtained,
- the $\chi^2$ value is finally obtained, using Eq. 8, for a given set of parameter values of interest, $\vec{\theta}$. If $\vec{\theta}$ is constrained, eq. (10) is used.

This procedure is followed for each point in the $\vec{\theta}$ grid, where the binning is so fine that it does not impact on the precision of the measurement. Finally the best fit is found by looking for the $\vec{\theta}$ grid point with the minimum $\chi^2_{bf}$ value. This point corresponds to the best-fit parameters $\vec{\theta}_{bf}$. The minimization can be also performed by using a discrete descent gradient algorithm implemented in the VALOR framework that allows the minimum $\chi^2$ to be found without the need of calculating the full $\chi^2$ distribution across all grid points.

This method is employed when a large number of fits is required, such as the calculation of critical values for Feldman-Cousins contours, as it drastically reduces the amount of CPU time required. A validation of the minimization algorithm will be shown in sec. 5 for the analyses where it is used.

### *4.3. Best-fit spectra with marginalization*

When profiling, the best fit spectra are defined by the MC prediction that gives the largest likelihood values.

However, since the nuisance parameters are marginalised, there does not exist a single set of parameters which corresponds to the best fit. Therefore a new method has been developed to represent the best fit spectra which is consistent with the oscillation results.

The following procedure is used. From each toy MC $i$, used to marginalise the nuisance parameters, a different spectrum, with $N^j(\vec{\theta}_{bf};\mathbf{a})$ number of events in each bin $j$, is produced as well as a likelihood $\lambda(\vec{\theta}_{bf},\mathbf{a_i})$ value, calculated using Eq. 7, where $\vec{\theta}_{bf}$ is the best-fit value of the free parameters vector and $\mathbf{a_i}$ is the vector of nuisance parameters corresponding to the toy $i$. The best-fit number of events in the bin $j$ of the spectrum is calculated as

$$N^j_{marg} = \frac{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf};\vec{f}_i)\cdot N^j(\vec{\theta}_{bf};\vec{f}_i)}{\sum_{i=1}^{n}\lambda(\vec{\theta}_{bf};\vec{f}_i)} \tag{11}$$

where $n$ is the number of toys MC.

One can easily find a relation between the profiled and marginalised best-fit spectra. In the limit of infinite number of toys MC, i.e. the full nuisance parameters space is scanned, the likelihood can be written as

$$\lambda_{marg}(\vec{\theta}_{bf}) = \frac{1}{n}\sum_{i=1,\ i\neq bf}^{n}\Big[\lambda(\vec{\theta}_{bf};\mathbf{a_i}) + \lambda_{\mathbf{prof}}\Big] \tag{12}$$

where $\lambda_{prof} = \lambda(\vec{\theta}_{bf};\mathbf{a_{bf}})$ is the profiled best-fit likelihood. If we denote as $N^j_{prof} = N^j_i(\vec{\theta}_{bf};\mathbf{a_{bf}})$ the number of events in the bin $j$ obtained with profiling, Eq. 11 can be written as

$$N^j_{marg} = \frac{\sum_{i=1,\ i\neq bf}^{n}\lambda(\vec{\theta}_{bf};\mathbf{a_i})\cdot\mathbf{N^j}(\tilde{\theta}_{\mathbf{bf}};\mathbf{a_i}) + \lambda_{\mathbf{prof}}\cdot\mathbf{N^j_{prof}}}{\sum_{i=1}^{n}\lambda(\vec{\theta}_{bf};\mathbf{a_i})} \tag{13}$$

### *4.4. Confidence intervals*

After the fit is completed, the full $\chi^2$ distribution is then shifted with respect to $\chi^2(\vec{\theta}_{bf})$:

$$\Delta\chi^2(\vec{\theta}) = \chi^2(\vec{\theta}) - \chi^2(\vec{\theta}_{bf}) \tag{14}$$

Confidence intervals are set on $\Delta\chi^2(\vec{\theta})$. If

$$\Delta\chi^2(\vec{\theta}) > \Delta\chi^2_{crit} \tag{15}$$

where $\Delta\chi^2_{crit}$ is the critical value for a certain confidence level $X$%(CL), that value of $\vec{\theta}$ is excluded at the $X$% CL. In the gaussian approximation constant values of $\Delta\chi^2_{crit}$ can be used. It will be called "constant $\Delta\chi^2$ method".

However in some cases the gaussian approximation is not reliable and the "Feldman-Cousins method" [54] must be used. In this method a different critical value for each point of the $\vec{\theta}$ grid $\Delta\chi^2(\vec{\theta})_{crit}$ is calculated. The procedure consists of the following steps:

1. many toys MC (e.g. 10k) are produced assuming the oscillation hypothesis of the grid point $\vec{\theta}$ and taking into account both the statistical and systematic uncertainties;
2. for each toy first $\chi^2(\vec{\theta}_{fix})$ is obtained by minimizing $\chi^2(\vec{\theta})$ where the oscillation parameters $\vec{\theta}$ are kept fixed to the values corresponding to the grid point and the nuisance parameters are marginalised. Then $\chi^2(\vec{\theta})$ is minimized, leaving the oscillation parameters $\vec{\theta}$ as free and $\chi^2(\vec{\theta}_{bf})$ is obtained.

3. The steps 2 and 3 are repeated for each toy MC and a distribution of $\Delta\chi^2(\vec{\theta})$ ($f(\Delta\chi^2)$) is obtained.

4. The critical value for $X\%$ CL, $\Delta\chi^2(\vec{\theta})_{crit}$, can be defined as:

$$\Delta\chi^2_{crit} : \int_{-\infty}^{\Delta\chi^2_{crit}} f(\Delta\chi^2) d(\Delta\chi^2) = X\% \tag{16}$$

5. This procedure is repeated for all the points of the $\vec{\theta}$ grid and several critical values are computed

Once a critical value, $\Delta\chi^2(\vec{\theta})_{crit}$, is computed for each oscillation hypothesis in the $\vec{\theta}$ grid, the confidence intervals are set by the condition of Eq. 15.

Several analyses are performed and confidence intervals are computed for:

- 1D $\delta_{CP}$ with reactor constraint: the parameter of interest is $\delta_{CP}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 1D $\delta_{CP}$ T2K only: the parameter of interest is $\delta_{CP}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the uniform prior, shown in Tab. 22, is used.
- 1D $\sin^2\theta_{13}$ T2K only: the parameter of interest is $\sin^2\theta_{13}$ and all the other oscillation parameters are marginalised;
- 1D $\Delta m^2_{32}$ ($\Delta m^2_{13}$) with reactor constraint: the parameter of interest is $\Delta m^2_{32}$ ($\Delta m^2_{13}$) and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 1D $\sin^2\theta_{23}$ with reactor constraint: the parameter of interest is $\sin^2\theta_{23}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 2D $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with reactor constraint: the parameters of interest are $\sin^2\theta_{13}$ and $\delta_{CP}$ while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is a free parameter, the gaussian prior of reactor measurements shown in Tab. 22, is used, through a penalty term as in Eq. 10.
- 2D $\sin^2\theta_{13}$ Vs $\delta_{CP}$ T2K only: the parameters of interest are $\sin^2\theta_{13}$ and $\delta_{CP}$ while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ Vs $\delta_{CP}$ T2K only: the parameters of interest are $\sin^2\theta_{23}$ and $\delta_{CP}$ while all the other oscillation parameters are marginalised;
- 2D $\Delta m^2_{32}$ Vs $\delta_{CP}$ T2K only: the parameters of interest are $\Delta m^2_{32}$ and $\delta_{CP}$ while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ Vs $\Delta m^2_{32}$ with reactor constraint: the parameters of interest are $\sin^2\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) while all the other oscillation parameters are marginalised;

Each confidence interval is calculated for both normal and inverted hierarchy, which is then profiled.
The 1D $\chi^2$ confidence intervals with "constant $\Delta\chi^2$ method" are used to set the $\pm1\sigma$ ranges.
Each point of the $\vec{\theta}$ grid corresponds to the center of the histogram bins, i.e. the $\chi^2$ values are calculated at each histogram bin center. The ranges and binnings used for the analyses listed above are shown in Tab. 22. All the 1-dimensional analyses, where the precision on the best-fit point is important and must be quoted, are performed with a fine binning. Since 2-dimensional analyses are performed only to set contours with "constant $\Delta\chi^2$ method" and a fit with a very high precision is not needed to find the best-fit point, courser binning are used, without affecting the contour. Furthermore a very fine 2-dimensional binning would require much more CPU time and storage space. In the future $\bar{\nu}/\nu$ joint analysis releases when the "Feldman-Cousins method" will be needed also for 2-dimensional fits, a finer binning will be used.

| Parameter | Min | Max | Step | Bins |
|---|---|---|---|---|
| | **1-dimensional fit** | | | |
| $\delta_{CP}$ | $-\pi$ | $+\pi$ | $\pi/200$ | 201 |
| $\sin^2\theta_{13}$ (T2K only) | 0 | 0.113 | 0.113 / 200 | 201 |
| $\sin^2\theta_{23}$ | 0.3 | 0.7 | 0.4 / 200 | 201 |
| $\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$) | $2\times10^{-3}$ | $3\times10^{-3}$ | $10^{-3}/200$ | 201 |
| | **2-dimensional fit** | | | |
| $\delta_{CP}$ | $-\pi$ | $+\pi$ | $\pi/50$ | 51 |
| $\sin^2\theta_{13}$ (reactors) | 0.015 | 0.036 | 0.021 / 81 | 81 |
| $\sin^2\theta_{13}$ (T2K only) | 0 | 0.113 | 0.113 / 81 | 81 |
| $\sin^2\theta_{23}$ | 0.3 | 0.7 | 0.4 / 81 | 81 |
| $\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$) | $2\times10^{-3}$ | $3\times10^{-3}$ | $10^{-3}/21$ | 21 |

Table 22: Grid limits and spacing used for all the 1-dimensional and 2-dimensional marginalization fits and confidence regions. The minimum and maximum values correspond to the center of the $\chi^2$ histogram first and last bin.

### *4.5. Contours with profiled mass hierarchy*

As already pointed out, the constant $\Delta\chi^2$ method gives the correct coverage when the gaussian approximation is valid and is very useful because it does not require a large amount of CPU time. However, we know that in some cases the constant $\Delta\chi^2$ method fails to give the correct coverage and can provide only approximate confidence intervals. An example of this is the case when the constraint on a parameter is not precise enough to keep it away from a physical boundary. For cases such as these a toy MC study is required in order to estimate the correct critical values (see section 4.4).

Also, when considering discrete parameters, such as the mass hierarchy, the $\Delta\chi^2$ is not expected to follow a $\chi^2$ distribution but more likely to follow a Bernoulli distribution as found in [59]. This is an important distinction as the exclusion significance one can interpret from a constant $\Delta\chi^2$ contour is heavily dependant on the distribution the $\Delta\chi^2$ is expected to take.

This means that when the constant $\Delta\chi^2$ method is used in order to set limits as a function of some oscillation parameters as well as the mass hierarchy, the constant critical values we usually use do not give the right coverage for the disfavored mass hierarchy hypothesis contour.

For this reason the data fit 2-dimensional confidence intervals, which are calculated for both the hierarchy hypotheses, will be computed in two different ways:

- normal and inverted hierarchy contours are obtained independently with respect to their own minimum ("independent")
- the global minimum between the normal and inverted hierarchy hypotheses is used to set both the contours ("global");

These different ways to show the confidence intervals correspond to the two extremes, where the "independent" and "global" contours provide respectively the most and the least conservative results.

Since for the 1-dimensional fits the $\Delta\chi^2$ distribution are shown, there is no need to provide results using both "independent" and "global" approaches. However the reader should remember that the constant critical values shown in the figures hold only for the $\Delta\chi^2$ distribution corresponding to the best-fit mass hierarchy hypothesis.

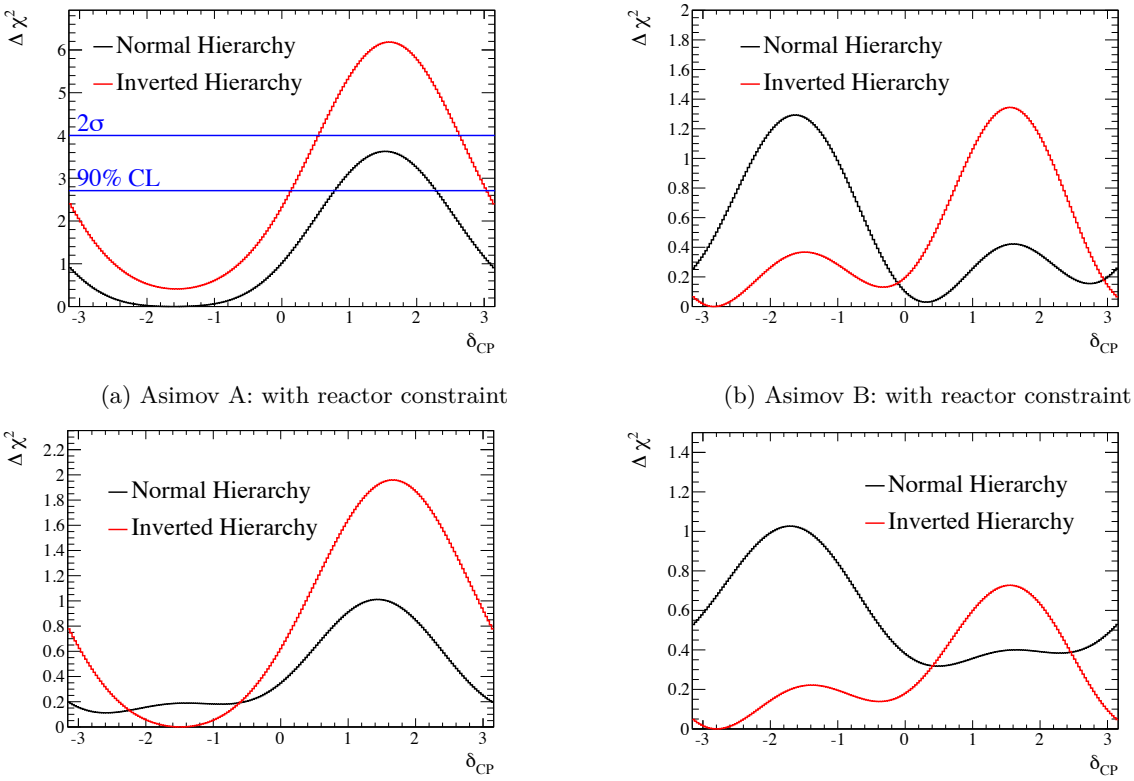
## 5. Validation of the $\bar{\nu}/\nu$ joint analysis

The validation of the fit is performed first by comparing the confidence intervals, obtained with Asimov data sets corresponding to the oscillation parameters in tab. 12 with the other group analyses [42]. Then the coverage study is performed for 1-dimensional $\delta_{CP}$ with reactor constraint fits, in order to verify if the "constant $\Delta\chi^2$" method is reliable or the "Feldman-Cousins" method is needed.

### *5.1. Sensitivity studies*

The critical values of constant $\Delta\chi^2$ method in 1-dimensional confidence intervals are given by $Z\sigma = \sqrt{\Delta\chi^2}$ where $Z$ is the number of standard deviations that defines the confidence level. For 90% CL the critical value is 2.71. The critical values for 2-dimensional fits are 2.3 and 4.61 respectively for 68% and 90% CL.

In Fig. 18 the sensitivity to $\delta_{CP}$ with reactor constraint and T2K only is shown for the Asimov data sets A and B.




(a) Asimov A: with reactor constraint

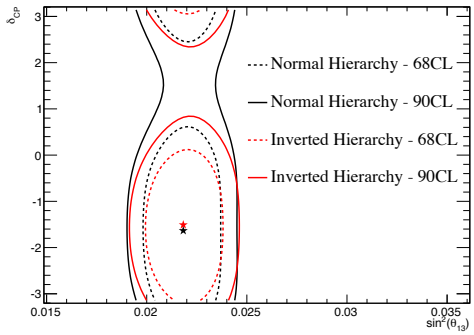
(b) Asimov B: with reactor constraint

(c) Asimov A: with T2K only

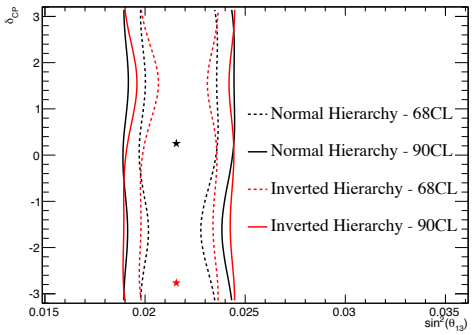
(d) Asimov B: with T2K only

Figure 18: The expected $\Delta\chi^2$ distribution as a function of $\delta_{CP}$ for the Asimov data sets A and B of Tab. 12 corresponding to the Run 1-6 statistics is compared to the result obtained by the other analysis groups. The mass hierarchy is minimized: both the normal and inverted hierarchy $\chi^2$ distributions are shifted to the same global best-fit $\chi^2$ value, which is taken as the minimum between normal and inverted hierarchy.

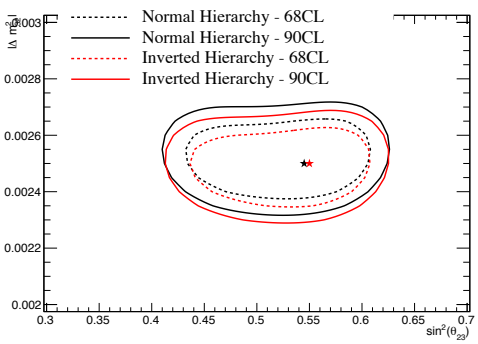
Also the sensitivity to $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with reactor constraint and T2K only data, and $\sin^2\theta_{23}$ Vs $m^2_{32}$ ($m^2_{13}$) with reactor constraint are shown in Fig. 19 and 20, where normal and inverted hierarchy contours are set independently, i.e. each calculated with respect to its own minimum $\chi^2$ value and not the global minimum as a function of the mass hierarchy.



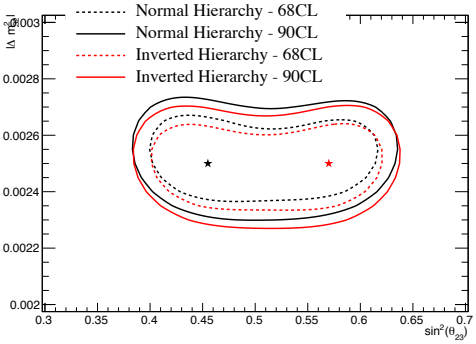

(a) Asimov A: $\sin^2\theta_{13}$ Vs $\delta_{CP}$




(b) Asimov B: $\sin^2\theta_{13}$ Vs $\delta_{CP}$

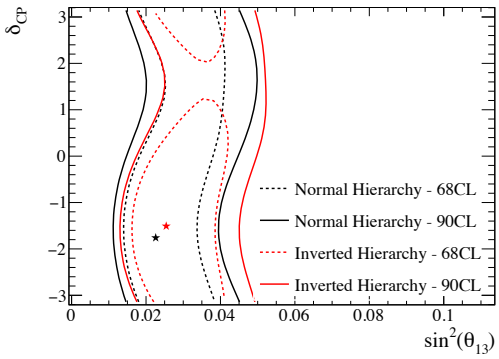



(c) Asimov A: $\sin^2\theta_{23}$ Vs $\Delta m^2_{32}$ $(m^2_{13})$

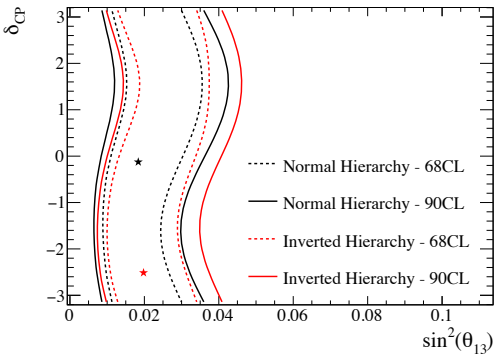



(d) Asimov B: $\sin^2\theta_{23}$ Vs $\Delta m^2_{32}$ $(m^2_{13})$

Figure 19: The contours at 68% and 90% CL with reactor constraint for the Asimov data sets A and B of tab. 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.




(a) Asimov A with reactor constraint




(b) Asimov B with reactor constraint

Figure 20: The contours at 68% and 90% CL for $\sin^2\theta_{13}$ Vs $\delta_{CP}$ for the Asimov data sets A and B of tab. 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.

| Gaussian approximation | |
|---|---|
| **CL** | **Critical value** |
| $1\sigma$ | 1 |
| 90% | 2.71 |
| $2\sigma$ | 4 |

| Feldman-Cousins - $\delta_{CP}^{true} = -\pi/2$ | | |
|---|---|---|
| **CL** | **Critical value** | **Coverage** |
| $1\sigma$ | 0.582 | 0.779 |
| 90% | 2.094 | 0.935 |
| $2\sigma$ | 3.242 | 0.972 |

| Feldman-Cousins - $\delta_{CP}^{true} = 0$ | | |
|---|---|---|
| **CL** | **Critical value** | **Coverage** |
| $1\sigma$ | 1.334 | 0.570 |
| 90% | 2.460 | 0.923 |
| $2\sigma$ | 3.198 | 0.981 |

| Feldman-Cousins - $\delta_{CP}^{true} = +\pi/2$ | | |
|---|---|---|
| **CL** | **Critical value** | **Coverage** |
| $1\sigma$ | 1.178 | 0.605 |
| 90% | 2.224 | 0.938 |
| $2\sigma$ | 3.080 | 0.977 |

Table 23: The critical values of $\Delta\chi^2$ used to set the confidence intervals are shown for both the gaussian approximation (i.e. constant $\Delta\chi^2$) as well as the ones obtained with the Feldman-Cousins method. 10k toy experiments were used for each true point of $\delta_{CP}$. Furthermore the coverage, defined by eq. (18), provided by the gaussian approximation for different true values $\delta_{CP}$ is shown. The true mass hierarchy is set to normal.

### *5.2. Coverage study of the simultaneous fit of $\delta_{CP}$ and mass hierarchy*

The fit validation is performed also by making coverage studies for the fit of the mass hierarchy (MH) and $\delta_{CP}$ with the reactor constraint, the main result of this release of the $\bar{\nu}/\nu$ joint fit, in order to understand if the "Feldman-Cousins method" is needed. The coverage is defined as the number of times the true values of $\delta_{CP}^{true}$ and $\mathrm{MH}^{true}$, used to generate the toys MC, is contained in the confidence interval defined by eq. 15:

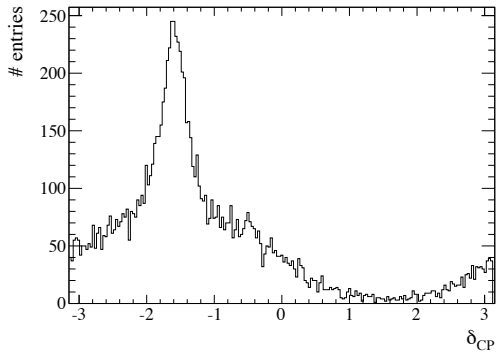
$$\Delta\chi^2(\delta_{CP}, \mathrm{MH}) = \chi^2(\delta_{CP}^{true}, \mathrm{MH}^{true}) - \chi^2(\delta_{CP}^{bf}, \mathrm{MC}^{bf}) \tag{17}$$

where $\delta_{CP}^{bf}$ and $\mathrm{MH}^{bf}$ are the best-fit values. Adopting the "constant $\Delta\chi^2$" method, corresponding to the gaussian approximation, the critical value $\Delta\chi^2_{crit} = \Delta\chi^2_{const}$, pre-computed analytically, at a certain $X\%$ confidence level is used for any value of $\delta_{CP}$ and MH. Many toys MC are performed (10k) and for each one eq. 17 is tested. The condition used to estimate the coverage is

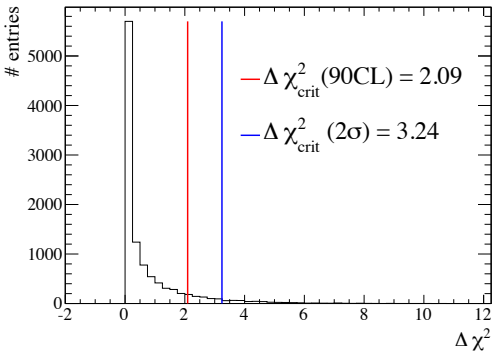
$$\Delta\chi^2(\delta_{CP}, \mathrm{MH}) < \Delta\chi^2_{crit} \tag{18}$$

The fit has a good coverage if the number of toys MC for which eq. 18 is satisfied ($N^X_{toys}$) is equal or close to $X\%$ within the statistical uncertainty of the number of toys MC. If the $N^X_{toys} > X\%$, then the fit has over-coverage, that means the confidence intervals should be tighter than what obtained with the "constant $\Delta\chi^2$ method", while if $N^X_{toys} < X\%$, then the fit has under-coverage and excludes a wider range of values than in the correct case. In Fig. 21 the $\Delta\chi^2$ distribution of $\delta_{CP}$ 1-dimensional fit with reactor constraint, obtained by using eq. (17), is shown for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$. The critical values computed using the Feldman-Cousins method are shown in table 23. It can be seen that the coverage given by the constant critical values is not correct, so, in order to provide correct confidence intervals, the Feldman-Cousins method is required.

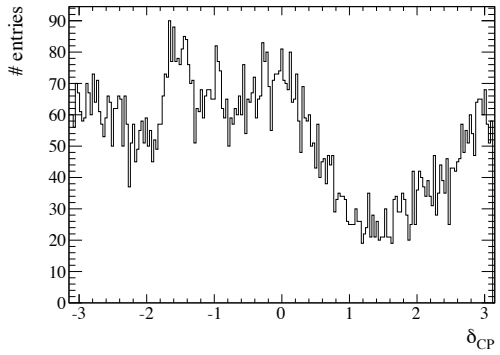
The constant $\Delta\chi^2$ method does not give the proper coverage when the parameter of interest is not gaussian distributed and when the relation with the likelihood is not linear. Another reason why the correct coverage could not be assured by the constant $\Delta\chi^2$ method is the presence of physical boundaries. $\delta_{CP}$ is



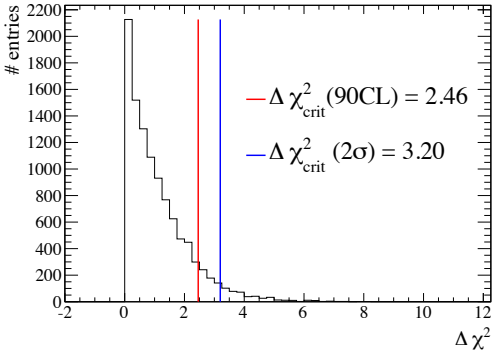

(a) Best-fit distribution of $\delta_{CP}$. $\delta_{CP}^{true} = -\pi/2$.



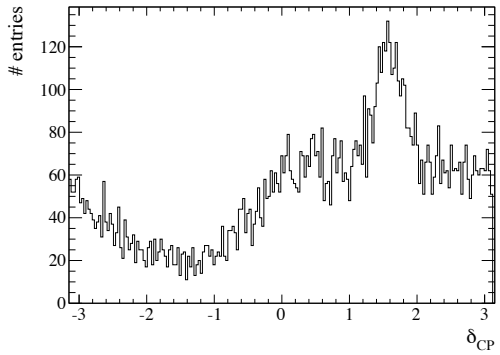

(b) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = -\pi/2$.



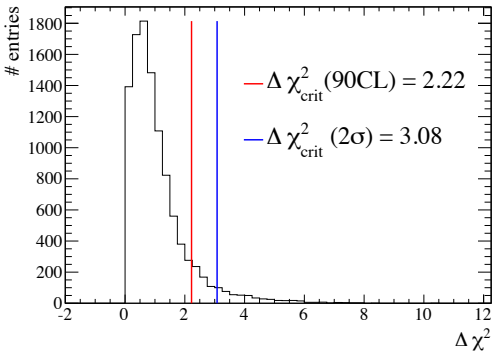

(c) Best-fit distribution of $\delta_{CP}$. $\delta_{CP}^{true} = 0$.




(d) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = 0$.




(e) Best-fit distribution of $\delta_{CP}$. $\delta_{CP}^{true} = +\pi/2$.



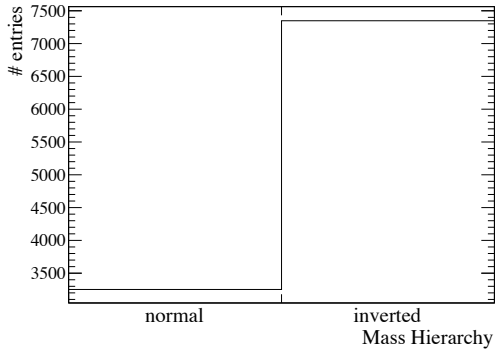

(f) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = +\pi/2$.

Figure 21: Distributions of best-fit $\delta_{CP}$ (left) and $\Delta\chi^2$ (right). $10k$ toy experiments are performed with true values of $\delta_{CP} = -\pi/2, 0, +\pi/2$, assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and $\delta_{CP}$ are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set.

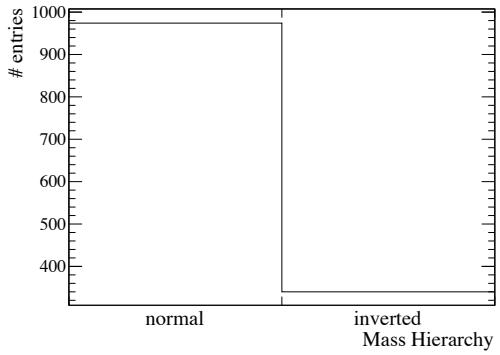
continuos because periodic, however it shows up physical boundaries at $\pm\pi/2$, since they correspond to the values that maximize or minimize the number of events in the $e$-like samples.

In general the gaussian approximation gives over-coverage for all the listed confidence intervals, except for $1\sigma$ CL of $\delta_{CP}^{true} = +\pi/2$ and 0 with $MH^{true} = NH$. This can be understood by analyzing the best-fit result of 10k toy experiments with $\delta_{CP}^{true} = +\pi/2$ and normal hierarchy. In this case the constant $\Delta\chi^2$ method gives under-coverage for $1\sigma$ CL (the constant critical value is larger than the one computed with Feldman-Cousins) but over-coverage for 90% and $2\sigma$ CL (the constant critical value is smaller than the one computed with Feldman-Cousins). In general a critical value estimated with Feldman-Cousins is larger (smaller) when the $\Delta\chi^2$ distribution is wider (tighter). Wider $\Delta\chi^2$ distributions are due to degeneracy with other param-

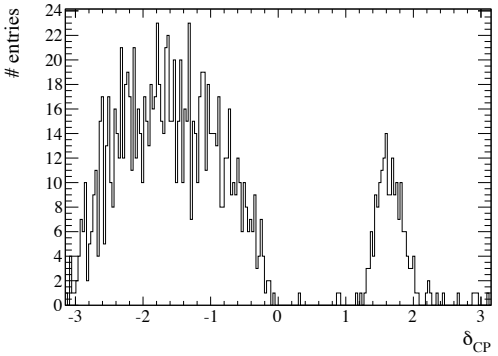
eters, in this case the mass hierarchy. Indeed in fig. 35 it is shown that the inverted hierarchy increases the expected number of events in neutrino mode $e$-like sample and reduces it in antineutrino mode $e$-like sample similarly to the effect of $\delta_{CP} = +\pi/2$. For this reason the best-fit mass hierarchy is chosen more often ($\sim 70\%$) as inverted than the other true points of $\delta_{CP}$, while the true mass hierarchy is normal, and make the $\Delta\chi^2$ distribution wider, as shown in fig. 22. This becomes more clear when looking to eq. (17). However for higher confidence intervals only the toy experiments less likely with the expectation matters, i.e. toys in the tail of the $\Delta\chi^2$ distribution. In fig. 22 the best-fit mass hierarchy and $\delta_{CP}$ distributions are shown only for those toy experiments which best-fit $\Delta\chi^2$ is larger than 2. This toy experiments correspond to fluctuations of the data that prefer values of $\delta_{CP} \sim -\pi/2$, removing consequently the degeneracy with the mass hierarchy which best-fit values is chosen to be equal to the true one, i.e. normal, about the 70% of the times. This effect does not make the $\Delta\chi^2$ distribution wider in the range of values which define the critical values for higher confidence levels. Another proof of this behavior can be given by looking to fig. 23, that shows the best-fit mass hierarchy obtained with 10k toy experiments with $\delta_{CP}^{true} = -\pi/2$ and normal hierarchy. In this case also for the $1\sigma$ CL the constant $\Delta\chi^2$ method gives over-coverage, because the $\Delta\chi^2$ distribution is tighter. The reason is that $\delta_{CP} = -\pi/2$ is degenerate with normal hierarchy, which is the true mass hierarchy. For this reason the inverted hierarchy is disfavored (chosen less than 70% of the times) and the eq. (17) does not provide a wider $\Delta\chi^2$ distribution.




(a) Best-fit mass hierarchy distribution.




(b) Best-fit mass hierarchy distribution. Only fits with best-fit $\Delta\chi^2 > 2$ are selected.




(c) Best-fit $\delta_{CP}$ distribution. Only fits with best-fit $\Delta\chi^2 > 2$ are selected.

Figure 22: Best-fit distribution of mass hierarchy and $\delta_{CP}$ from the simultaneous fit of $\delta_{CP}$ and mass hierarchy. $10k$ toy experiments are produced with $\delta_{CP}^{true} = +\pi/2$ and normal hierarchy.

From the best-fit distributions of $\delta_{CP}$ it is clear that T2K is not able to measure $\delta_{CP}$ with the current statistics. However if $\delta_{CP}^{true} = \pm\pi/2$ the best-fit distribution shows a peak around the true value, while for $\delta_{CP}^{true} = 0$ the fit prefers values smaller than 0. This is, again, due to the degeneracy between the mass hierarchy and $\delta_{CP}$. Indeed if the best-fit mass hierarchy is inverted, the event rates is decreased and $\delta_{CP} = -\pi/2$ is preferred in order to balance it in the neutrino mode sample. While when the normal hierarchy is preferred,

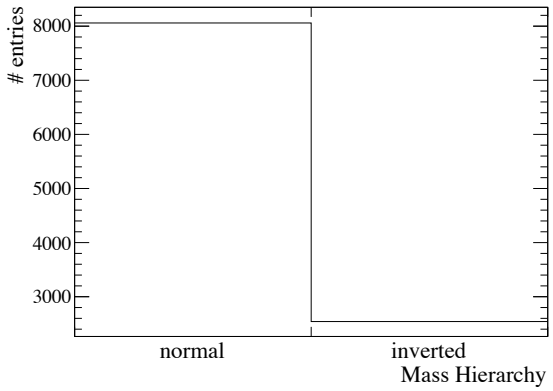



Figure 23: Best-fit distribution of mass hierarchy from the simultaneous fit of $\delta_{CP}$ and mass hierarchy. $10k$ toy experiments are produced with $\delta_{CP}^{true} = -\pi/2$ and normal hierarchy.

i.e. the true value, a fluctuation of the data could fit any value of $\delta_{CP}$ symmetrically distributed around 0. This becomes more clear when looking to the best-fit distribution of $\delta_{CP}$ when the mass hierarchy is not fitted but fixed to normal (the true value). As shown in fig. 24, the best-fit $\delta_{CP}$ distribution becomes quite uniform, with a wide peak around $\delta_{CP} = 0$.

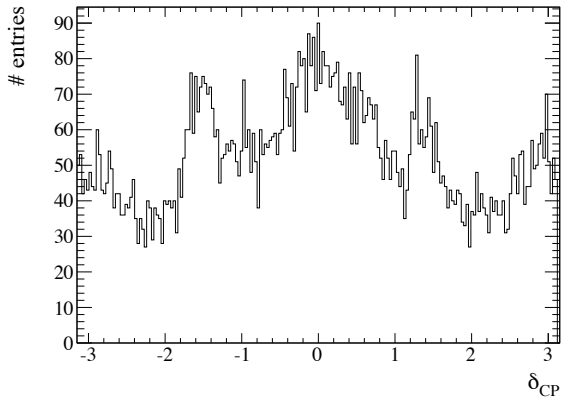



Figure 24: Best-fit distribution of $\delta_{CP}$ obtained by fixing the mass hierarchy to normal and not profiled. $10k$ toy experiments are produced with $\delta_{CP}^{true} = 0$ and normal hierarchy.

The $\Delta\chi^2$ distribution corresponding to $\delta_{CP}^{true} = -\pi/2$ shows a spike at 0. This comes from the intrinsic physical boundary present at $-\pi/2$, since the $\sin\delta_{CP}$ term in the oscillation probability has a upper boundary at $\pm\pi/2$. Since the true mass hierarchy is normal and the $\delta_{CP}^{true} = -\pi/2$ a downward fluctuation in the antineutrino $e$-like sample and a upward fluctuation in the neutrino mode $e$-like sample cannot be accomodated by changing the hierarchy to inverted, since it have an opposite effect, and further need to choose $-\pi/2$ as best-fit value. A similar behavior is shown in the best-fit distribution of $\delta_{CP}^{true} = +\pi/2$, though the effect is smaller because a downward fluctuation in the neutrino $e$-like sample and a upward fluctuation in the antineutrino $e$-like sample can be accomodated also by choosing the hierarchy as inverted, since it has a degenerate effect as $\delta_{CP} = +\pi/2$.

In general most of the strange behaviors present in the best-fit and $\Delta\chi^2$ distributions can be explained with a still low sensitivity to both $\delta_{CP}$ and mass hierarchy and the strong degeneracy between the two oscillation parameters.

### *5.3. Validation of the minimization algorithm*

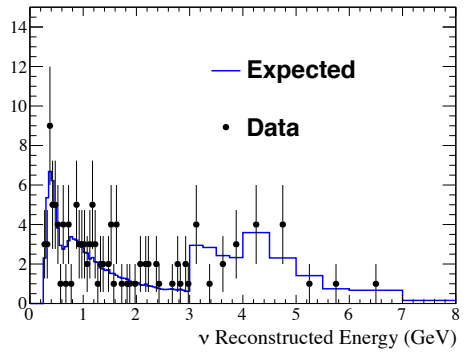
As explained in section 4.4, the best-fit parameters are found by setting up a grid of points of the parameters of interest and calculating the $\Delta\chi^2$ for each of them. The grid is fine enough in order to have a very small variation of the $\Delta\chi^2$ distribution as a function of the parameter of interest. The best-fit point can be found by computing the entire $\chi^2$ distribution and looking for the lowest $\chi^2$ value. However since the Feldman-Cousins implies a large number of toy experiments and fits for several points of $\delta_{CP}$, this can be very CPU time consuming. For this reason a minimization algorithm has been developed. Similar to MINUIT [53], this algorithm uses the descent gradient: at each point of the parameter of interests it calculates the $\chi^2$ in the adjacent points along all the possible directions (depending on the number of dimensions) and move toward the lower one, until it stops in a minimum. It is worth to note the MINUIT cannot be used because the grid of points of the free parameters is discrete, while it needs to work in a continuos space.

However the descent gradient algorithm is strongly affected by the presence of local minima and depends on the point of the free parameters from where the fit starts. In order to avoid this problem the fit is performed 4 times starting from 4 different starting points: $-\pi$, $-\pi/2$, $+\pi/2$ and $+\pi$. The method has been tested by performing $10k$ toys for normal hierarchy and $4k$ toys for inverted hierarchy for the true points $\delta_{CP} = -\pi/2, 0, +\pi/2, +\pi$. For all the tests the number of times where the fit stopped in a local minimum is $\leq 1/10000$. Similar numbers have been found for inverted hierarchy. For all the critical values the minimization algorithm is used helping to reduce the CPU time of about 30% - 50%, depending on the toy experiment.

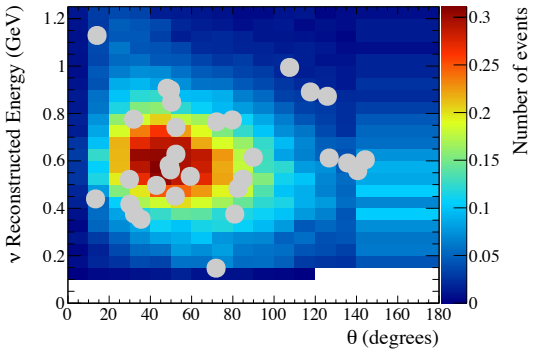
## 6. Results of the $\bar{\nu}/\nu$ joint analysis with the Run 1-6 dataset

The results of the Run 1-6 data set fit for an exposure of $6.914\times10^{20}$POT in neutrino mode and of $4.011\times10^{20}$POT in antineutrino mode is shown in this chapter. In fig. 25 the predicted spectra of $\mu$-like and $e$-like samples are compared to the observed data. In table 24 the expected and observed number of events in Run 1-6 data set are shown. The predicted number of events are shown for the oscillation parameters in table 12 of Asimov data sets A and B. All the predicted and observed events are in good agreement. A small excess of events with respect to the expectation is observed in the neutrino mode $e$-like samples, though it is still within $1\sigma$ of the statistical uncertainty. This trend is mostly inherited from the Run 1-4 data set, since only 2 neutrino mode $e$-like events are observed in Run 5+6. Furthermore the latest global fit of the reactor experiments gives a lower value of $\sin^2\theta_{13}$ with a much smaller uncertainty (see Sec. 3.1 and Tab. 12). It is interesting to see that the number of events of antineutrino mode $e$-like events is in perfect agreement with the assumption of $\delta_{CP} = -1.601$, which is the best-fit value of the Run 1-4 data set. This means that, for what concerns the event rate, we expect to see agreement between neutrino and antineutrino mode data in the fit of $\delta_{CP}$, though the statistics are not expected to be enough to reject $\delta_{CP} = 0$ at any significant confidence level.

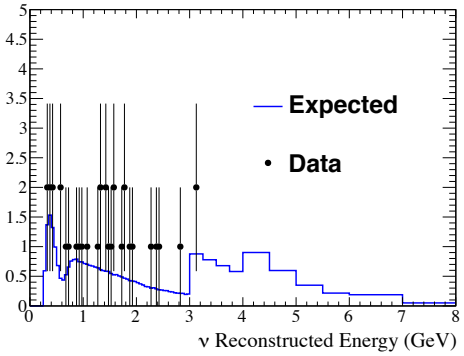
The comparison of the confidence intervals obtained by the fit of the Run 1-6 data set among all the oscillation analysis groups is shown in [42].



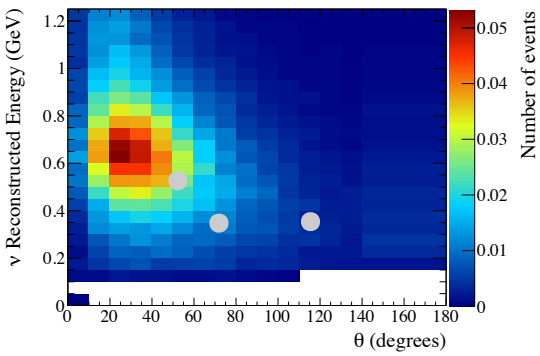

(a) Muon-like, neutrino mode




(b) Electron-like, neutrino mode




(c) Muon-like, antineutrino mode




(d) Electron-like, antineutrino mode

Figure 25: The predicted spectra compared to the Run 1-6 dataset (dots), which corresponds to an exposure of $6.914\times10^{20}$POT in neutrino mode and of $4.011\times10^{20}$POT in antineutrino mode, are shown. The expected distribution corresponds to the oscillation parameters of the Asimov data set A shown in table 12. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

| Beam mode | Sample | Exp. Asimov A | Exp. Asimov B | Exp. Not Osc | Observed | Best fit |
|---|---|---|---|---|---|---|
| neutrino | $\mu$-like | 125.914 | 130.464 | 484.737 | 124 | 125.688 |
| neutrino | $e$-like | 26.594 | 19.954 | 5.698 | 31 | 26.969 |
| antineutrino | $\mu$-like | 34.284 | 35.032 | 99.02 | 34 | 34.114 |
| antineutrino | $e$-like | 3.218 | 3.355 | 1.253 | 3 | 3.344 |

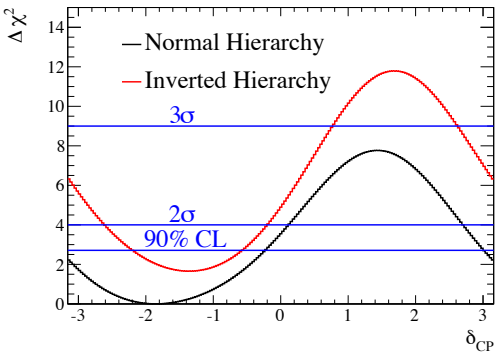
Table 24: The observed, expected and best-fit number of events in Run 1-6 data set for an exposure of $6.914\times10^{20}$POT in neutrino mode and of $4.011\times10^{20}$POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning and the oscillation parameters shown in Tab. 12. The total number of events of the best-fit spectra is obtained by using Eq. 11.

### *6.1. 1-dimensional $\chi^2$ distributions and best-fit*

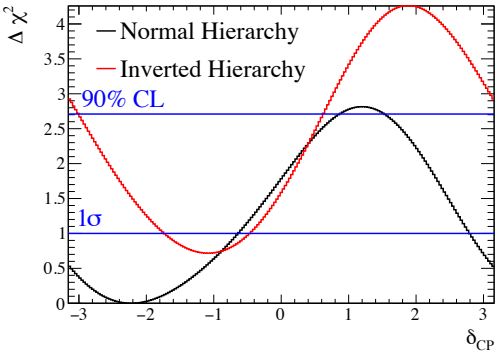
The same 1-dimensional $\chi^2$ distributions as in sec. 5.1 are produced and shown in Fig. 26. The best-fit value of $\delta_{CP}$ when the reactor constraint is applied is -1.854 not as close to $-\pi/2$ as the Run 1-4 fit result, which best-fit value was -1.601 (see Tab. 12). However the difference of $\chi^2$ between $\delta^{bf}_{CP}$ (best-fit value) and $\delta_{CP} = -\pi/2$ is smaller than 0.1. So the result is consistent with $\delta_{CP} = -\pi/2$ as well as the Run 1-4 data fit.

Looking at the $\chi^2$ distribution of $\sin^2\theta_{13}$ with T2K only data we can see that the reactor measurements are in agreement with the T2K data at about $1\sigma$ level if $\delta_{CP} \sim -\pi/2$ and normal hierarchy is assumed.

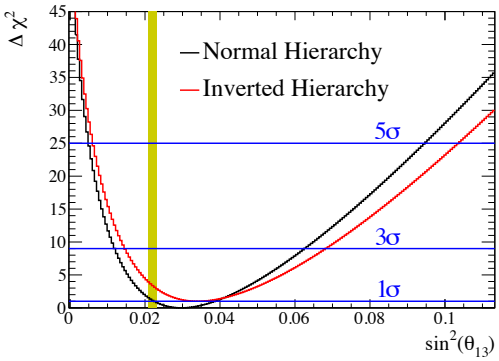
In Fig. 27 the $\chi^2$ distributions as a function of $\delta_{CP}$ with reactor constraint is obtained by running the analysis using either only neutrino or antineutrino mode samples for the Run 1-6 data set, in order to check the compatibility of neutrino and antineutrino mode data. The result is mainly driven by the excess of the observed number of events in $e$-like neutrino mode sample compared to what is expected. Though the



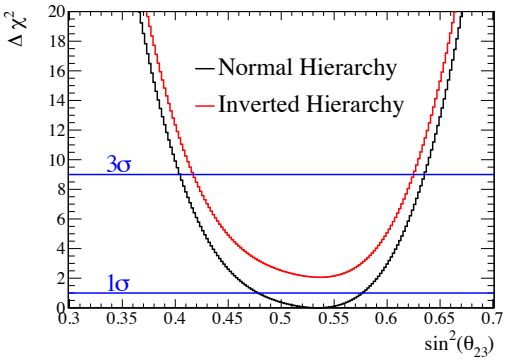

(a) $\delta_{CP}$ with reactor constraint



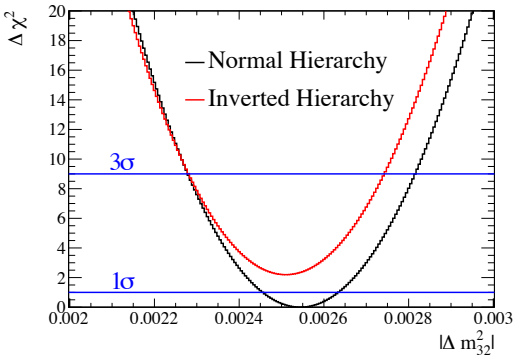

(b) $\delta_{CP}$ with T2K data only




(c) $\sin^2\theta_{13}$ with T2K data only




(d) $\sin^2\theta_{23}$ with reactor constraint




(e) $\Delta m^2_{32}$ ($\Delta m^2_{13}$) with reactor constraint

Figure 26: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.

significance given by the antineutrino mode samples is still low, also given by the fact that $\sin^2\theta_{23}$ is not constrained by the antineutrino mode $\mu$-like sample as much as by the neutrino mode $\mu$-like sample, it is interesting to note that the antineutrino data also prefer a value of $\delta_{CP}$ close to $-\pi/2$.

In fig. 28 the $\chi^2$ distributions as a function of $\delta_{CP}$ is obtained by performing a rate-only analysis; instead of using the shape information of the $E_{reco}$ Vs $\theta_l$ spectra, the total number of events of each sample is fitted, reducing the $E_{reco}$ Vs $\theta_l$ template to 1 bin. Then the best-fit value of $\delta_{CP}$ is -1.602 closer to $-\pi/2$ with respect to the case where the shape information is used.

In Tab. 25 the best-fit values as well as the $1\sigma$ ranges for each oscillation parameter are shown, for

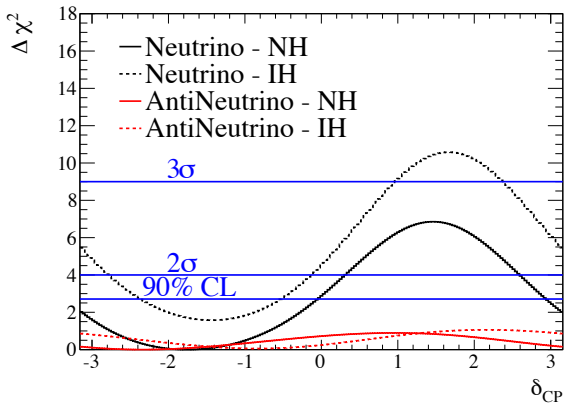



Figure 27: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set as a function of $\delta_{CP}$ with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22.

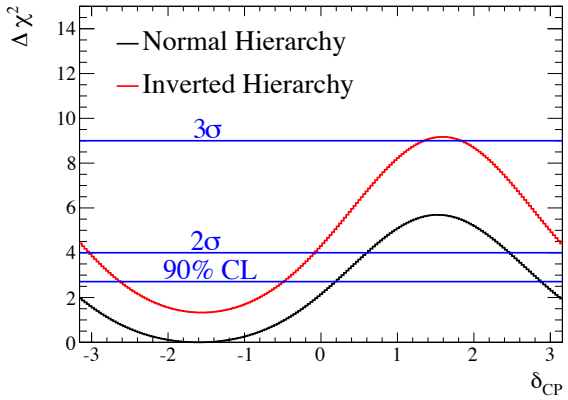



Figure 28: Measured $\Delta\chi^2$ distributions with a rate-only analysis using the full run 1-6 data set as a function of $\delta_{CP}$ with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22.

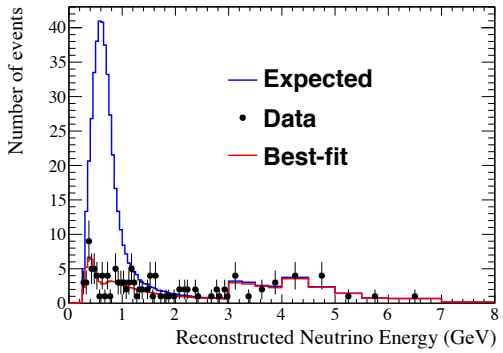
both normal and inverted hierarchy. The $1\sigma$ ranges are extracted for each oscillation parameter from the 1-dimensional $\Delta\chi^2$ distributions shown in Fig. 26, where one oscillation parameter is free and all the other oscillation as well as the systematic parameters are marginalised. The mass hierarchy is minimized: for both normal and inverted hierarchy the $1\sigma$ ranges are calculated with respect to the same global minimum. Since this approach is not entirely correct and there does not exist any procedure to set more reliable confidence intervals which does not consist of toy MC studies (see section 4.5), the advice to the reader is to give more importance to the difference in $\chi^2_{marg}$ between normal and inverted hierarchies in the respective best-fit points, instead of the $1\sigma$ ranges of inverted hierarchy, which are smaller by construction. It is also worth to notice that the $1\sigma$ ranges for normal hierarchy (best-fit) are set correctly with the constant $\Delta\chi^2$ method in the gaussian approximation.

The best-fit spectra, shown in fig. 29 are computed as described in sec. 4.3 and are compared to the unoscillated nominal spectra. For $e$-like samples, where 2-dimensional templates are used, the projection onto the reconstructed neutrino energy axis is shown. All the systematic and oscillation parameters are marginalised following the procedure described in sec. 4.3, while $\sin^2\theta_{23}$, $\delta_{CP}$ and the mass hierarchy were fixed to their best-fit values shown in table 25. Indeed these oscillation parameters have the strongest effect on the measurement, since $\sin^2\theta_{13}$ is constrained with the reactor measurement.

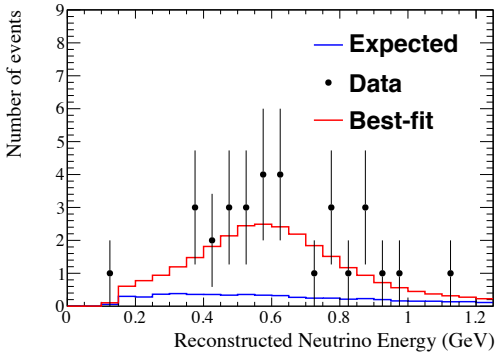
A goodness of test fit is performed by comparing data with the best fit spectra shown in fig. 29. In order

| | Reactors | Normal Hierarchy | | | Inverted Hierarchy | | |
|---|---|---|---|---|---|---|---|
| **Parameter** | | Best-fit | $\chi^2_{min}$ | $\pm1\sigma$ interval | Best-fit | $\chi^2_{min}$ | $\pm1\sigma$ interval |
| $\delta_{CP}$ | YES | -1.854 | 296.51 | [-2.737; -0.859] | -1.351 | 298.16 | - |
| $\delta_{CP}$ | NO | -2.231 | 299.49 | [-$\pi$; -0.640] & [2.793;+$\pi$] | 1.100 | 300.20 | [-1.735;-0.469] |
| $\sin^2\theta_{13}$ | NO | 0.0299 | 297.09 | [0.0224; 0.0387] | 0.0339 | 298.09 | - |
| $\sin^2\theta_{23}$ | YES | 0.534 | 295.92 | [0.478; 0.577] | 0.536 | 297.98 | - |
| $\Delta m^2_{32}$ or $\Delta m^2_{13}$ ($10^{-3}$ eV$^2/c^4$) | YES | 2.545 | 295.36 | [2.455; 2.634] | 2.510 | 297.56 | - |

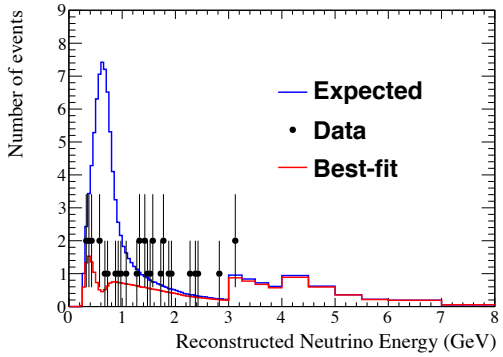
Table 25: The measured oscillation parameter best-fit and minimum $\chi^2$ values as well as the $1\sigma$ CL intervals are shown, for both normal and inverted hierarchy. The $\pm1\sigma$ interval is obtained by using the *constant* $\Delta\chi^2$ method. All the parameters are measured using the reactor constraint $\sin^2 2\theta_{13} = 0.085 \pm 0.005$ [66], except for $\delta_{CP}$ and $\sin^2\theta_{13}$. As explained in section 4.5, the gaussian approximation does not work well for discrete parameters like the mass hierarchy however, since there is not a clear way to compute $1\sigma$ ranges for both mass hierarchies without performing CPU intensive toy MC studies, the constant $\Delta\chi^2$ method is used.



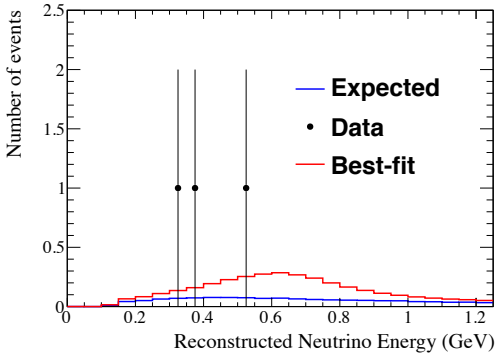

(a) Muon-like, neutrino mode




(b) Electron-like, neutrino mode




(c) Muon-like, antineutrino mode




(d) Electron-like, antineutrino mode

Figure 29: The best-fit spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the run 1-6 data set. They correspond to the fit of $\sin^2\theta_{23}$, $\delta_{CP}$ and mass hierarchy with reactor constraint where all the nuisance parameters are marginalised. The method described in sec. 11 is used. The best-fit distributions are compared with the predicted unoscillated spectrum (all the oscillation parameters are set to 0), generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

to determine how compatible the data is with our model the log likelihood is calculated for each sample as in eq. (3) and the $\chi^2/ndof$ is used to estimate the fit probability using the Poisson statistic, given the number of degrees of freedom equal to $ndof = N_{bins} - N_{free}(8-5=3)$, where $N_{bins}$ is the number of template bins and $N_{free}$ is the number of free parameters in the fit. The free parameters are $\sin^2\theta_{23}$, $\delta_{CP}$ and the mass hierarchy in this case. In this analysis, a coarse binning scheme is used for the goodness of fit test as in previous analyses. This is used so that there are enough events in each bin to calculate p-values in the

gaussian approximation. 5 bins (0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV) are used for the neutrino mode muon-like spectrum, 1 bin is used for each of the muon-like antineutrino and electron-like spectra in order to enter the gaussian regime, where the $\chi^2$ distribution is well defined. The $\chi^2$ and resulting p-value for each spectrum is shown in Tab. 26. We see for all the spectra, the data and best fit spectra are very compatible.

| | $\chi^2$ / ndf | p-value |
|---|---|---|
| MH fixed | 2.08917 / 6 | 0.84 |
| MH fitted | 2.08917 / 5 | 0.91 |

Table 26: Goodness of fit for the 2-dimensional fit of $\sin^2\theta_{23}$, $\delta_{CP}$ and mass hierarchy (both fitted and fixed) where all the other parameters are marginalised.

### *6.2. Feldman-Cousins and significance for the simultaneous fit of $\delta_{CP}$ and mass hierarchy*

In Fig. 30 the confidence intervals obtained with the "Feldman-Cousins" method are shown for $\delta_{CP}$ with reactor constraint. The critical values at both 90% and $2\sigma$ CL are computed for both the mass hierarchies and the following true values of $\delta_{CP}$: $-\pi$, -2.356, $-\pi/2$, -0.785, 0, +0.785, $+\pi/2$, -2.356, $+\pi$. The procedure described in section 4.4 is used. From the zoom on a small range of $\Delta\chi^2$ it is clear that $\delta_{CP} = 0$ is excluded with a significance of $2\sigma$, while $\delta_{CP} = \pi$ is not excluded at 90% CL. Between 8k and 10k toy experiments were used for the computation of the critical values.

The $\pm1\sigma$ error on the critical values are shown as well. They are computed assuming that the critical value follows a binomial p.d.f., which becomes clear when looking to eq. (18). The critical values corresponding to respectively $\Delta\chi^2_{crit} \pm 1\sigma$ are computed for the $X$% CL using the following equation:

$$\Delta\chi^2_{crit} \pm 1\sigma : \int_{-\infty}^{\Delta\chi^2_{crit}\pm1\sigma} f(\Delta\chi^2)d(\Delta\chi^2) = X\% \pm \sqrt{N \cdot X\% \cdot (1 - X\%)} \tag{19}$$

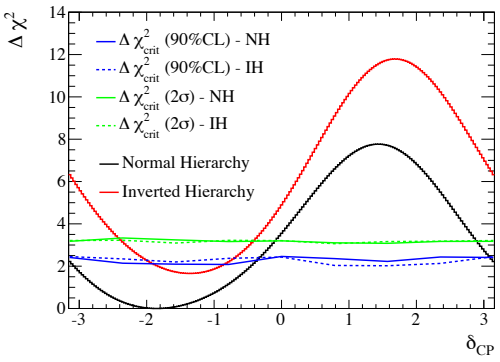
where $N$ is the number of toy experiments. The higher is the confidence level to be computed the more toy experiments are needed. Indeed higher confidence levels corresponds to higher critical values, for which the tail of the $\Delta\chi^2$ distribution, which is less populated, becomes more important. However from fig. 30 it is shown that even 4k toy experiments would be enough for the computation of the critical values, since the $\pm1\sigma$ variation of the critical values does not change significantly the confidence intervals.

The significance to $\delta_{CP} = 0$ and $\pi$, shown in fig. 31 and table 27, is computed by performing the p-value using the same $\Delta\chi^2$ distributions obtained with the Feldman-Cousins method and 10k toy experiments, as explained in section E. The p-value for $\delta_{CP} = 0$ is 0.032, which corresponds to a significance of Z = 2.14 standard deviations, assuming the errors to be gaussian distributed. Instead the p-value for $\delta_{CP} = \pi$ is 0.122 which corresponds to only Z = 1.55 standard deviations.

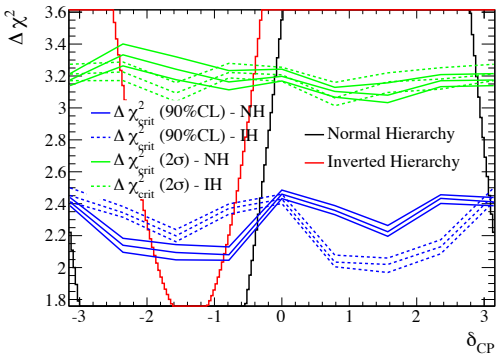
$\delta_{CP} = 0$ is favored by the data with respect to $\delta_{CP} = \pi$. Indeed $\delta_{CP} = \pi$ produces an event migration toward higher energy, which is disfavored by the $e$-like Run 1-6 data, for both neutrino and antineutrino mode. The variation of $\delta_{CP}$ from 0 to $\pi$ in the best-fit spectra, keeping $\sin^2\theta_{23}$ and the mass hierarchy fixed to the global best-fit values, produces an event migration to lower energy values with respect to 0.6 GeV as shown in table 28 and fig. 32. In particular the data set of the $e$-like antineutrino mode sample shows all the events with reconstructed energy smaller than 0.6 GeV.

| $\delta_{CP}$ | $\Delta\chi^2_{data}$ | p-value | # of std. deviations |
|---|---|---|---|
| 0 | 3.552 | 0.032 | 2.14 |
| $\pi$ | 2.221 | 0.122 | 1.55 |

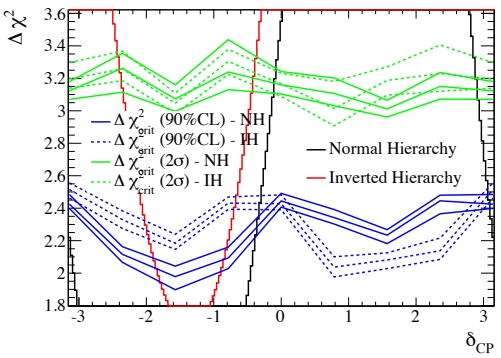
Table 27: Summary of best-fit $\Delta\chi^2$ value measured from the Run 1-6 data set, p-value and number of standard deviations with respect to the shown value of $\delta_{CP}$. The best-fit mass hierarchy is normal. All the other systematic parameters are marginalised.




(a) Confidence intervals with more than 8k toys MC in each point of $\delta_{CP}$.

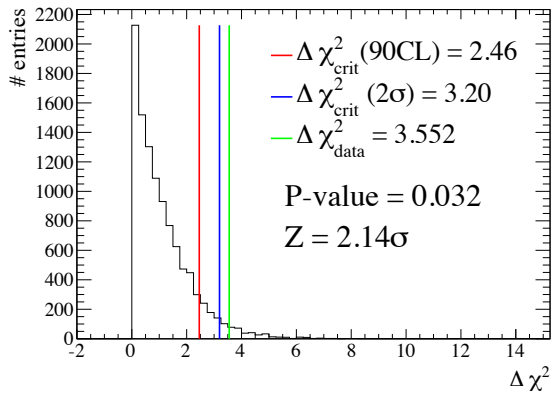



(b) Confidence intervals with more than 8k toys MC in each point of $\delta_{CP}$ with a zoom on the critical values. The $\pm 1\sigma$ errors on the critical values are shown as well.




(c) Confidence intervals with 4k toys MC in each point of $\delta_{CP}$ with a zoom on the critical values. The $\pm 1\sigma$ errors on the critical values are shown as well.

Figure 30: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. The "Feldman-Cousins" method is used to set the $\Delta\chi^2$ critical values in order to provide confidence intervals with the proper coverage. The ranges of $\delta_{CP}$ which correspond to $\Delta\chi^2$ values larger than the critical values (blue for 90% CL and green for $2\sigma$ CL) are excluded at a certain confidence level. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22. The $\pm 1\sigma$ error on the critical values is shown as well, where specified, and is represented by the lowest and highest set of critical values, with the same color of the central ones.



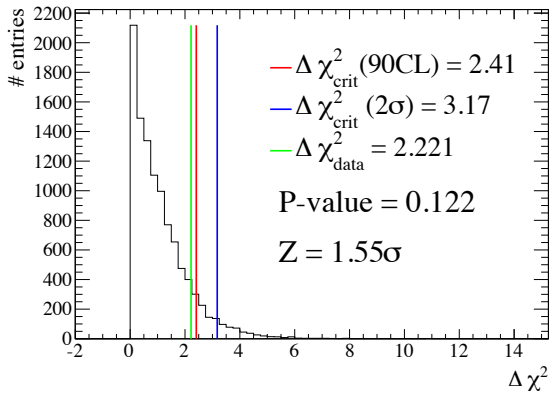
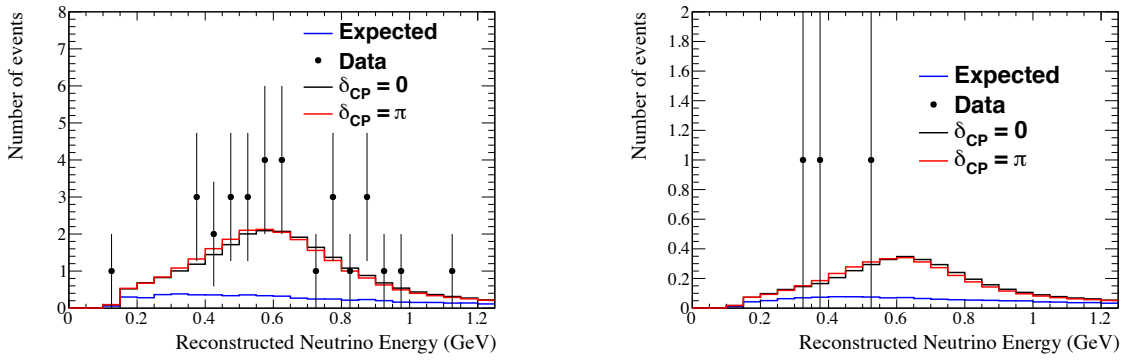




Figure 31: Distributions of $\Delta\chi^2$. $10k$ toy experiments are performed with true values of $\delta_{CP} = 0$ (left) and $\pi$ (right), assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and $\delta_{CP}$ are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set. The $\Delta\chi^2$ value (green line) measured from the Run 1-6 data set as well as the p-value and the significance as number of standard deviations (Z), are shown as well.

| $e$-like neutrino mode | | | |
|---|---|---|---|
| | Best-fit, $\delta_{CP} = 0$ | Best-fit, $\delta_{CP} = \pi$ | Data |
| Events with $E_{reco} < 0.6$ GeV | 58.4% | 61.1% | 64.5% |
| $< E_{reco} >$ | 0.616 | 0.604 | 0.627 |
| $e$-like antineutrino mode | | | |
| | Best-fit, $\delta_{CP} = 0$ | Best-fit, $\delta_{CP} = \pi$ | Data |
| Events with $E_{reco} < 0.6$ GeV | 53.3% | 56.3% | 100% |
| $< E_{reco} >$ | 0.644 | 0.631 | 0.408 |

Table 28: The fraction of events with reconstructed neutrino energy smaller than 0.6 GeV and the mean reconstructed neutrino energy is shown for the best-fit ($\sin^2\theta_{23}$ Vs $\delta_{CP}$ Vs mass hierarchy) spectra where $\delta_{CP}$ is kept fixed to either 0 or $\pi$, for both neutrino and antineutrino mode $e$-like samples, and for the Run 1-6 data set. The other two free oscillation parameters, $\sin^2\theta_{23}$ and mass hierarchy, as set to the same best-fit values as in fig. 29. The reactor constraint is applied and all the nuisance parameters are marginalised. The method described in sec. 11 is used.




(a) Muon-like, neutrino mode

(b) Electron-like, neutrino mode

Figure 32: The best-fit spectra as a function of the neutrino reconstructed energy where $\delta_{CP}$ is fixed to either 0 (red) or $\pi$ (black) are shown for both neutrino (left) and antineutrino (right) mode $e$-like samples of the run 1-6 data set. and for the Run 1-6 data set. The other two free oscillation parameters, $\sin^2\theta_{23}$ and mass hierarchy, as set to the same best-fit values as in fig. 29. The reactor constraint is applied and all the nuisance parameters are marginalised. The method described in sec. 11 is used as well as the same best-fit values for $\sin^2\theta_{23}$ and mass hierarchy. The best-fit distributions are compared with the predicted unoscillated spectrum (all the oscillation parameters are set to 0), generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

### 6.3. 2-dimensional confidence intervals

The same 2-dimensional confidence intervals as in sec. 5.1 are produced. In Fig. 33 the corresponding 2-dimensional contours produced with the Run 1-6 data set are shown drawing the normal and inverted hierarchy contours independently, i.e. each contour is produced with respect to the global minimum under that hierarchy hypothesis. In Fig. 34 both normal and inverted hierarchy contours are produced with respect to the same global minimum, i.e. the normal hierarchy hypothesis.

As already explained in section 4.5, "global" and "independent" contours correspond respectively to the least and most conservative approach.

As in 1-dimensional analysis results, looking at contours of $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with T2K only data, in Fig 33a, it is clear that the reactor measurements are in agreement with the T2K data at $1\sigma$ level if $\delta_{CP} \sim -\frac{\pi}{2}$. The tension between the $\sin^2\theta_{13}$ reactor constraint and the T2K data constraint is clear at $\sim \frac{+\pi}{2}$ where the T2K data prefer a lager mixing angle with both hierarchy hypotheses. This is the cause of the closed 90% contour in Fig 33b which is also in good agreement with the 1D $\delta_{CP}$ fit in Fig.26a. This effect is much greater for the inverted hierarchy fits and thus leads to a much tighter constraint on $\delta_{CP}$. As a result, we see that all $\delta_{CP}$ constraints in Fig. 33b, 33c, 33d are tighter for the inverted hierarchy hypothesis than for the normal. Fig 33c also shows a slight anti-correlation between $\delta_{CP}$ and $\Delta m^2_{32}$ in the normal hierarchy fit and a correlation in the inverted most likely coming from the fact that $\Delta m^2_{32}$ and $cos(\delta_{CP})$ have similar effects on the shape of the appearance spectra. In Fig 33d we see correlation where the constraint on $\delta_{CP}$ is narrower for smaller values of $\sin^2\theta_{23}$ as they both appear in the appearance term. In Fig 33e we see that $\sin^2\theta_{23}$ and $\Delta m^2_{32}$ are not very correlated however the inverted hierarchy contour is shifted down in $\Delta m^2_{32}$ with respect to the normal hierarchy contour. This effect is due to the correlation between $\Delta m^2_{32}$ and the mass hierarchy shown in Fig. 35. Indeed the inverted hierarchy hypothesis produces an event migration toward higher energy which is balanced in the fit by a lower fit value of $\Delta m^2_{13}$ with respect to the normal hierarchy case.

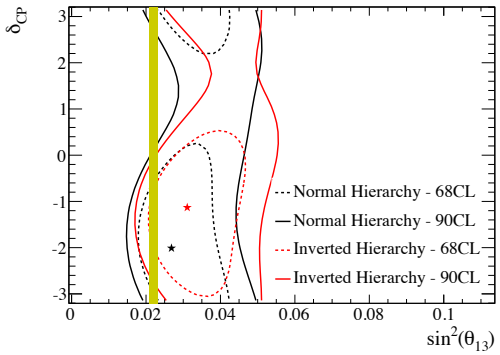
The *constant* $\Delta\chi^2$ contours with respect to the global minimum in Fig. 34 show similar trends. As the normal hierarchy was preferred in all fits, the normal hierarchy contours are identical to those in Fig. 33. The inverted hierarchy contours are noticeably tighter in all cases. In Fig 34a we see that there is a closed $\Delta\chi^2 = 4.61$ contour, and in all others the $\Delta\chi^2 = 4.61$ contour is of similar size to the $\Delta\chi^2 = 2.3$ contour for the normal hierarchy.

### 6.4. Discussion of the results

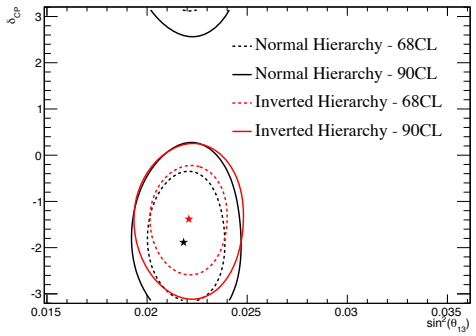
The value of $\delta_{CP}$ favored by the joint fit analysis when the reactor constraint is applied is -1.854 and corresponds to normal hierarchy. $\delta_{CP} = 0$ is excluded at 2.14 standard deviations and a large fraction of the range between $\delta_{CP} = 0$ and $+\pi$ is excluded with a significance of $2\sigma$ (with the constant $\Delta\chi^2$ method), while $\delta_{CP} = \pi$ is not excluded at 90% CL. Basically with the Run 1-6 data set we cannot exclude the hypothesis of no CP violation in the leptonic sector at 90% CL.

The inverted hierarchy is disfavored at $\sim 1\sigma$ level and it is partially excluded at $3\sigma$ for values of $\delta_{CP}$ around $+\pi/2$. It has been shown this exclusion, stronger than what is expected from the fit of the Asimov data sets for $\delta_{CP} = -1.601$, is due to the small excess of events in the neutrino mode $e$-like sample. Indeed the impact of antineutrino mode samples on the result is quite small. If the reactor constraint is not applied, the best-fit value of $\delta_{CP}$ is still quite close to $-\pi/2$. However the exclusion is much weaker than when the reactor constraint is applied. The contour of $\sin^2\theta_{13}$ Vs $\delta_{CP}$ shows an agreement at the level of $1\sigma$ with the measurement of the reactors if $\delta_{CP} \sim -\pi/2$.

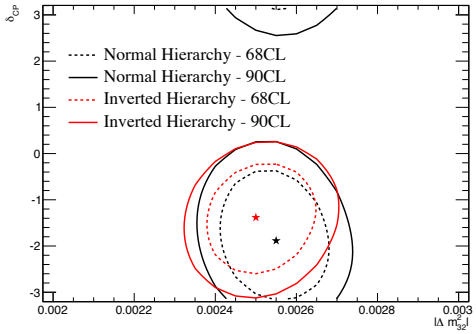
The result shows a stronger exclusion of $\delta_{CP}$ compared to the 2014 joint fit analysis [8]. This is mainly due to the new reactor constraint which has lower $\sin^2\theta_{13}$ (0.0217 instead of 0.0250) and a smaller uncertainty: $0.085 \pm 0.005$ instead of $0.095 \pm 0.01$.



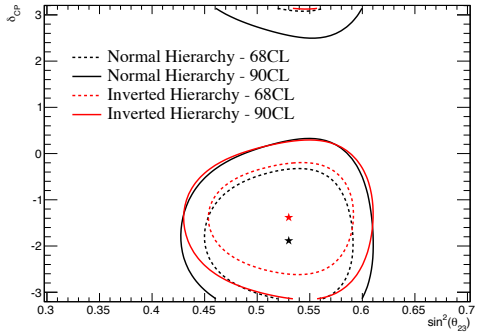

(a) $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with T2K data only



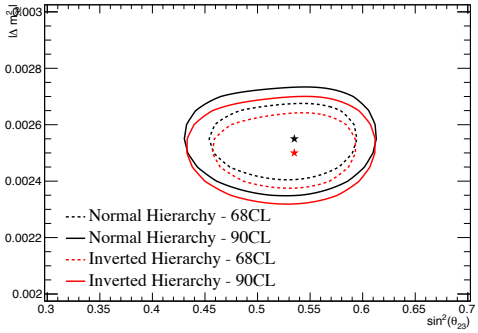

(b) $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with reactor constraint




(c) $\Delta m^2_{32}$ ($\Delta m^2_{13}$) Vs $\delta_{CP}$ with reactor constraint

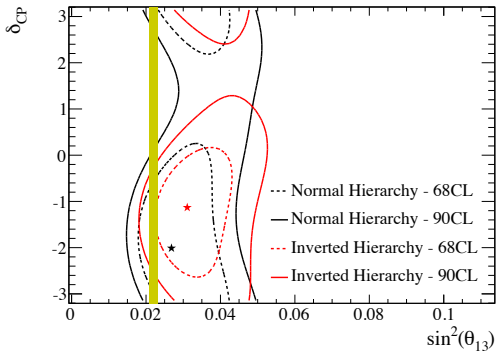



(d) $\sin^2\theta_{23}$ Vs $\delta_{CP}$ with reactor constraint

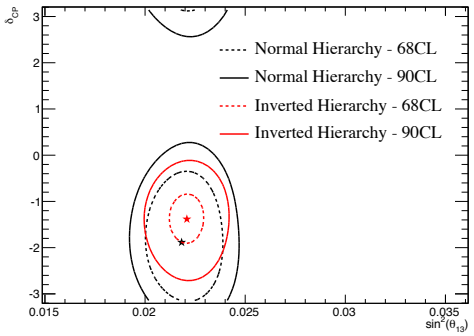



(e) $\sin^2\theta_{23}$ Vs $\Delta m^2_{32}$ ($\Delta m^2_{13}$) with reactor constraint

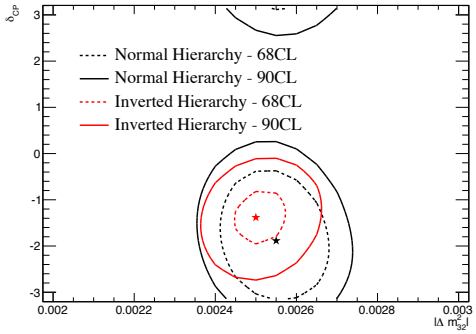
Figure 33: Measured confidence intervals with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed independently with respect to the global minimum of each hierarchy (star). All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured contour of $\sin^2\theta_{13}$ Vs $\delta_{CP}$ without reactor constraint, in Fig 33a is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.



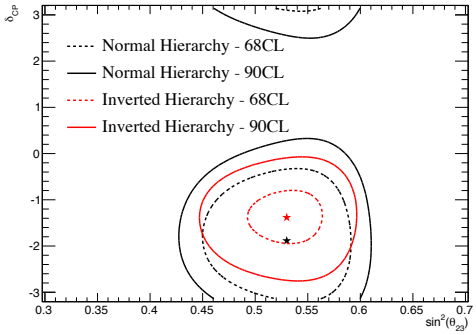

(a) $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with T2K data only



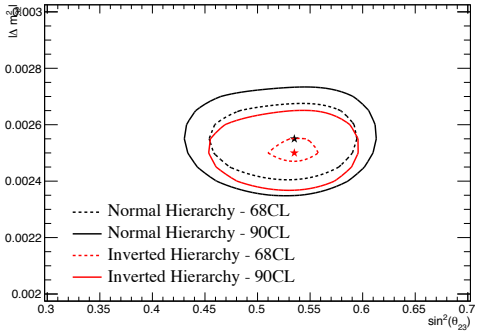

(b) $\sin^2\theta_{13}$ Vs $\delta_{CP}$ with reactor constraint




(c) $\Delta m^2_{32}$ ($\Delta m^2_{13}$) Vs $\delta_{CP}$ with reactor constraint

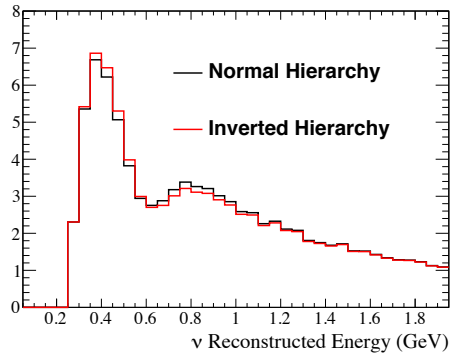



(d) $\sin^2\theta_{23}$ Vs $\delta_{CP}$ with reactor constraint

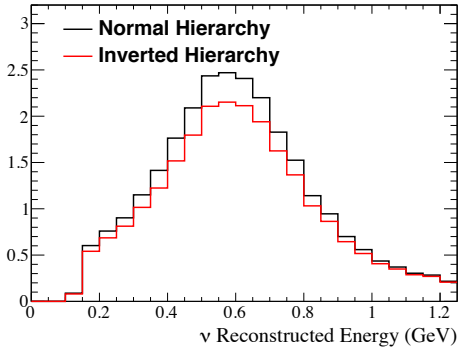



(e) $\sin^2\theta_{23}$ Vs $\Delta m^2_{32}$ ($\Delta m^2_{13}$) with reactor constraint

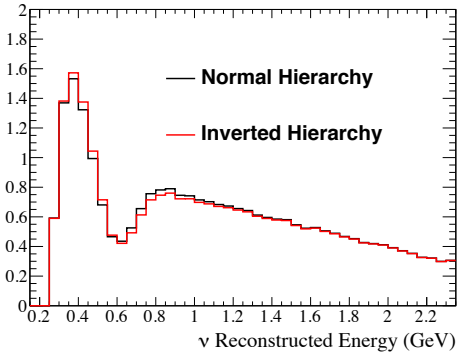
Figure 34: Measured confidence intervals with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. Contours for normal (black) and inverted (red) hierarchy are computed with respect to the minimum of the best-fit hierarchy (star), which corresponds to normal hierarchy. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured contour of $\sin^2\theta_{13}$ Vs $\delta_{CP}$ without reactor constraint is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [66], shown by the yellow bar.



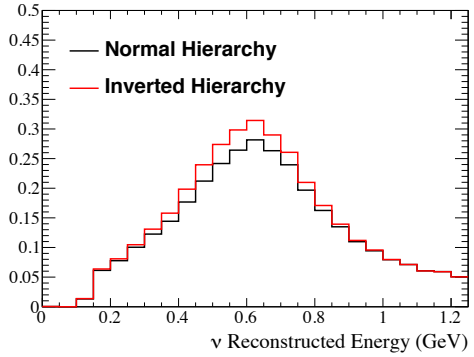

(a) Muon-like, neutrino mode




(b) Electron-like, neutrino mode




(c) Muon-like, antineutrino mode




(d) Electron-like, antineutrino mode

Figure 35: The expected spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the run 1-6 data set for both normal (black) and inverted (red) hierarchies. They correspond to the Asimov data set A in table 12 and are generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

## 7. Summary

In this technical note, we have presented the results of the 3-flavour $\bar{\nu}/\nu$ joint analysis performed by the *VALOR* group, on the combined Run 1-6 data set, for an exposure of $6.914\times10^{20}$ POT in neutrino mode and $4.011\times10^{20}$ POT in antineutrino mode. The goal of the analysis is to search for CP violation in the leptonic and put limits on $\delta_{CP}$ as well as measure all the other oscillation parameters to which T2K is sensitive: $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, $\Delta m^2_{32}$ ($\Delta m^2_{13}$) and partially mass hierarchy, to which T2K has little sensitivity.

This analysis predicts 125.914 ± 6.459 (syst) for neutrino $\mu$-like, 34.284 ± 1.755 (syst) for antineutrino $\mu$-like, 26.594 ± 1.808 (syst) for neutrino $e$-like, and 3.218 ± 0.238 (syst) for antineutrino $e$-like single ring events in Super K and observes respectively 124 for neutrino $\mu$-like, 34 for antineutrino $\mu$-like, 31 for neutrino $e$-like, 3 for antineutrino $e$-like.

The $\bar{\nu}/\nu$ analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single $\mu$-like ring events and reconstructed energy versus lepton angle of single $e$-like ring events was used to compute the confidence intervals. All 101 systematic parameters as well as the oscillation parameters which were not parameters of interest considered in this analysis were marginalised.

From the fit of the Run 1-6 data we can see that the favored value of $\delta_{CP}$ is -1.854, close to $-\pi/2$. The correct coverage is provided by using the "Feldman-Cousins" method with which the significance to $\delta_{CP} = 0$ becomes 2.14 standard deviations, while $\delta_{CP} = \pi$ is not excluded at 90% CL. The Run 1-6 data set does not exclude the hypothesis of no CP violation in the leptonic sector at 90% CL.

In case of inverted hierarchy a not negligible range of $\delta_{CP}$ around $+\pi/2$ is excluded with a significance of $3\sigma$.

From the contour of $\sin^2\theta_{13}$ Vs $\delta_{CP}$ without the reactor constraint it is clear that the T2K data are in agreement with the reactor measurements within $1\sigma$ if $\delta_{CP} \sim -\pi/2$.

It has also been shown that the result is dominated by the excess of event in the neutrino mode $e$-like sample, though the antineutrino mode $e$-like also prefers values of $\delta_{CP}$ close to $-\pi/2$, though the significance is not high.

## References


[1] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K $3.23\times10^{19}$-POT Muon-Neutrino Disappearance Analysis, T2K-TN-036(v4), http://www.t2k.org/docs/technotes/036

[2] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K $1.431\times10^{20}$-POT Muon-Neutrino Disappearance Analysis T2K-TN-064(v3), http://www.t2k.org/docs/technotes/064

[3] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K $1.431\times10^{20}$-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-087(v1), http://www.t2k.org/docs/technotes/087

[4] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K $3.010\times10^{20}$-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-141(v10), http://www.t2k.org/docs/technotes/141

[5] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K $6.3933\times10^{20}$-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-155(v2), http://www.t2k.org/docs/technotes/155

[6] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K $6.57\times10^{20}$-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-183(v3), http://www.t2k.org/docs/technotes/183

[7] C.Andreopoulos et al., T2K $3.010\times10^{20}$-POT Joint 3-Flavour Oscillation Analysis T2K-TN-154(v3), http://www.t2k.org/docs/technotes/154

[8] C.Andreopoulos et al., T2K $6.57\times10^{20}$-POT Joint 3-Flavour Oscillation Analysis T2K-TN-175, http://www.t2k.org/docs/technotes/175

[9] C.Andreopoulos et al., T2K $2.315\times10^{20}$-POT 3-Flavour Muon-Antineutrino Disappearance Analysis T2K-TN-243, http://www.t2k.org/docs/technotes/243

[10] C.Andreopoulos et al., T2K $4.0108\times10^{20}$-POT 3-Flavour Electron-Antineutrino Appearance Analysis T2K-TN-252, http://www.t2k.org/docs/technotes/252

[11] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K $6.5700\times10^{20}$-POT 3-Flavour Muon-Neutrino Disappearance Analysis: Supplementary material on the effect of systematics, T2K-TN-183 supplement, http://www.t2k.org/docs/technotes/183

[12] T2K Collaboration (K.Abe et al.), First Muon-Neutrino Disappearance Study with an Off-Axis Beam, Phys.Rev.D85:031103, 2012; arXiv:1201.1386

[13] T2K Collaboration (K.Abe et al.), Measurement of Neutrino Oscillation Parameters from Muon Neutrino Disappearance with an Off-axis Beam, arXiv:1308.0465 [hep-ex], 2013.

[14] T2K Collaboration (K.Abe et al.), Measurement of neutrino oscillation in appearance and disappearance channels by the T2K experiment with $6.6\times10^{20}$ protons on target http://www.t2k.org/comm/pubboard/JournalPapers/oa2014prd/draft3

[15] XXX

[16] Y.Hayato, A neutrino interaction simulation program library NEUT, Acta Phys.Polon.B40:2477-2489,2009.

[17] Beam WG, Flux tuning (11b v3.2), http://www.t2k.org/beam/NuFlux/FluxRelease/11brelease/11btuned-v3.2

[18] M.Otani and J.Kameda, Super-K events and Updated systematic errors for $\nu_\mu$ disappearance analysis with T2K $1.431\times10^{20}$ POT T2K-TN-065(v6), http://www.t2k.org/docs/technotes/065

[19] P. de Perio et al., NEUT nuclear effects (FSI) T2K-TN-033(v2.0), http://www.t2k.org/docs/technotes/033

[20] P. de Perio et al., Cross section parameters for the 2012a oscillation analysis T2K-TN-108(v1.3), http://www.t2k.org/docs/technotes/108

[21] P. de Perio et al., Implementation of the NIWG cross-section parametrization, T2K-TN-113(v1.2)

[22] Y.Hayato, K.McFarland, and M.Wascko (Private communication)

[23] P. de Perio et al., Pion Hadronic Secondary Interactions in Super-Kamiokande, T2K-TN-105(v1.1) http://www.t2k.org/docs/technotes/105

[24] Y. Ashie et al., Phys. Rev. Lett. 93, 101801 (2004); Y. Ashie et al., Phys. Rev. D71, 112005 (2005); J. Hosaka et al., Phys. Rev. D 74, 032002 (2006).

[25] Beam group, http://www.t2k.org/beam/NuFlux/FluxRelease/11arelease/11atuned-v2.3.

[26] M. Wilking, Status of the 2013 ND280 $\nu_\mu$ analysis, http://www.t2k.org/asg/meeting/20130411/nd280numustatus/view.

[27] M. Hartz, A. Kaboth, and K. Mahn, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2013 Oscillation Analysis, T2K-TN-166(v3.0) http://www.t2k.org/docs/technotes/166.

[28] M. Hartz, A. Kaboth, K. Mahn, Jordan Myslik, Christine Nielsen and Mark Scott, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2014/2015 Oscillation Analysis, T2K-TN-220(v1.0) http://www.t2k.org/docs/technotes/220.

[29] M. Hartz, A. Kaboth, K. Mahn, Jordan Myslik, Christine Nielsen and Mark Scott, ND280 Flux and Cross Section Constraint (BANFF) for 2015/16, T2K-TN-230(v.1.0) http://www.t2k.org/docs/technotes/230.

[30] J. Kameda, "Updated study of the systematic error in $\nu_\mu$ disappearance analysis from Super-Kamiokande", T2K-TN-159

[31] J. Albert et al., "Study on SK $\nu_e$ candidates and systematic errors with T2K $3.23 \times 10^{19}$ POT data" T2K-TN-157

[32] J. Hignight et al., Super-Kamiokande events and data quality studies for T2K Run4 T2K-TN-148(v2r3), http://www.t2k.org/docs/technotes/148.

[33] 14a data and MC info http://www.t2k.org/t2ksk/documentation-page/MCWeighting/MCReweighting14a

[34] E. Wang, L. Alvarez-Ruso, Y. Hayato, K. Mahn, J. Nieves "Photon emission in neutral current interactions at the T2K experiment" http://arxiv.org/pdf/1507.02446v1.pdf

[35] 2015 SK splines http://www.t2k.org/asg/oagroup/inputs/201415/2015SKsplines

[36] Super-K 13a flux release http://www.t2k.org/beam/NuFlux/FluxRelease/13arelease

[37] A. Bercellie et al. Cross section parameters for 2014 oscillation analysis. T2K-TN-192 (v2.1) http://www.t2k.org/docs/technotes/192/ .

[38] T. Wongjirad, A. Minamino and J. Kameda, Muon neutrino disappearance analysis with T2K $6.57 \times 10^{20}$ POT data, T2K-TN-164(v4), http://www.t2k.org/docs/technotes/164

[39] C. Bronner et al., Run1-4 p-theta electron neutrino appearance analysis, T2K-TN-161(v2.6), http://www.t2k.org/docs/technotes/161

[40] Y. Nishimura, J. Hignight, and S. Nakayama $\nu_e$ appearance analysis of RUN1-4 data with neutrino energy spectrum, T2K-TN-162(v2.2), http://www.t2k.org/docs/technotes/162

[41] A. Kaboth, R. Calland and D. Payne (Mach3) A Joint ND280-SK $1R_\mu$-SK $1R_e$ Fit using MCMC T2K-TN-171 http://www.t2k.org/docs/technotes/171

[42] K. Duffy and A. Kaboth (Mach3) A Joint ND280-SK $1R_\mu$-SK $1R_e$ of neutrino and antineutrino-mode data using MCMC T2K-TN-269 http://www.t2k.org/docs/technotes/269

[43] K. Duffy for the Mach3 group, http://www.t2k.org/asg/meeting/2015-06-18/mach3_nuebarapp

[44] H. Tanaka for the p-$\theta$ group, http://www.t2k.org/asg/meeting/2015-06-18/nuebar_appearance_study

[45] N Abgrall et al. Measurements of production properties of $K^0_S$ mesons and $\Lambda$ hyperons in proton-carbon interactions at 31 GeV/c Phys. Rev. C 89, 025205

[46] N Abgrall et al. Measurements of Cross Sections and Charged Pion Spectra in Proton-Carbon Interactions at 31 GeV/c. Phys.Rev., C84:034604, 2011; N. Abgrall et al. Measurement of production properties of positively charged kaons in proton-carbon interactions at 31 GeV/c. arXiv:hep-ex/1112.0150, 2011.

[47] J. Beringer et al. (Particle Data Group), Phys. Rev. D86, 010001 (2012)

[48] Lorena Escudero Sanchez for the *VALOR* group, http://www.t2k.org/asg/oagroup/meeting/20121219/VaLOR

[49] Tom Dealtry for the *VALOR* group, http://www.t2k.org/asg/meeting/2012/20121015/valornumudisapp/view

[50] Davide Sgalaberna for the *VALOR* group, http://www.t2k.org/asg/meeting/2015-05-14/view

[51] G.Cowan, K.Cranmer, E.Gross, O.Vitells, "Asymptotic formulae for likelihood-based tests of new physics" arXiv:1007.1727v2

[52] G. Cowan (1998) Statistical data analysis, Oxford University Press, ISBN 0-19-850155-2.

[53] F.James, M.Roos, Minuit: A System for Function Minimization and Analysis of the Parameter Errors and Correlations, Comput.Phys.Commun. 10:343 (1975)

[54] G.J. Feldman and R.D. Cousins, A Unified approach to the classical statistical analysis of small signals, Phys.Rev.D57:3873-3889,1998; physics/9711021

[55] R.D. Cousins and V.L.Highland, Nucl.Instr. and Meth. A320:331035, 1992.

[56] J. Stuart, A. Ord and S.Arnold, Kendall's Advanced Theory of Statistics, Vol 2A (6th Ed.) (Oxford University Press, New York, 1994).

[57] N.Grant for the *VALOR* group, http://www.t2k.org/asg/meeting/2011/20111124/pvalues/view

[58] D.Karlen, p-values, distributions and low statistics, http://www.t2k.org/asg/oagroup/meeting/20120416/lowstat/view

[59] X. Qian, A. Tan, W. Wang, J. J. Ling, R. D. McKeown and C. Zhang, Statistical evaluation of experimental determinations of neutrino mass hierarchy, Phys.Rev.D86, 113011 (2012)

[60] K. McFarland for the T2K $\nu_\mu$-disappearance contour committee (C.Andreopoulos, N.Grant, J.Kameda, D.Karlen, E.Kearns, K.McFarland, M.Otani, M.Zito) Report to ASG conveners meeting, 17 May 2011, http://www.t2k.org/asg/contour/ASG-Conveners-Report/view

[61] R.Nichol, Talk at Neutrino 2012

[62] P. Adamson, et al., Measurement of Neutrino and Antineutrino Oscillations Using Beam and Atmospheric Data in MINOS, arXiv:1304.6335 [hep-ex], 2013

[63] A. Radovic, Talk at DPF 2013

[64] Y. Itow, Recent results in atmospheric neutrino oscillations in the light of large $\theta_{13}$, Nucl. Phys. B (Proc. Suppl.) 235 79 (2013).

[65] J. Beringer et al. (Particle Data Group with 2013 update), Phys. Rev. D86, 010001 (2012)

[66] K.A. Olive et al. (Particle Data Group 2014 with 2015 update), Chin. Phys. C, 38, 090001

[67] G.L.Fogli, E.Lisi et al., Evidence of theta(13)>0 from global neutrino data analysis, Phys.Rev.D84:053007, 2011; arXiv:1106.6028 [hep-ph]

[68] M.C.Gonzalez-Garcia, M.Maltoni, J.Salvado and T.Schwetz, Global fit to three neutrino mixing: critical look at present precision, arXiv:1209.3023 [hep-ph]

[69] Prob3++ 3-flavour oscillation probability software, http://www.phy.duke.edu/ raw22/public/Prob3++/

[70] P. Adamson *et al.* [MINOS Collaboration], Phys. Rev. Lett. **110** (2013) 25, 251801 [arXiv:1304.6335 [hep-ex]].

# Appendices

## A. Predicted number of events

In Tab. 29, 30 and 31, 32 the total number of events expected for the Asimov data set B (see table 12) is shown.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 69.747 | 0.035 | 4.335 | 0.002 | 0.151 | 0.001 | 74.270 |
| CC1pi | 17.206 | 0.017 | 1.572 | 0.001 | 0.049 | 0.000 | 18.845 |
| CCcoherent | 0.330 | 0.000 | 0.080 | 0.000 | 0.001 | 0.000 | 0.412 |
| MEC | 19.725 | 0.010 | 0.852 | 0.000 | 0.036 | 0.000 | 20.623 |
| CCother | 7.742 | 0.008 | 0.494 | 0.001 | 0.002 | 0.000 | 8.247 |
| NC1pi0 | 0.537 | 0.016 | 0.017 | 0.001 | N/A | N/A | 0.572 |
| NC1piPM | 3.944 | 0.089 | 0.142 | 0.008 | N/A | N/A | 4.183 |
| NCcoherent | 0.016 | 0.000 | 0.001 | 0.000 | N/A | N/A | 0.018 |
| NCother | 2.993 | 0.121 | 0.165 | 0.012 | N/A | N/A | 3.290 |
| NC1gamma | 0.004 | 0.000 | 0.000 | 0.000 | N/A | N/A | 0.004 |
| Total | 122.243 | 0.297 | 7.659 | 0.025 | 0.238 | 0.002 | 130.464 |

Table 29: Predicted number of events in the neutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $6.914\times10^{20}$ POT. The Asimov data set B in tab. 12 is used.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 7.088 | 0.004 | 13.306 | 0.005 | 0.004 | 0.016 | 20.422 |
| CC1pi | 2.718 | 0.003 | 3.066 | 0.002 | 0.002 | 0.003 | 5.794 |
| CCcoherent | 0.054 | 0.000 | 0.173 | 0.000 | 0.000 | 0.000 | 0.228 |
| MEC | 2.525 | 0.001 | 2.118 | 0.001 | 0.001 | 0.002 | 4.649 |
| CCother | 1.282 | 0.001 | 0.886 | 0.000 | 0.000 | 0.000 | 2.170 |
| NC1pi0 | 0.060 | 0.002 | 0.050 | 0.001 | N/A | N/A | 0.114 |
| NC1piPM | 0.378 | 0.015 | 0.380 | 0.010 | N/A | N/A | 0.784 |
| NCcoherent | 0.001 | 0.000 | 0.006 | 0.000 | N/A | N/A | 0.008 |
| NCother | 0.530 | 0.025 | 0.296 | 0.012 | N/A | N/A | 0.862 |
| NC1gamma | 0.000 | 0.000 | 0.000 | 0.000 | N/A | N/A | 0.000 |
| Total | 14.637 | 0.051 | 20.282 | 0.032 | 0.008 | 0.022 | 35.032 |

Table 30: Predicted number of events in the antineutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $4.011\times10^{20}$ POT. The Asimov data set B in tab. 12 is used.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.062 | 2.129 | 0.001 | 0.088 | 10.811 | 0.121 | 13.211 |
| CC1pi | 0.017 | 0.348 | 0.000 | 0.027 | 1.236 | 0.020 | 1.648 |
| CCcoherent | 0.000 | 0.002 | 0.000 | 0.002 | 0.011 | 0.002 | 0.017 |
| MEC | 0.009 | 0.710 | 0.000 | 0.022 | 2.750 | 0.022 | 3.513 |
| CCother | 0.001 | 0.045 | 0.000 | 0.003 | 0.025 | 0.001 | 0.076 |
| NC1pi0 | 0.524 | 0.012 | 0.017 | 0.001 | N/A | N/A | 0.553 |
| NC1piPM | 0.099 | 0.002 | 0.003 | 0.000 | N/A | N/A | 0.105 |
| NCcoherent | 0.174 | 0.004 | 0.018 | 0.001 | N/A | N/A | 0.197 |
| NCother | 0.126 | 0.005 | 0.008 | 0.001 | N/A | N/A | 0.139 |
| NC1gamma | 0.464 | 0.007 | 0.021 | 0.001 | N/A | N/A | 0.494 |
| Total | 1.476 | 3.264 | 0.069 | 0.145 | 14.833 | 0.167 | 19.954 |

Table 31: Predicted number of events in the neutrino mode $e$-like sample obtained after applying the BANFF weight for a total exposure of $6.914\times10^{20}$ POT. The Asimov data set B in tab. 12 is used.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.002 | 0.189 | 0.003 | 0.339 | 0.269 | 1.376 | 2.178 |
| CC1pi | 0.002 | 0.040 | 0.001 | 0.060 | 0.045 | 0.163 | 0.312 |
| CCcoherent | 0.000 | 0.000 | 0.000 | 0.006 | 0.000 | 0.022 | 0.029 |
| MEC | 0.001 | 0.069 | 0.000 | 0.063 | 0.087 | 0.219 | 0.439 |
| CCother | 0.001 | 0.008 | 0.000 | 0.004 | 0.003 | 0.003 | 0.019 |
| NC1pi0 | 0.045 | 0.002 | 0.049 | 0.001 | N/A | N/A | 0.098 |
| NC1piPM | 0.011 | 0.000 | 0.008 | 0.000 | N/A | N/A | 0.020 |
| NCcoherent | 0.021 | 0.001 | 0.081 | 0.001 | N/A | N/A | 0.103 |
| NCother | 0.022 | 0.001 | 0.012 | 0.000 | N/A | N/A | 0.035 |
| NC1gamma | 0.037 | 0.002 | 0.081 | 0.002 | N/A | N/A | 0.121 |
| Total | 0.142 | 0.313 | 0.235 | 0.478 | 0.404 | 1.783 | 3.355 |

Table 32: Predicted number of events in the antineutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $4.011\times10^{20}$ POT. The Asimov data set B in tab. 12 is used.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 338.683 | 0.037 | 8.672 | 0.002 | 0.000 | 0.000 | 347.394 |
| CC1pi | 39.668 | 0.018 | 2.276 | 0.001 | 0.000 | 0.000 | 41.963 |
| CCcoherent | 0.678 | 0.000 | 0.136 | 0.000 | 0.000 | 0.000 | 0.815 |
| MEC | 74.638 | 0.011 | 1.548 | 0.000 | 0.000 | 0.000 | 76.198 |
| CCother | 8.732 | 0.009 | 0.560 | 0.001 | 0.000 | 0.000 | 9.301 |
| NC1pi0 | 0.537 | 0.016 | 0.017 | 0.001 | N/A | N/A | 0.572 |
| NC1piPM | 3.944 | 0.089 | 0.142 | 0.008 | N/A | N/A | 4.183 |
| NCcoherent | 0.016 | 0.000 | 0.001 | 0.000 | N/A | N/A | 0.018 |
| NCother | 2.993 | 0.121 | 0.165 | 0.012 | N/A | N/A | 3.290 |
| NC1gamma | 0.004 | 0.000 | 0.000 | 0.000 | N/A | N/A | 0.004 |
| Total | 469.893 | 0.301 | 13.518 | 0.025 | 0.000 | 0.000 | 483.737 |

Table 33: Predicted number of events in the neutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $6.914\times10^{20}$ POT. Oscillations are not applied.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 15.420 | 0.004 | 55.525 | 0.005 | 0.000 | 0.000 | 70.954 |
| CC1pi | 4.071 | 0.003 | 6.536 | 0.002 | 0.000 | 0.000 | 10.611 |
| CCcoherent | 0.074 | 0.000 | 0.612 | 0.000 | 0.000 | 0.000 | 0.687 |
| MEC | 4.721 | 0.001 | 7.811 | 0.001 | 0.000 | 0.000 | 12.534 |
| CCother | 1.457 | 0.001 | 1.008 | 0.000 | 0.000 | 0.000 | 2.467 |
| NC1pi0 | 0.060 | 0.002 | 0.050 | 0.001 | N/A | N/A | 0.114 |
| NC1piPM | 0.378 | 0.015 | 0.380 | 0.010 | N/A | N/A | 0.784 |
| NCcoherent | 0.001 | 0.000 | 0.006 | 0.000 | N/A | N/A | 0.008 |
| NCother | 0.530 | 0.025 | 0.296 | 0.012 | N/A | N/A | 0.862 |
| NC1gamma | 0.000 | 0.000 | 0.000 | 0.000 | N/A | N/A | 0.000 |
| Total | 26.713 | 0.052 | 72.225 | 0.032 | 0.000 | 0.000 | 99.021 |

Table 34: Predicted number of events in the antineutrino mode $\mu$-like sample obtained after applying the BANFF weight for a total exposure of $4.011\times10^{20}$ POT. Oscillations are not applied.

In Tab. 33, 34 and 35, 36 the total number of unoscillated events is shown.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.413 | 2.294 | 0.002 | 0.094 | 0.000 | 0.000 | 2.803 |
| CC1pi | 0.069 | 0.371 | 0.001 | 0.028 | 0.000 | 0.000 | 0.469 |
| CCcoherent | 0.002 | 0.003 | 0.000 | 0.002 | 0.000 | 0.000 | 0.006 |
| MEC | 0.095 | 0.759 | 0.000 | 0.023 | 0.000 | 0.000 | 0.878 |
| CCother | 0.004 | 0.046 | 0.000 | 0.003 | 0.000 | 0.000 | 0.054 |
| NC1pi0 | 0.524 | 0.012 | 0.017 | 0.001 | N/A | N/A | 0.553 |
| NC1piPM | 0.099 | 0.002 | 0.003 | 0.000 | N/A | N/A | 0.105 |
| NCcoherent | 0.174 | 0.004 | 0.018 | 0.001 | N/A | N/A | 0.197 |
| NCother | 0.126 | 0.005 | 0.008 | 0.001 | N/A | N/A | 0.139 |
| NC1gamma | 0.464 | 0.007 | 0.021 | 0.001 | N/A | N/A | 0.494 |
| Total | 1.970 | 3.504 | 0.071 | 0.154 | 0.000 | 0.000 | 5.698 |

Table 35: Predicted number of events in the neutrino mode $e$-like sample obtained after applying the BANFF weight for a total exposure of $6.914\times10^{20}$ POT. Oscillations are not applied.

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu_\mu}$ | $\bar{\nu_e}$ | Osc. $\nu_e$ | Osc. $\bar{\nu_e}$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.012 | 0.203 | 0.018 | 0.361 | 0.000 | 0.000 | 0.593 |
| CC1pi | 0.004 | 0.043 | 0.003 | 0.064 | 0.000 | 0.000 | 0.114 |
| CCcoherent | 0.000 | 0.000 | 0.000 | 0.006 | 0.000 | 0.000 | 0.007 |
| MEC | 0.003 | 0.074 | 0.003 | 0.067 | 0.000 | 0.000 | 0.147 |
| CCother | 0.001 | 0.009 | 0.000 | 0.004 | 0.000 | 0.000 | 0.015 |
| NC1pi0 | 0.045 | 0.002 | 0.049 | 0.001 | N/A | N/A | 0.098 |
| NC1piPM | 0.011 | 0.000 | 0.008 | 0.000 | N/A | N/A | 0.020 |
| NCcoherent | 0.021 | 0.001 | 0.081 | 0.001 | N/A | N/A | 0.103 |
| NCother | 0.022 | 0.001 | 0.012 | 0.000 | N/A | N/A | 0.035 |
| NC1gamma | 0.037 | 0.002 | 0.081 | 0.002 | N/A | N/A | 0.121 |
| Total | 0.156 | 0.334 | 0.255 | 0.507 | 0.000 | 0.000 | 1.253 |

Table 36: Predicted number of events in the antineutrino mode $e$-like sample obtained after applying the BANFF weight for a total exposure of $4.011\times10^{20}$ POT. Oscillations are not applied.

## B. Predicted *e*-like spectra

In Fig. 36 each flavor component of the neutrino and antineutrino mode $e$-like samples is shown separately in order to compare the different shapes as a function of $E_{reco}$. The NC spectrum distribution, which is the largest component of the background, is quite flat over all the energy range and is quite difficult to separate. The $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events populate the same energy region, around 0.6 GeV, with a very similar shape.

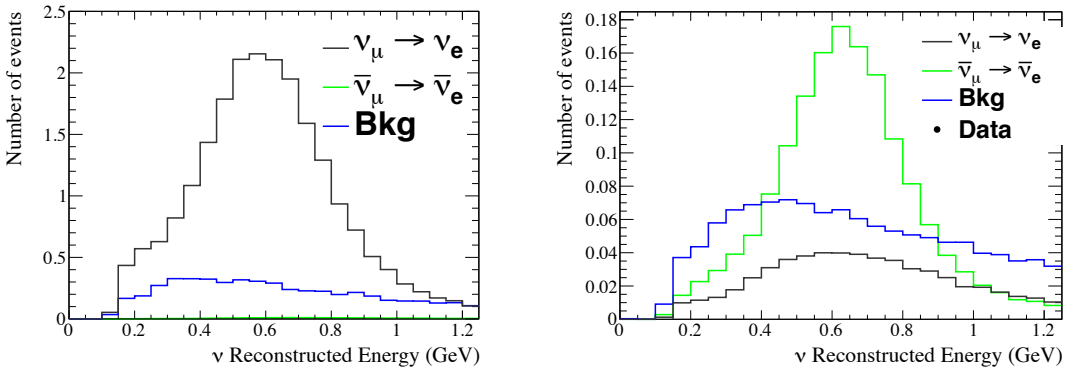



Figure 36: Predicted spectra of reconstructed neutrino energy for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of $6.914 \times 10^{20}$ POT in neutrino mode and of $4.011 \times 10^{20}$ POT in antineutrino mode. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), background, which includes NC, $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), $\nu_\mu$ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

Another kinematical variable is thus needed in order to better separate different flavour / reaction compoonents. In Fig. 40 and 41 the 2-dimensional $E_{reco}$ Vs $\theta$ spectra are shown for each flavour component and NC for both neutrino and antineutrino mode $e$-like samples. In particular the shapes of $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ distributions are quite different, thus this templates have a better capability to discriminate signal from background. In general this template parametrization produces quite different shapes for all the flavour and NC components.

In Fig. 39 the projection of the 2-dimensional $e$-like templates onto the $E_{reco}$ and $\theta$ axes are shown.

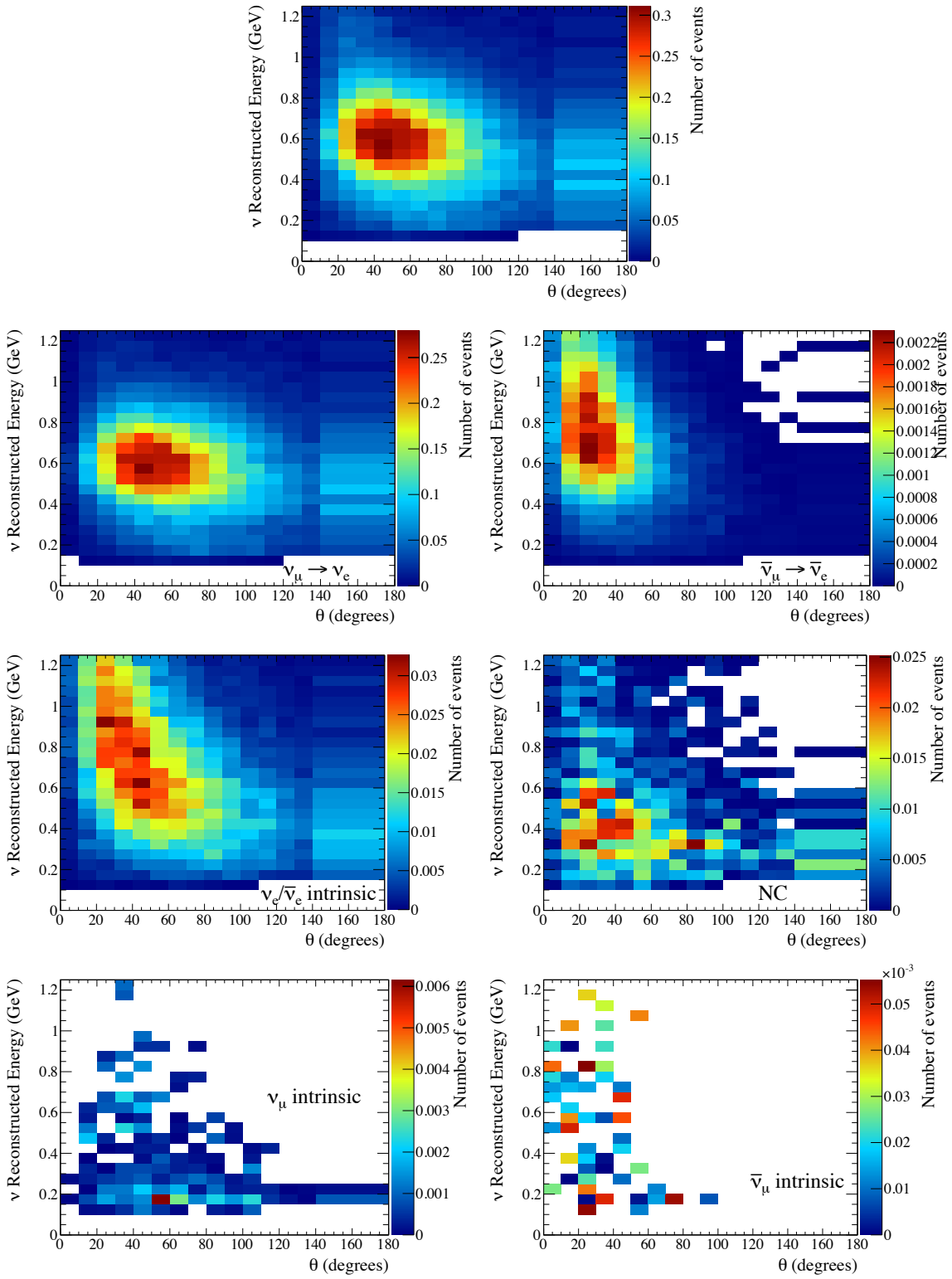



Figure 37: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of $6.914 \times 10^{20}$ POT in neutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, $\nu_\mu$ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

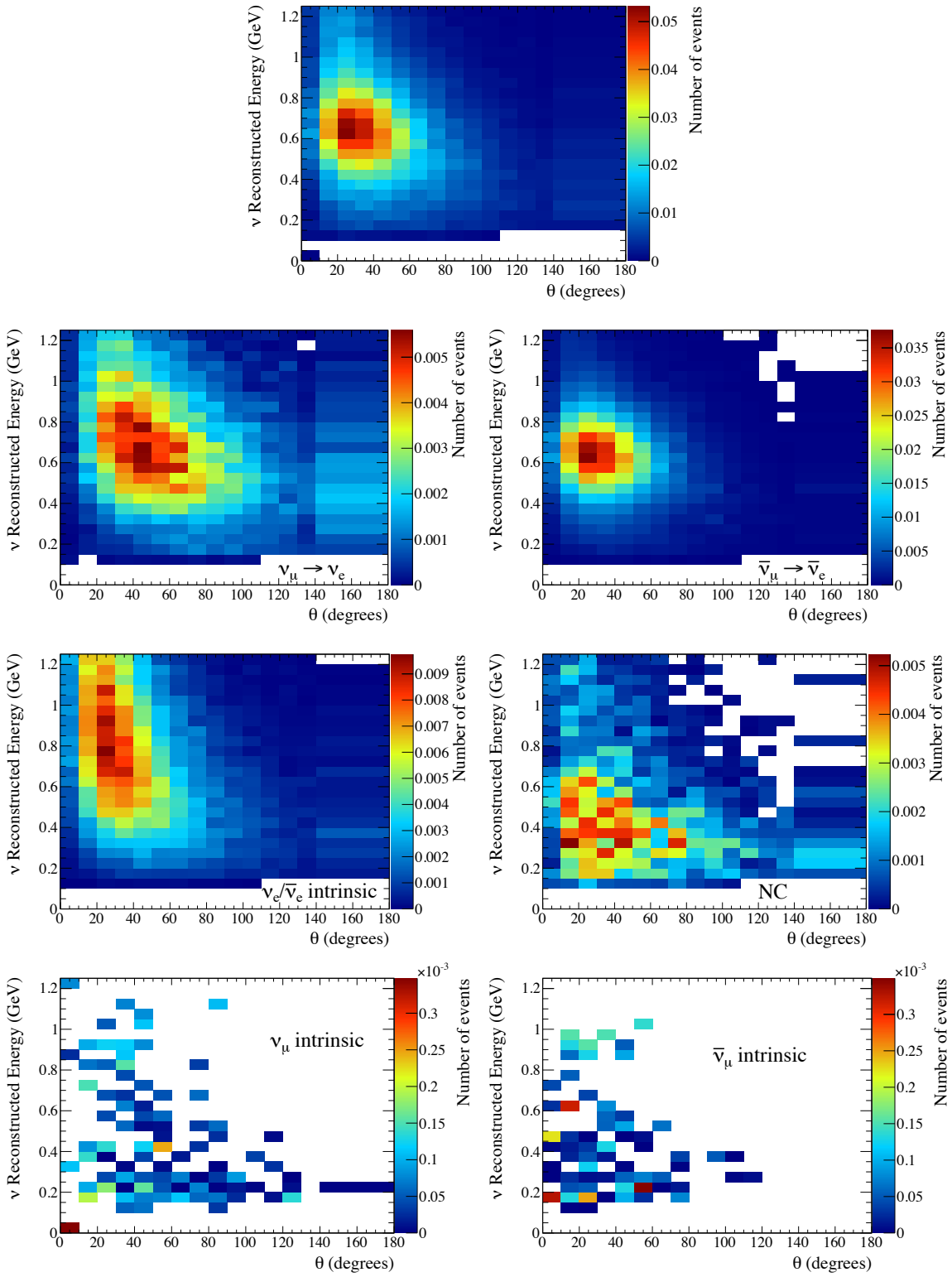



Figure 38: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of $4.011 \times 10^{20}$ POT in antineutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, $\nu_\mu$ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

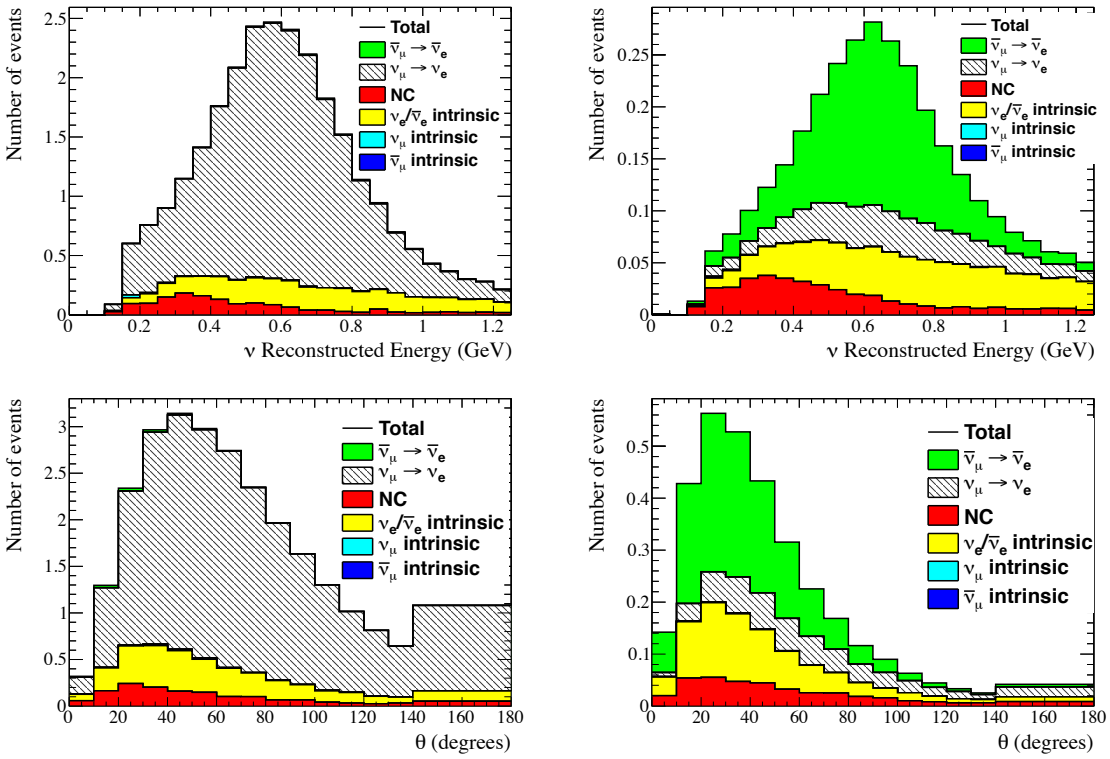



Figure 39: Predicted spectra of the reconstructed neutrino energy (top) and the angle between the neutrino and the outgoing lepton directions (bottom) for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of $6.914 \times 10^{20}$ POT in neutrino mode and of $4.011 \times 10^{20}$ POT in antineutrino mode. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), $\nu_\mu$ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

## C. Observed *e*-like spectra

In Fig. 40 and 41 the predicted $e$-like 2-dimensional spectra are compared to data for both neutrino and antineutrino beam mode.

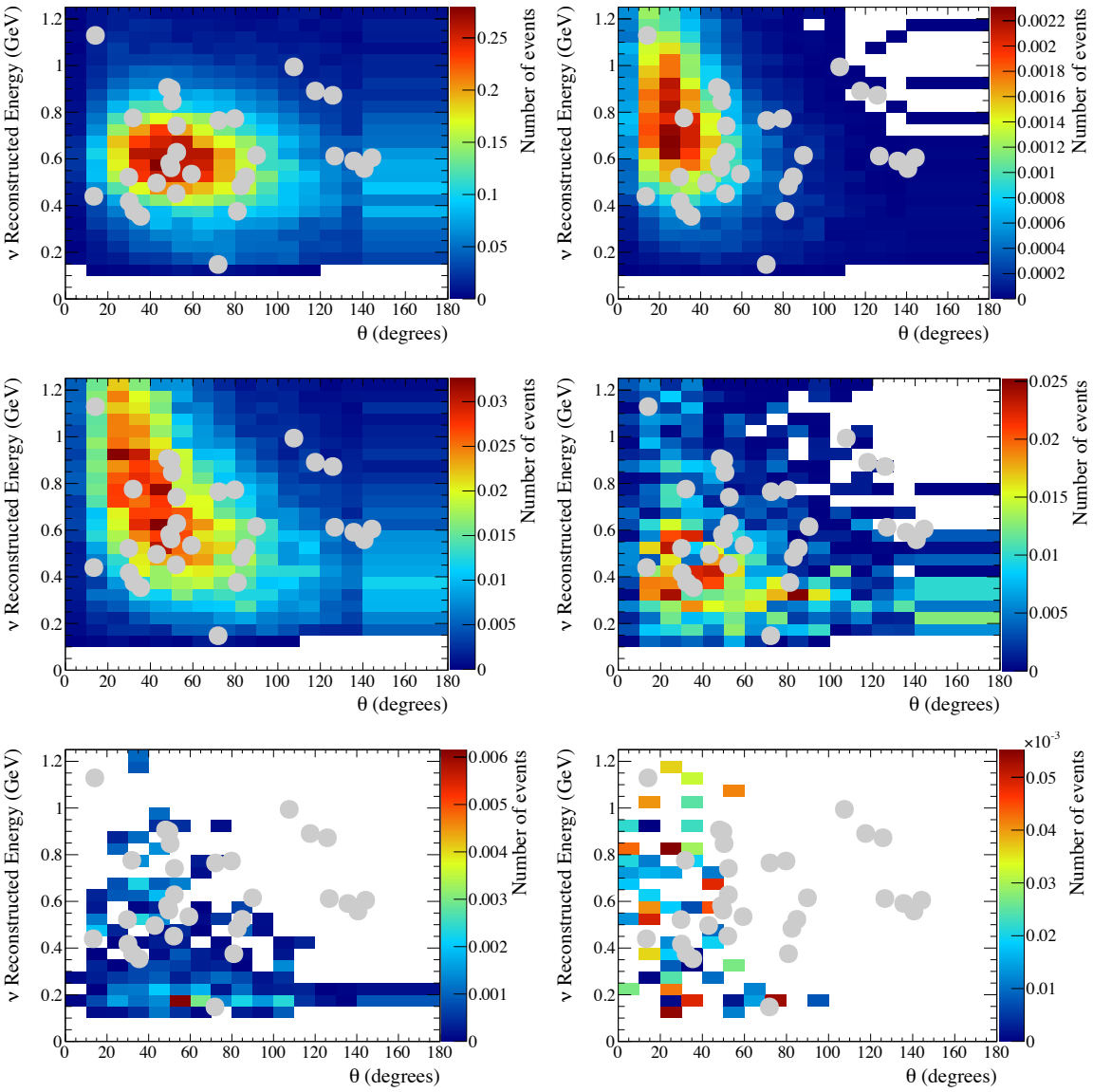



Figure 40: Predicted and observed (grey dots) spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of $6.914 \times 10^{20}$ POT in neutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, $\nu_\mu$ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

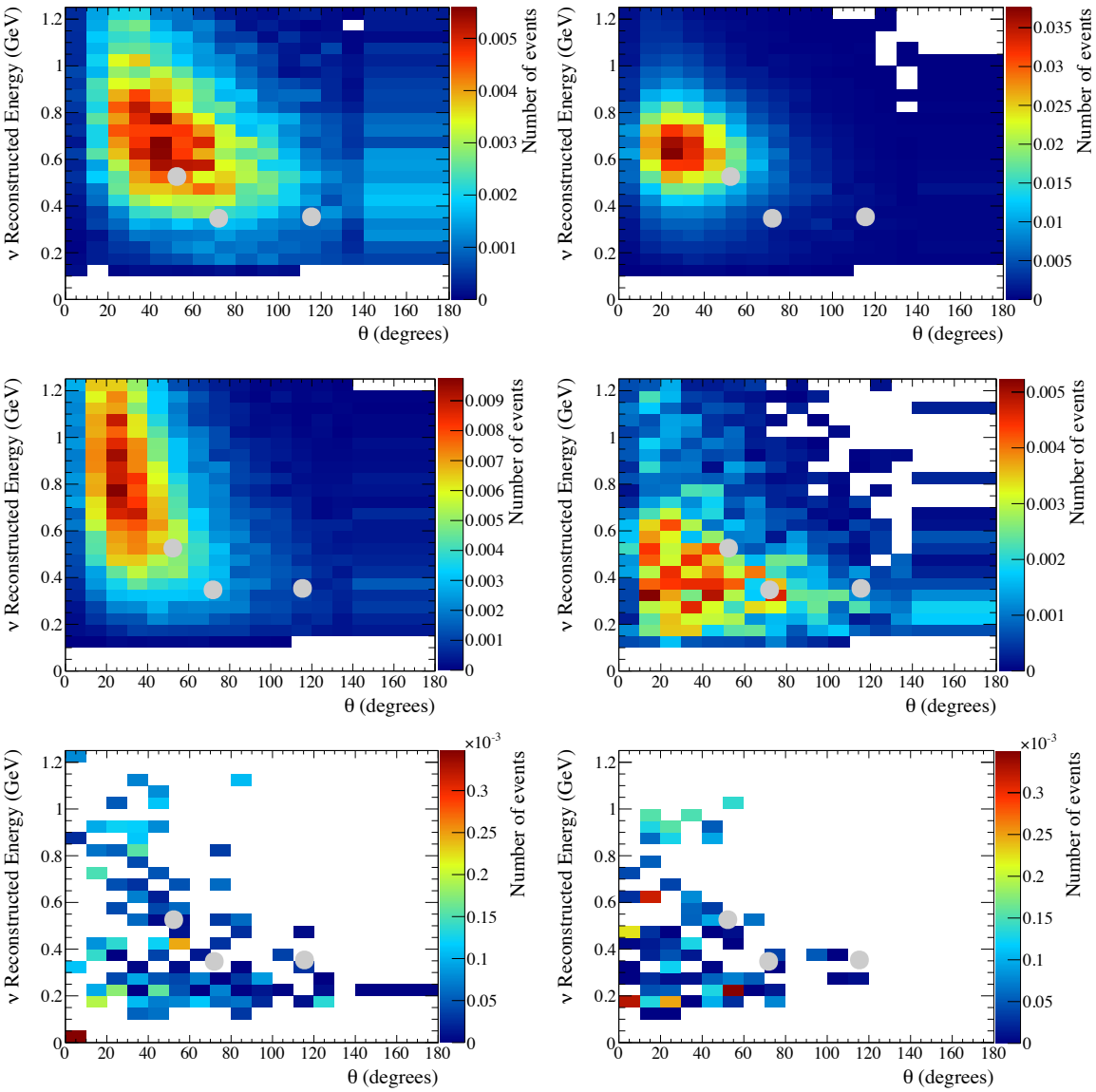



Figure 41: Predicted and observed (grey dots) spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure of $4.011 \times 10^{20}$ POT in antineutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, $\nu_\mu$ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

## D. Effect of systematic uncertainties

In table 37, table 38, table 39 and table 38 it is shown which systematic parameter is applied to which reaction and flavour modes for all the Super-K samples.

| | $\sin^2(\theta_{12})$ | $\sin^2(\theta_{13})$ | $\Delta m^2_{21}$ | $\delta_{CP}$ | $f^{SK}_{E;r}$ | $f^{SK+FSI}_{0;t,r}$ | $f^{SK+FSI}_{1;t,r}$ | $f^{SK+FSI}_{2;t,r}$ | $f^{SK+FSI}_{3;t,r}$ | $f^{SK+FSI}_{4;t,r}$ | $f^{SK+FSI}_{5;t,r}$ | $f^{banff}_{0;t,r}$ | $f^{banff}_{1;t,r}$ | $f^{banff}_{2;t,r}$ | $f^{banff}_{3;t,r}$ | $f^{banff}_{4;t,r}$ | $f^{banff}_{5;t,r}$ | $f^{banff}_{6;t,r}$ | $f^{banff}_{7;t,r}$ | $f^{banff}_{8;t,r}$ | $f^{banff}_{9;t,r}$ | $f^{banff}_{10;t,r}$ | $f^{banff}_{11;t,r}$ | $f^{banff}_{12;t,r}$ | $f^{banff}_{13;t,r}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\nu_\mu$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu/\nu_\tau$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu/\nu_\tau$ NC $1\pi^0$ | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu/\nu_\tau$ NC coherent | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_\mu/\nu_\tau$ NC other | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\nu_e$ NC $1\gamma$ | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| $\bar{\nu}_\mu$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu$ NC $1\gamma$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ |
| $\nu_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | | |
| $\nu_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ NC $1\pi^0$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ NC coherent | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\nu_e$ NC other | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\bar{\nu}_\mu$ NC $1\gamma$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\nu_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| Osc. $\nu_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| Osc. $\nu_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| Osc. $\nu_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| Osc. $\nu_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | |
| Osc. $\bar{\nu}_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\bar{\nu}_e$ CC $1\pi^{+/-}$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\bar{\nu}_e$ CC $1\pi^0$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\bar{\nu}_e$ CC coherent | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\bar{\nu}_e$ CC other | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| $\bar{\nu}_e$ NC $1\gamma$ | | | | | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ | | | | | | | | | | | | | | |

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| $f^{banff}_{7;t,r}$ | $f^{banff}_{6;t,r}$ | $f^{banff}_{5;t,r}$ | $f^{banff}_{4;t,r}$ | $f^{banff}_{3;t,r}$ | $f^{banff}_{2;t,r}$ | $f^{banff}_{1;t,r}$ | $f^{banff}_{0;t,r}$ | $f^{SK+FSI}_{17;t,r}$ | $f^{SK+FSI}_{16;t,r}$ | $f^{SK+FSI}_{15;t,r}$ | $f^{SK+FSI}_{14;t,r}$ | $f^{SK+FSI}_{13;t,r}$ | $f^{SK+FSI}_{12;t,r}$ | $f^{SK+FSI}_{11;t,r}$ | $f^{SK+FSI}_{10;t,r}$ | $f^{SK+FSI}_{9;t,r}$ | $f^{SK+FSI}_{8;t,r}$ | $f^{SK+FSI}_{7;t,r}$ | $f^{SK+FSI}_{6;t,r}$ | $f^{SK}_{E;r}$ | $\delta_{CP}$ | $\Delta m^2_{21}$ | $\sin^2(\theta_{13})$ | $\sin^2(\theta_{12})$ | |
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| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CCQE |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC $1\pi$ |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC coherent |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC MEC |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC other |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_\mu/\nu_\tau$ NC $1\pi^{+/-}$ |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_\mu/\nu_\tau$ NC $1\pi^0$ |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_\mu/\nu_\tau$ NC coherent |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_\mu/\nu_\tau$ NC other |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_e$ NC $1\gamma$ |
| | | | | | | | | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CCQE |
| | | | | | | | | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC $1\pi$ |
| | | | | | | | | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC coherent |
| | | | | | | | | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC MEC |
| | | | | | | | | | | | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC other |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$ |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$ |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_\mu$ NC $1\gamma$ |
| | | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CCQE |
| | | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CC $1\pi$ |
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| | | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CC other |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_e$ NC $1\pi^{+/-}$ |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_e$ NC $1\pi^0$ |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_e$ NC coherent |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\nu_e$ NC other |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_\mu$ NC $1\gamma$ |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CCQE |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CC $1\pi$ |
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| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CC other |
| | | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CCQE |
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| | | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CC coherent |
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| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC $1\pi^{+/-}$ |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC $1\pi^0$ |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC coherent |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC other |
| | | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC $1\gamma$ |
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| | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\bar{\nu}_e$ CC MEC |
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| $f^{banff\ RHC}_{13;t,r}$ | $f^{banff\ RHC}_{12;t,r}$ | $f^{banff\ RHC}_{11;t,r}$ | $f^{banff\ RHC}_{10;t,r}$ | $f^{banff\ RHC}_{9;t,r}$ | $f^{banff\ RHC}_{8;t,r}$ | $f^{banff\ RHC}_{7;t,r}$ | $f^{banff\ RHC}_{6;t,r}$ | $f^{banff\ RHC}_{5;t,r}$ | $f^{banff\ RHC}_{4;t,r}$ | $f^{banff\ RHC}_{3;t,r}$ | $f^{banff\ RHC}_{2;t,r}$ | $f^{banff\ RHC}_{1;t,r}$ | $f^{banff\ RHC}_{0;t,r}$ | $f^{SK+FSI\ RHC}_{5;t,r}$ | $f^{SK+FSI\ RHC}_{4;t,r}$ | $f^{SK+FSI\ RHC}_{3;t,r}$ | $f^{SK+FSI\ RHC}_{2;t,r}$ | $f^{SK+FSI\ RHC}_{1;t,r}$ | $f^{SK+FSI\ RHC}_{0;t,r}$ | $f^{SK}_{E;r}$ | $\delta_{CP}$ | $\Delta m^2_{21}$ | $\sin^2(\theta_{13})$ | $\sin^2(\theta_{12})$ | |
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| | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CCQE |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | ✓ | ✓ | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC $1\pi$ |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | ✓ | ✓ | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC coherent |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC MEC |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | ✓ | ✓ | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_\mu$ CC other |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | | | | $\nu_\mu/\nu_\tau$ NC $1\pi^{+/-}$ |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | | | | $\nu_\mu/\nu_\tau$ NC $1\pi^0$ |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | | | | $\nu_\mu/\nu_\tau$ NC coherent |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | | | | $\nu_\mu/\nu_\tau$ NC other |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | | | | $\nu_e$ NC $1\gamma$ |
| ✓ | ✓ | ✓ | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CCQE |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC $1\pi$ |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC coherent |
| ✓ | ✓ | ✓ | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC MEC |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_\mu$ CC other |
| ✓ | ✓ | ✓ | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$ |
| ✓ | ✓ | ✓ | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$ |
| ✓ | ✓ | ✓ | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent |
| ✓ | ✓ | ✓ | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other |
| ✓ | ✓ | ✓ | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\nu_\mu$ NC $1\gamma$ |
| | | | | | | | | | | | | | | | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CCQE |
| | | | | | | | | | | | | | | | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CC $1\pi$ |
| | | | | | | | | | | | | | | | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CC coherent |
| | | | | | | | | | | | | | | | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CC MEC |
| | | | | | | | | | | | | | | | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\nu_e$ CC other |
| | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\nu_e$ NC $1\pi^{+/-}$ |
| | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\nu_e$ NC $1\pi^0$ |
| | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\nu_e$ NC coherent |
| | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\nu_e$ NC other |
| | | | | | | | | | | | | | | ✓ | | | | | | ✓ | | | | | $\bar{\nu}_\mu$ NC $1\gamma$ |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CCQE |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CC $1\pi$ |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CC coherent |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CC MEC |
| | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\nu_e$ CC other |
| | | | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CCQE |
| | | | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CC $1\pi$ |
| | | | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CC coherent |
| | | | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CC MEC |
| | | | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | $\bar{\nu}_e$ CC other |
| | | | | | | | | | | | | | | ✓ | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC $1\pi^{+/-}$ |
| | | | | | | | | | | | | | | ✓ | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC $1\pi^0$ |
| | | | | | | | | | | | | | | ✓ | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC coherent |
| | | | | | | | | | | | | | | ✓ | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC other |
| | | | | | | | | | | | | | | ✓ | | | | | | | ✓ | | | | | $\bar{\nu}_e$ NC $1\gamma$ |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\bar{\nu}_e$ CCQE |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\bar{\nu}_e$ CC $1\pi$ |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\bar{\nu}_e$ CC coherent |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\bar{\nu}_e$ CC MEC |
| ✓ | ✓ | ✓ | | | | | | | | | | | | | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Osc. $\bar{\nu}_e$ CC other |

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| | $\sin^2(\theta_{12})$ | $\sin^2(\theta_{13})$ | $\Delta m^2_{21}$ | $\delta_{CP}$ | $f^{SK}_{E;r}$ | $f^{SK+FSI\,RHC}_{6;t,r}$ | $f^{SK+FSI\,RHC}_{7;t,r}$ | $f^{SK+FSI\,RHC}_{8;t,r}$ | $f^{SK+FSI\,RHC}_{9;t,r}$ | $f^{SK+FSI\,RHC}_{10;t,r}$ | $f^{SK+FSI\,RHC}_{11;t,r}$ | $f^{SK+FSI\,RHC}_{12;t,r}$ | $f^{SK+FSI\,RHC}_{13;t,r}$ | $f^{SK+FSI\,RHC}_{14;t,r}$ | $f^{SK+FSI\,RHC}_{15;t,r}$ | $f^{SK+FSI\,RHC}_{16;t,r}$ | $f^{SK+FSI\,RHC}_{17;t,r}$ | $f^{banff\,RHC}_{0;t,r}$ | $f^{banff\,RHC}_{1;t,r}$ | $f^{banff\,RHC}_{2;t,r}$ | $f^{banff\,RHC}_{3;t,r}$ | $f^{banff\,RHC}_{4;t,r}$ | $f^{banff\,RHC}_{5;t,r}$ | $f^{banff\,RHC}_{6;t,r}$ | $f^{banff\,RHC}_{7;t,r}$ |
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| $\nu_\mu$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu/\nu_\tau$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu/\nu_\tau$ NC $1\pi^0$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu/\nu_\tau$ NC coherent | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_\mu/\nu_\tau$ NC other | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\nu_e$ NC $1\gamma$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_\mu$ CCQE | | | | | ✓ | | | | ✓ | | ✓ | | | | | | | | | | | | | |
| $\bar{\nu}_\mu$ CC $1\pi$ | | | | | ✓ | | | | ✓ | | ✓ | | | | | | | | | | | | | |
| $\bar{\nu}_\mu$ CC coherent | | | | | ✓ | | | | ✓ | | ✓ | | | | | | | | | | | | | |
| $\bar{\nu}_\mu$ CC MEC | | | | | ✓ | | | | ✓ | | ✓ | | | | | | | | | | | | | |
| $\bar{\nu}_\mu$ CC other | | | | | ✓ | | | | ✓ | | ✓ | | | | | | | | | | | | | |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\nu_\mu$ NC $1\gamma$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\nu_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\nu_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\nu_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\nu_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\nu_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\nu_e$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\nu_e$ NC $1\pi^0$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\nu_e$ NC coherent | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\nu_e$ NC other | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_\mu$ NC $1\gamma$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| Osc. $\nu_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Osc. $\nu_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Osc. $\nu_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Osc. $\nu_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Osc. $\nu_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| $\bar{\nu}_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\bar{\nu}_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\bar{\nu}_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\bar{\nu}_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\bar{\nu}_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | ✓ | ✓ | | | | | | | | | | |
| $\bar{\nu}_e$ NC $1\pi^{+/-}$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_e$ NC $1\pi^0$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_e$ NC coherent | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_e$ NC other | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| $\bar{\nu}_e$ NC $1\gamma$ | | | | | ✓ | | | | | | | | | | ✓ | ✓ | ✓ | | | | | | | |
| Osc. $\bar{\nu}_e$ CCQE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC $1\pi$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC coherent | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC MEC | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| Osc. $\bar{\nu}_e$ CC other | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $f^{banff}_{8;t,r}RHC$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | |
| $f^{banff}_{9;t,r}RHC$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | |
| $f^{banff}_{10;t,r}RHC$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | |
| $f^{banff}_{11;t,r}RHC$ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| $f^{banff}_{12;t,r}RHC$ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| $f^{banff}_{13;t,r}RHC$ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| $f^{banff}_{14;t,r}RHC$ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| $f^{banff}_{15;t,r}RHC$ | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| $f^{banff}_{16;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{17;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{18;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{19;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{20;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{21;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{22;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | | | | | | | |
| $f^{banff}_{23;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | |
| $f^{banff}_{24;t,r}RHC$ | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | |
| $f^{banff}_{M^{QE}_A}RHC$ | ✓ | | | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | |
| $f^{banff}_{M^{RES}_A}RHC$ | | ✓ | | | | ✓ | ✓ | | | | | ✓ | | | | ✓ | ✓ | | | | | ✓ | | | ✓ | ✓ | | | | ✓ | | | | ✓ | ✓ | | | | ✓ | | | | | |
| $f^{banff}_{Norm_{MEC}}RHC$ | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | | ✓ | | | | | | | | | | | | | | | | | | ✓ | | | |
| $f^{banff}_{C^A_5}RHC$ | | ✓ | | | | ✓ | ✓ | | | | | ✓ | | | | ✓ | ✓ | | | | | ✓ | | | ✓ | ✓ | | | | ✓ | | | | ✓ | ✓ | | | | ✓ | | | | | |
| $f^{banff}_{Bg_{RES}}RHC$ | | ✓ | | | | ✓ | ✓ | | | | | ✓ | | | | ✓ | ✓ | | | | | ✓ | | | ✓ | ✓ | | | | ✓ | | | | ✓ | ✓ | | | | ✓ | | | | | |
| $f^{banff}_{p_f}RHC$ | ✓ | | | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | |
| $f^{banff}_{Shape_{CCoth}}RHC$ | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | | | | | ✓ | | | | | | | | | | | | | | | | ✓ |
| $f^{banff}_{E_B}RHC$ | ✓ | | | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | |
| $f^{banff}_{Norm_{CCcoh}}RHC$ | | | ✓ | | | | | | | | | | ✓ | | | | | | | | | ✓ | | | | | | | | ✓ | | | ✓ | | | | | | | | ✓ | | | |
| $f^{banff}_{Norm_{NCcoh}}RHC$ | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | ✓ | | | | | | | | | | | | ✓ | | | | | | |
| $f^{banff}_{Norm_{NCother}}RHC$ | | | | | | | | | ✓ | | | | | | | | | | ✓ | | | | | | | | ✓ | | | | | | | | | | | | ✓ | | | | | |
| $f^{banff}_{Norm_{\nu_e\to\nu_\mu}}RHC$ | | | | | | | | | | | | | | | | | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | | | | |

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In tab. 41, 42, 43, 44 and 45 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set B.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov B) |
|---|---|
| SKDet+FSI+SI | 4.02% |
| Flux | 3.56% |
| XSec Corr Tot | 4.03% |
| MEC (corr) | 3.49% |
| MECbar (corr) | 0.20% |
| NC other (uncorr) | 0.77% |
| NC 1gamma (uncorr) | 0.00% |
| XSec nue/numu | 0.01% |
| XSec Tot | 4.10% |
| Flux+XSec | 2.85% |
| Flux+XSec+SKDet+FSI+SI | 4.96% |
| Oscillations | 0.10% |
| All | 4.96% |

Table 41: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov B) |
|---|---|
| SKDet+FSI+SI | 3.67% |
| Flux | 3.61% |
| XSec Corr Tot | 4.16% |
| MEC (corr) | 3.94% |
| MECbar (corr) | 0.07% |
| NC other (uncorr) | 0.21% |
| NC 1gamma (uncorr) | 1.94% |
| XSec nue/numu | 2.55% |
| XSec Tot | 5.26% |
| Flux+XSec | 4.29% |
| Flux+XSec+SKDet+FSI+SI | 5.68% |
| Oscillations | 4.18% |
| All | 7.19% |

Table 42: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

In fig. 42 and 43 respectively the absolute error envelope and the fractional error envelopes expressed as a ratio of event count are shown for the Asimov data set B (see table 12).

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov B |
|---|---|
| SKDet+FSI+SI | 3.83% |
| Flux | 3.70% |
| XSec Corr Tot | 4.10% |
| MEC (corr) | 2.95% |
| MECbar (corr) | 1.83% |
| NC other (uncorr) | 0.73% |
| NC 1gamma (uncorr) | 0.00% |
| XSec nue/numu | 0.00% |
| XSec Tot | 4.17% |
| Flux+XSec | 3.31% |
| Flux+XSec+SKDet+FSI+SI | 5.04% |
| Oscillations | 0.06% |
| All | 5.11% |

Table 43: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov B) |
|---|---|
| SKDet+FSI+SI | 3.88% |
| Flux | 3.81% |
| XSec Corr Tot | 4.52% |
| MEC (corr) | 2.92% |
| MECbar (corr) | 2.56% |
| NC other (uncorr) | 0.32% |
| NC 1gamma (uncorr) | 2.81% |
| XSec nue/numu | 1.68% |
| XSec Tot | 5.59% |
| Flux+XSec | 4.62% |
| Flux+XSec+SKDet+FSI+SI | 6.18% |
| Oscillations | 3.59% |
| All | 7.09% |

Table 44: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

| Source of uncertainty | $\mu$-like RMS/Mean | e-like RMS/Mean |
|---|---|---|
| SKDet | 0.6% | 1.2% |
| FSI+SI | 2.5% | 3.6% |
| SKDet+FSI+SI | 2.6% | 3.7% |
| Flux | 1.6% | 1.7% |
| XSec NC other (uncorr) | 0.0% | 0.1% |
| XSec NC 1$\gamma$ (uncorr) | 0.0% | 0.9% |
| XSec $\nu_e$ / $\nu_\mu$ (uncorr) | 0.0% | 3.3% |
| XSec Tot | 1.7% | 4.4% |
| Flux+XSec | 1.8% | 4.1% |
| Flux+XSec+SDet+FSI+SI | 3.1% | 5.6% |
| Oscillations | 0.0% | 0.6% |
| All | 3.2% | 5.6% |

Table 45: Effect of $1\sigma$ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

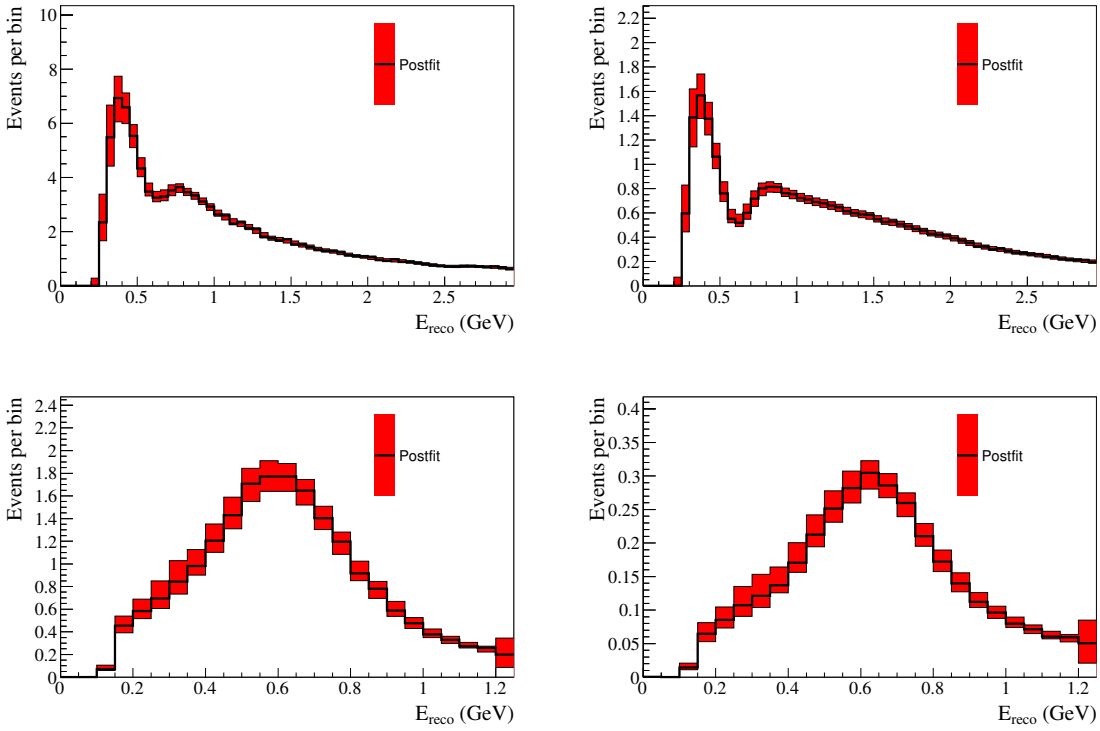



Figure 42: Error envelopes for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set B shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

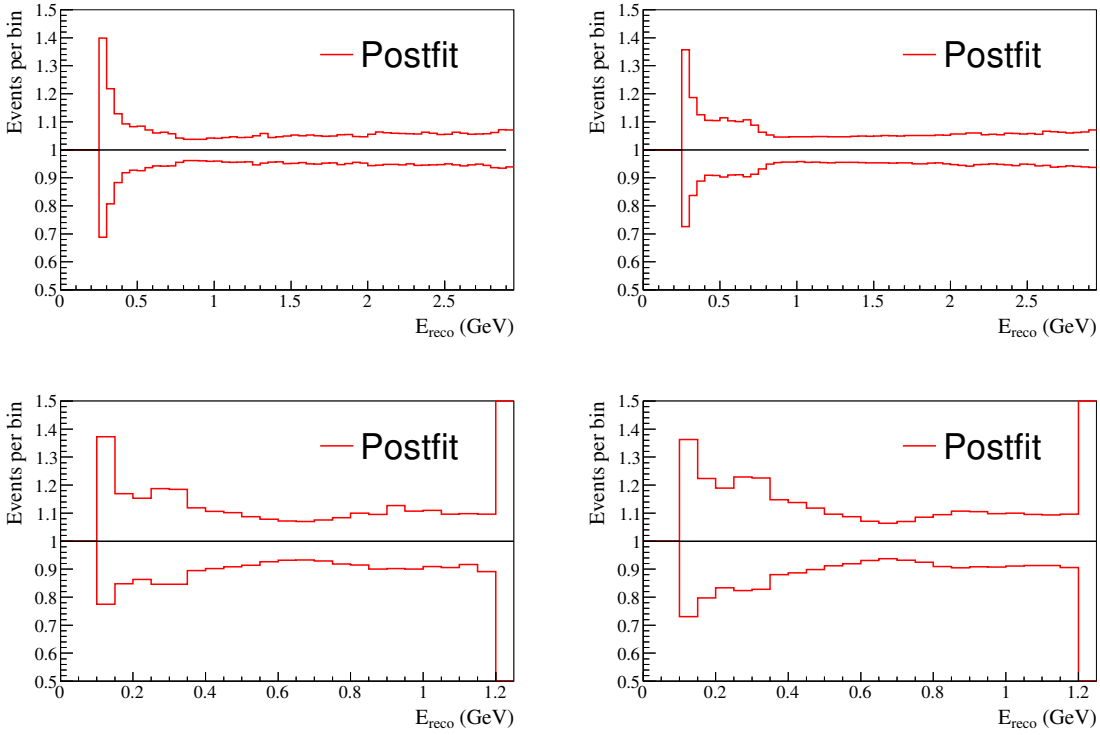



Figure 43: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set B shown in tab. 12 with randomized systematic parameters and the $1\sigma$ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the $1\sigma$ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account.

In tab. 46, 47, 48 and 49 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set A, but without taking into account the correction of the post-fit BANFF systematic parameters as well the additional covariance matrix that take into account the bias in the flux parameters of the BANFF fit. The impact of the uncertainty due to the bias of the BANFF fit is small and has a small impact compared to the other systematic uncertainties.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 4.11% |
| Flux | 3.26% |
| XSec Corr Tot | 3.96% |
| MEC (corr) | 3.53% |
| MECbar (corr) | 0.20% |
| NC other (uncorr) | 0.79% |
| NC 1gamma (uncorr) | 0.00% |
| XSec nue/numu | 0.01% |
| XSec Tot | 4.04% |
| Flux+XSec | 2.75% |
| Flux+XSec+SKDet+FSI+SI | 5.03% |
| Oscillations | 0.03% |
| All | 5.00% |

Table 46: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 3.47% |
| Flux | 3.40% |
| XSec Corr Tot | 4.17% |
| MEC (corr) | 3.95% |
| MECbar (corr) | 0.05% |
| NC other (uncorr) | 0.16% |
| NC 1gamma (uncorr) | 1.49% |
| XSec nue/numu | 2.63% |
| XSec Tot | 5.15% |
| Flux+XSec | 4.08% |
| Flux+XSec+SKDet+FSI+SI | 5.51% |
| Oscillations | 4.19% |
| All | 6.93% |

Table 47: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 3.94% |
| Flux | 3.40% |
| XSec Corr Tot | 4.20% |
| MEC (corr) | 3.00% |
| MECbar (corr) | 1.76% |
| NC other (uncorr) | 0.76% |
| NC 1gamma (uncorr) | 0.00% |
| XSec nue/numu | 0.00% |
| XSec Tot | 4.27% |
| Flux+XSec | 2.90% |
| Flux+XSec+SKDet+FSI+SI | 4.90% |
| Oscillations | 0.03% |
| All | 4.91% |

Table 48: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

| **Source of uncertainty** | $\delta N_{SK}/N_{SK}$ (Asimov A) |
|---|---|
| SKDet+FSI+SI | 3.94% |
| Flux | 3.53% |
| XSec Corr Tot | 4.44% |
| MEC (corr) | 3.11% |
| MECbar (corr) | 2.24% |
| NC other (uncorr) | 0.33% |
| NC 1gamma (uncorr) | 2.98% |
| XSec nue/numu | 1.52% |
| XSec Tot | 5.57% |
| Flux+XSec | 4.39% |
| Flux+XSec+SKDet+FSI+SI | 6.11% |
| Oscillations | 3.98% |
| All | 7.24% |

Table 49: Effect of $1\sigma$ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters $\theta_{23}$ and $\Delta m^2_{32}$ ($\Delta m^2_{13}$) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and $\Delta m^2_{21}$ are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

## E. Method for significance calculation

The significance for a certain oscillation hypothesis given by the measured data is provided as following. First the $\Delta\chi^2$ distribution is obtained by many toy experiments (e.g. $10k$), then the $\Delta\chi^2_{data}$ value is measured with the data set. This quantities are shown in fig. 44.

First the p-value is calculated as:

$$p - value = \int_{-\infty}^{\Delta\chi^2_{data}} \mathcal{F}(\Delta\chi^2) d(\Delta\chi^2) \tag{20}$$

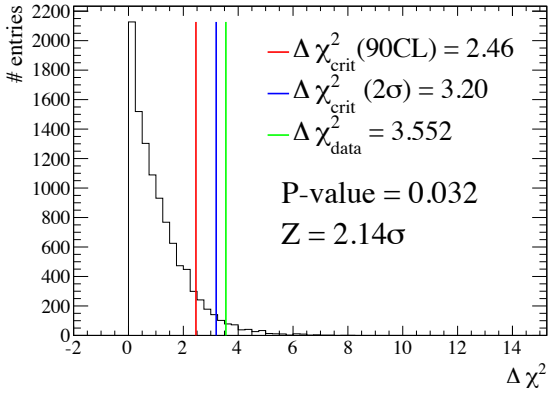



Figure 44: Distributions of $\Delta\chi^2$. $10k$ toy experiments are performed with true values of $\delta_{CP} = 0$ (left) and $\pi$ (right), assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and $\delta_{CP}$ are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set. The $\Delta\chi^2$ value (green line) measured from the Run 1-6 data set as well as the p-value and the significance as number of standard deviations (Z), are shown as well.

Then the significance can be shown as the number of standard deviations where the measured quantity is assumed to be gaussian distributed. In fig. 45 the gaussian distribution of the measured variable $\mu$ is shown as a function of the $Z$-score, i.e. how many standard deviations a certain measurement $x$ is far from the expected value:

$$Z = \frac{x - \mu}{\sigma} \tag{21}$$

If an experiment measures $x$, $\alpha$ is the area of the p.d.f. corresponding to a certain CL (90% in the example). The quantity $\sigma$ is the bias of the measured value with respect to the expectation that contains $1 - \alpha = 68.27\%$ of the total area.

The significance can be reported with the $Z$-score in the following way:

- the p-value is measured using eq. (20)
- Assuming the p-value to be 1-sided (see fig. 44) and $\alpha$ 2-sided (see fig. 45), we set $\alpha$ = p-value
- $Z$, shown in eq. (21), is computed using the quantile distribution. Given the cumulative distribution function of the normal standard distribution with Mean=0 and RMS=1 ($\sigma = 1$),

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_Z^{+\infty} e^{-t^2/2} dt = \alpha/2 \tag{22}$$

the quantile is defined as

$$Z = \Phi^{-1}(1 - \alpha/2). \tag{23}$$

The ROOT function Math::gaussian_quantile, that takes $\alpha$ as input, and calculate the cumulative distribution function of the lower tail of the normal (Gaussian) distribution is defined as following:

TF1 quantile("quantile","ROOT::Math::gaussian_quantile(1.-($\alpha$/2.),1.)",0.,1.);
double Zscore = quantile.Eval(pvalue);

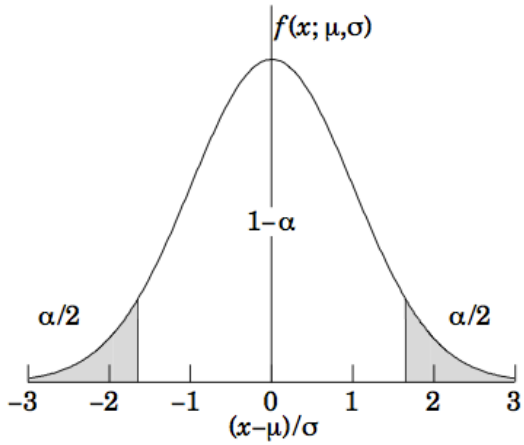



Figure 45: Illustration of a symmetric 90% confidence interval (unshaded) for a measurement of a single quantity with Gaussian errors. Integrated probabilities, defined by $\alpha = 0.1$, are shown.